Lecture notes for GGD 2021

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Abstract

This file gives an overview of the topics covered in the 2021 Gauge-Gravity Duality (GGD) course at NBI. More material will be added as the course progresses. Please contact us with any typos, comments or suggestions.

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1 Field theory

We start off by reviewing and introducing some aspects of quantum field theory. Following Ammon and Erdmenger [1], and the conventions of the field, we'll work in 'mostly plus' signature, so some signs might be different than what you're used to. (One reason for this is that it's a bit easier to generalizes to higher dimensions.) In this section and the next, we'll also denote the total number of spacetime dimensions by d.

1.1 Ward identities

You've seen the 'original' Ward identity associated to $U(1)$ global symmetry in QED in your QFT course. In fact, you showed how it can be derived from the Schwinger-Dyson equation for the fermion two-point function.

We will now review the derivation of the general Schwinger-Dyson equation and then use it to derive a Ward identity for general symmetries.

Schwinger-Dyson equation Let's assume we have some general action $S[\phi]$ for a field $\phi(x)$. A generating function for its correlators in terms of a source $J(x)$ is given by

$$Z[J] = \int \mathcal{D}\phi \, e^{i\left(S[\phi] + \int d^d x J(x)\phi(x)\right)} \tag{1.1}$$

From this, for example the two-point function is given by

$$\langle \phi(x_1)\phi(x_2)\rangle = (-i)^2 \frac{\delta^2}{\delta J(x_1)\delta J(x_2)} Z[J]\bigg|_{J=0}. \tag{1.2}$$

Now let's consider a general field transformation

$$\phi' = \phi + \delta\phi. \tag{1.3}$$

Since the path integral integrates over all field configurations anyway, we might as well write it in terms of the transformed field ϕ',

$$Z[J] = \int \mathcal{D}\phi' \, e^{i\left(S[\phi'] + \int d^d x J(x)\phi'(x)\right)}. \tag{1.4}$$

Now if we assume that $\mathcal{D}\phi' = \mathcal{D}\phi$, we can expand the RHS in $\delta\phi$ and rewrite it as

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi e^{iS[\phi]} \left(1 + i\int d^d x \frac{\delta S[\phi]}{\delta\phi(x)}\delta\phi(x)\right) && \text{(1.5a)} \\ &\qquad \times e^{i\int d^d y J(y)\phi(y)} \left(1 + i\int d^d x J(x)\delta\phi(x)\right) + \mathcal{O}(\delta\phi^2) \\ &= Z[J] + i\int \mathcal{D}\phi e^{i\left(S[\phi] + \int d^d y J(y)\phi(y)\right)} && \text{(1.5b)} \\ &\qquad \times \int d^d x \left(\frac{\delta S[\phi]}{\delta\phi(x)} + J(x)\right)\delta\phi(x) + \mathcal{O}(\delta\phi^2). \end{aligned}$$

Cancelling the $Z[J]$'s on both sides and dropping higher-order terms we get

$$i \int \mathcal{D}\phi e^{i\left(S[\phi]+\int d^d y J(y)\phi(y)\right)} \int d^d x \left(\frac{\delta S[\phi]}{\delta\phi(x)} + J(x)\right)\delta\phi(x) = 0. \tag{1.6}$$

This is the general Schwinger–Dyson equation that results from a small perturbation.

You can think of this equation as a generating function for 'equations of motion' for correlators. For example, deriving it twice using

$$(-i)^2 \frac{\delta^2}{\delta J(x_1)\delta J(x_2)} \tag{1.7}$$

we get the identity

$$i \int d^d x \left\langle \frac{\delta S[\phi]}{\delta\phi(x)}\delta\phi(x)\phi(x_1)\phi(x_2)\right\rangle + \langle\phi(x_1)\delta\phi(x_2)\rangle + \langle\delta\phi(x_1)\phi(x_2)\rangle = 0. \tag{1.8}$$

Solving the Schwinger-Dyson equation in an interacting theory is ultimately equivalent to doing the corresponding problem in terms of Feynman diagrams. However, this way of writing things gives a useful perspective on symmetries of the quantum theory.

Noether currents Let's recall how we can associate a conserved current to a symmetry $\delta_\epsilon\phi(x)$ of an action $S[\phi]$. One way to derive the corresponding Noether current is by allowing the parameter of this symmetry transformation to depend on spacetime, $\epsilon = \epsilon(x)$, so that we can write

$$\delta_{\epsilon(x)} S[\phi] = \int d^d x \frac{\delta S}{\delta\phi(x)}\delta_{\epsilon(x)}\phi(x) = -\int d^d x\, \partial_\mu \epsilon(x) j^\mu(x) \tag{1.9a}$$

$$= +\int d^d x\, \epsilon(x)\partial_\mu j^\mu(x). \tag{1.9b}$$

The first equality is a variational identity, and the second equality follows from the fact that the LHS vanishes if we take ϵ to be constant, since we know that $\delta_\epsilon\phi$ is a symmetry of the action for constant ϵ. In the last equality, we just integrate by parts (and assume that the boundary term vanishes).

Generally, such a spacetime-dependent transformation will no longer be a symmetry of the action. However, if $\phi(x)$ satisfies the equations of motion (also known as being 'on-shell'), the LHS vanishes by definition, so that

$$0 = \int d^d x \frac{\delta S}{\delta\phi(x)}\delta_{\epsilon(x)}\phi(x) = +\int d^d x\, \epsilon(x)\partial_\mu j^\mu(x) \tag{1.10}$$

Since $\epsilon(x)$ can be an arbitrary function, this means that $\partial_\mu j^\mu(x) = 0$, so we can identify $j^\mu(x)$ with the Noether current corresponding to our symmetry.

Ward identities The above is a purely 'classical' argument: in the path integral, we need to be more careful, and $\partial_\mu j^\mu(x)$ actually does not vanish identically in general correlators. Instead, symmetries manifest themselves in the quantum theory as *Ward identities* that correlators have to satisfy.

We can derive these identities using the Schwinger–Dyson equation, for example in the form of Equation (1.8). If we replace the general variations in (1.8) by the transformations $\delta_{\epsilon(x)}\phi(x)$, we can use (1.9b) to write it as

$$i\int d^dx\epsilon(x)\langle\partial_\mu j^\mu(x)\phi(x_1)\phi(x_2)\rangle + \langle\phi(x_1)\delta_{\epsilon(x_2)}\phi(x_2)\rangle + \langle\delta_{\epsilon(x_1)}\phi(x_1)\phi(x_2)\rangle = 0. \quad (1.11)$$

If we now restrict ourselves to the symmetry transformations $\epsilon(x) = \epsilon$, the first term is the integral of a total derivative with respect to x^μ and hence vanishes (assuming there are no boundary terms), so we get

$$\langle\phi(x_1)\delta_\epsilon\phi(x_2)\rangle + \langle\delta_\epsilon\phi(x_1)\phi(x_2)\rangle = 0. \quad (1.12)$$

Let's work out what this means for the two-point function of a scalar field in a translation-invariant theory. Then the infinitesimal transformation is

$$\delta_\epsilon\phi(x) = \epsilon a^\mu\partial_\mu\phi(x), \quad (1.13)$$

for a fixed but arbitrary vector a^μ. As a result, the two-point function satisfies

$$\epsilon a^\mu\left(\frac{\partial}{\partial x_1^\mu} + \frac{\partial}{\partial x_2^\mu}\right)\langle\phi(x_1)\phi(x_2)\rangle = 0. \quad (1.14)$$

This is the Ward identity for the two-point function of $\phi(x)$ that follows from translation symmetry. We can solve this equation for the two-point function $f(u,v) = \langle\phi(x_1)\phi(x_2)\rangle$ as follows. If we introduce the null coordinates $u^\mu = x_1^\mu + x_2^\mu$ and $v^\mu = x_1^\mu - x_2^\mu$, we get

$$\frac{\partial}{\partial u^\mu}f(u,v) = 0 \quad \Longrightarrow \quad \langle\phi(x_1)\phi(x_2)\rangle = f(x_1 - x_2). \quad (1.15)$$

So translation symmetry means the two-point function only depends on the difference between the two insertion coordinates. This is perhaps not a particularly spectacular result, but Ward identities are very useful to determine the consequences of more complicated symmetry transformations. We'll see some examples later on in the course.

1.2 Non-abelian Yang–Mills action

Recall that the action for quantum electrodynamics (QED) is given by

$$S_{\text{QED}} = \int d^dx\left[-\frac{1}{4g^2}F_{\mu\nu}F^{\mu\nu} + i\bar\psi\not{D}\psi - m\bar\psi\psi\right] \quad (1.16)$$

Here, the covariant derivative $D_\mu = \partial_\mu + iA_\mu$ is given in terms of the $U(1)$ gauge field A_μ, and $\not{D} = \gamma^\mu D_\mu$. This action is invariant under the gauge symmetry

$$\psi(x) \to e^{i\alpha(x)}\psi(x), \qquad A_\mu(x) \to A_\mu(x) - \partial_\mu\alpha(x), \quad (1.17)$$

where $\alpha(x)$ is a spacetime-dependent function. The field strength (or 'curvature') is given by $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and is related to the familiar electric and magnetic fields. This is an interacting Lagrangian due to the $-\gamma^\mu \bar{\psi} A_\mu \psi$ term, which corresponds to electromagnetic force on fermions ψ that is mediated by gauge bosons A_μ.

It's also possible to couple the $U(1)$ gauge field to a scalar field, which leads to an action that one could call 'scalar QED':

$$S_{\text{sQED}} = -\int d^d x \left[\frac{1}{4g^2} F_{\mu\nu} F^{\mu\nu} + D_\mu \phi D^\mu \phi^* + m^2 \phi \phi^* \right] \tag{1.18}$$

Here, the gauge symmetry acts on the complex scalar field $\phi(x)$ in the same way that it did on the fermion above. To illustrate some of the general features of gauge theory that we want to discuss, it will often be a bit easier not to have to deal with fermions.

Global non-abelian symmetries You've also already encountered non-abelian gauge symmetries in the QCD action, but since they will play an essential role in much of what is to follow, let's quickly review how they are constructed. First, consider the Lagrangian

$$\mathcal{L} = -\partial_\mu \vec{\phi} \cdot \partial^\mu \vec{\phi}^\dagger - m^2 |\vec{\phi}|^2, \tag{1.19}$$

where $\vec{\phi}(x) \in \mathbb{C}^N$ is an N-vector of complex fields. It has the global symmetry

$$\vec{\phi}(x) \to U \vec{\phi}(x), \tag{1.20}$$

where $U \in U(N)$ is a constant unitary $N \times N$ matrix, so that $U^\dagger U = 1$. Recall that we can parametrize the Lie group elements as $U = \exp iX$, using generators X of the Lie algebra $u(N)$. Note that the factor of i mean that we're using Hermitian generators, so $X^\dagger = X$. We can write the generators in terms of a basis T_a, so that $X = X^a T_a$. Then the Lie bracket is encoded by the structure constants

$$[T_a, T_b] = i f_{ab}{}^c T_c. \tag{1.21}$$

As a concrete example, for the $su(2)$ Lie algebra, where the generators have to satisfy the additional requirement $\operatorname{Tr} X = 0$, it's common to use the basis $T_a = \sigma_a/2$. Here, $\sigma_a = (\sigma_1, \sigma_2, \sigma_3)$ are the Pauli matrices, and we find the structure constants $f_{ab}{}^c = \epsilon_{ab}{}^c$, where indices are raised and lowered using the Kronecker δ_{ab} and its inverse. Adding the 2×2 unit matrix then gives a basis for $u(2)$.

The action of the unitary matrix U on the vector-valued fields $\vec{\phi}(x)$ in Equation (1.20) corresponds to the *fundamental* representation of the (special) unitary group/algebra. There's also a natural action of the Lie algebra on itself, given by the bracket (1.21), which is known as the *adjoint* representation.

Local non-abelian symmetries We can gauge the non-abelian symmetry of the Lagrangian (1.19) by introducing the covariant derivative

$$D_\mu \vec{\phi}(x) = \partial_\mu \vec{\phi}(x) + i A_\mu(x) \vec{\phi}(x). \tag{1.22}$$

Here, $A_\mu(x)$ takes values in the non-abelian Lie algebra, and we can consider either $u(N)$ or $su(N)$. If we now allow the global symmetries of (1.20) to depend on spacetime,

$$\vec{\phi}(x) \to U(x)\vec{\phi}(x), \qquad U(x) = e^{i\alpha(x)^a T_a}, \tag{1.23}$$

the derivative (1.22) transforms covariantly if we let

$$A_\mu(x) \to U(x)A_\mu(x)U(x)^\dagger + i\left(\partial_\mu U(x)\right)U(x)^\dagger. \tag{1.24}$$

so that infinitesimally, in terms of the components $A_\mu = A_\mu^a T_a$, we have

$$\delta A_\mu^a(x) = -\partial_\mu \alpha(x)^a - f_{bc}{}^a \alpha(x)^b A_\mu^c(x). \tag{1.25}$$

Likewise, the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]$ transforms as

$$F_{\mu\nu}(x) \to U(x)F_{\mu\nu}U(x)^\dagger, \tag{1.26}$$

so that the scalar $\operatorname{Tr} F_{\mu\nu}F^{\mu\nu}$ is gauge-invariant. With that, our full gauge-invariant Lagrangian is

$$\mathcal{L} = -\frac{1}{4g^2}\operatorname{Tr} F_{\mu\nu}F^{\mu\nu} - D_\mu\vec{\phi}\cdot D^\mu\vec{\phi}^\dagger - m^2|\vec{\phi}|^2. \tag{1.27}$$

The prefactor of the first term depends on the normalization of the Lie algebra generators. It's also common to redefine $A_\mu \to gA_\mu$ using the coupling constant g. With the latter convention, the non-Abelian Yang–Mills action $\operatorname{Tr} F_{\mu\nu}F^{\mu\nu}$ contains a three-point vertex proportional to g, and a four-point vertex proportional to g^2.

QCD Lagrangian and beta function In general one can construct further gauge-invariant Lagrangians from a Lagrangian with global symmetry by replacing partial by covariant derivatives. For example, starting from fermions ψ in the fundamental representation of $SU(3)$, one can construct

$$\mathcal{L} = -\frac{1}{4g^2}\operatorname{Tr} F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi, \tag{1.28}$$

This is the QCD Lagrangian, which has $SU(3)$ gauge symmetry. In addition to the cubic and quartic couplings from the pure Yang–Mills action, its interactions contain the fermion-gauge boson vertex that we also saw in QED.

Consider a general $SU(N)$ non-Abelian Yang–Mills theory coupled to N_f fundamental fermions. Its beta function turns out to be

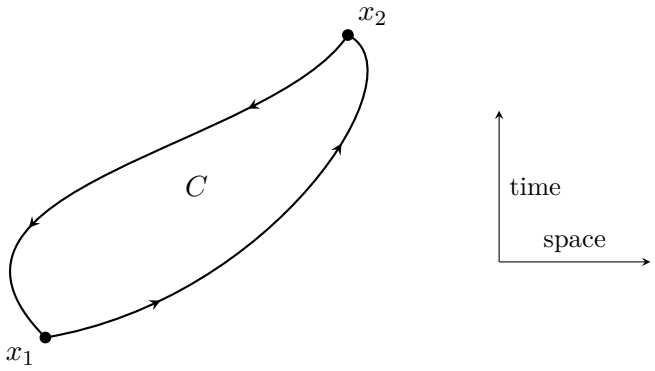
$$\beta(g) = \frac{\partial g}{\partial \log(\mu)} = -\frac{g^3}{48\pi^2}\left(11N - 2N_f.\right) \tag{1.29}$$

This is very different from QED, where the beta function is positive so that the coupling decreases at low energy. For QCD, we get a negative beta function, so the coupling *increases* at low energy!

As a result, we cannot use perturbation theory to extract predictions from QCD at low energies. Instead, we bang particles together at LHC to test the perturbative high-energy regime. Much of the strongly-coupled low-energy physics in the Standard Model (responsible for the structure of nuclei) is still not understood.

1.3 Wilson loops

Apart from correlators of elementary fields, there are other interesting observables that we can study in gauge theory. For example, imagine transporting a heavy matter particle along a loop through spacetime,



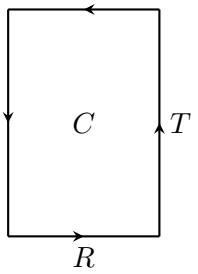
If this particle is charged under a particular gauge symmetry with potential A_μ, the total Lagrangian will contain a term of the form $j^\mu A_\mu$, where j^μ is the charge current associated to the matter field of our particle. Since we are thinking of a heavy particle, we can ignore any perturbative corrections to the particle and its trajectory — in other words, we consider the particle as a 'probe' traveling through the gauge field background. As a result, we can think of the matter current as being localized at the curve of our probe particle, $j^\mu(x) \sim \delta(x - x_c)$, where x_c is any point on the curve. For this reason, we can assign the following quantity to the probe particle traversing the curve C,

$$\langle \mathcal{W}(C) \rangle = \left\langle \frac{1}{N} \operatorname{Tr} \left[\mathcal{P} \exp \left(i \oint_C A_\mu dx^\mu \right) \right] \right\rangle . \tag{1.30}$$

This is known as a *Wilson loop*.

Let's explain the notation. First, if our gauge symmetry is non-abelian, the gauge potential A_μ is matrix-valued, so to get a number we take the trace, as we do in the QCD Lagrangian. The exponential of a matrix is defined through its Taylor series, and the result is potentially ambiguous since we are integrating over the path C. This is fixed through the 'path-ordering' operator $\mathcal{P}$, which, similar to the time-ordering operator, orders all operators, now based on their proper length parameter along the curve.

So what can this object do for us? Consider a very long rectangular spacetime path that traverses a distance T in time and R in space on each of its sides,



Schematically, you can think of this as a 'Feynman diagram' where we start in the vacuum, two (heavy) charged particles are created at a distance R apart, and they are annihilated after a time T. Since the loop is closed, the particles are oppositely charged. Additionally, since we don't consider any other matter contributions, only the gauge field contributes to this diagram. As a result, you would expect (1.30) to tell us something about the force between the two particles that is mediated by the gauge field.

We can check this in the perturbative regime (high energy for non-abelian, low energy for abelian). The computation is a bit involved, but after some work one obtains

$$\langle \mathcal{W}(C) \rangle \simeq e^{iV(R)T}, \tag{1.31}$$

where $V(R)$ is the gauge potential between the charged particles. In the limit $T \gg R$, where the particles exist for a relatively long time, we get

$$V(R) \approx -\frac{g^2}{4\pi R}, \tag{1.32}$$

which corresponds to the usual Coulomb potential. (This is the result for $d = 4$.)

On the other hand, in the non-perturbative regime, the behavior of the Wilson loop will be very different. At strong coupling, it is commonly believed that charged particles such as quarks experience a *linear* potential,

$$V(R) = \sigma R, \tag{1.33}$$

where σ is the tension of the hypothetical gluon 'strings'. As a result, the energy between the two quarks increases linearly in their separation, which results in confinement. Note that, with the potential (1.33), the expectation value of the Wilson line would be proportional to the (exponential of) TR, which corresponds to the *area* it encloses. On the other hand, in the weakly-coupled regime and in the limit $T \gg R$ where we get (1.32), the Wilson loop's expectation value is proportional to T, which roughly corresponds to the *perimeter* of the loop. This is taken to be a sign of the *deconfinement* of charged particles associated to the Coulomb potential (1.32).

Of course, checking whether the confining potential (1.33) actually arises at strong coupling is impossible using perturbation theory. One can study Wilson loops numerically using lattice simulations, and this has led to evidence that QCD at strong coupling indeed leads to an area law in Wilson loops. On the other hand, as we will see later in this course, the AdS/CFT correspondence provides a mechanism to relate the expectation value of a Wilson loop to the area of a corresponding *string worldsheet* in Anti-de Sitter spacetime!

1.4 Large N

Previously, we saw that the coupling constant of non-Abelian Yang–Mills flows to a large value in the infrared. As a result, we cannot use the regular small coupling expansion and Feynman diagrams to describe the interacting theory at low energies.

Schematically, having a coupling constant also means that the force mediated by the gauge bosons is strong (hence we refer to it as the 'strong force' for $SU(3)$ in Standard Model). In contrast to the electromagnetic force corresponding to $U(1)$ fields, which decreases between two charged particles as they are separated in space, the strong force increases in separation. This should be responsible for the fact that we only see 'color-neutral' objects (which are singlets under $SU(3)$ transformations) such as mesons and baryons at low energies.

To be able to study Yang–Mills using a perturbative expansion in the coupling constant, we must therefore look at high energies such as at the LHC. Alternatively, we can use computers to simulate the low-energy, strongly-coupled theory on a discretized spacetime grid. Both are active research directions.

But now we will focus on a different approach. We want to find another parameter to expand in, and to do so, we'll take the rank N of the $SU(N)$ gauge group to be large.

As an analogy you can think of thermodynamics: if we have $N_A \sim 10^{23}$ interacting particles, we usually don't want to describe each one individually. Instead, we can use thermodynamics as an effective description, which reveals new physical phenomena that we'd probably never be able to understand using the individual particle picture. So if we have N interacting particles, their dynamics as $N \to \infty$ is described by a theory that is qualitatively very different from the underlying microscopic theory. In fact, this emergent description is more appropriate for the physical questions that we want to ask in those settings, such as how a gas behaves when it heats or expands.

One of the main ideas of this course is that gravity can be thought of as a large N effective description of strongly coupled field theory. At first sight (and perhaps also at second sight), this sounds like a crazy idea. We will spend the majority of this course building up the ingredients that go in to making the statement of this idea more precise. In this lecture, we'll start with the large N limit in gauge theory.

Coupling constant For simplicity, let us look at $SU(N)$ Yang–Mills theory, with no matter. This theory describes $N^2 - 1$ interacting gauge bosons and the idea is to study it in the $N \to \infty$ limit. To see why this could be a good idea, consider its beta function

$$\beta(g) = -\frac{11 g^3}{48\pi^2} N \qquad \Longrightarrow \qquad \mu \frac{\partial g}{\partial \mu} = -\frac{11 g^3}{48\pi^2} N \tag{1.34}$$

Taking $N \to \infty$ with the explicit factor of N in this equation does not seem like a very good idea. So instead, let's define a *new* coupling constant,

$$\lambda = g^2 N. \tag{1.35}$$

This λ is known as the 't Hooft parameter. Its RG flow equation is

$$\mu \frac{\partial \lambda}{\partial \mu} = -\frac{11}{48\pi^2} \lambda^2 \tag{1.36}$$

In terms of the new coupling constant λ, the RG flow still makes sense even in the limit $N \to \infty$, as long as we keep $\lambda = g^2 N$ fixed.

The beta function of λ is still negative, so at first sight it looks like we haven't gained much. However, if we study the theory in the perturbative regime of λ, it turns out that the corresponding Feynman diagram organise themselves in a very suggestive way.

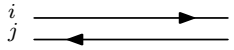
Double line notation In the $SU(N)$ Yang–Mills gauge theories we looked at previously, the gauge field A_μ transformed in the adjoint. To avoid having to do gauge fixing to obtain the propagators, we can model the theory using a bosonic field $\Phi = \Phi^a T_a$ that also transforms in the adjoint of the $SU(N)$ gauge group. Mimicking the interaction structure in the Yang–Mills action, we consider the Lagrangian

$$\mathcal{L} = \frac{1}{g^2}\left[-\frac{1}{2}\operatorname{Tr}\partial_\mu\Phi\partial^\mu\Phi + \operatorname{Tr}\Phi^3 + \operatorname{Tr}\Phi^4\right]. \tag{1.37}$$

If we write the field as $\Phi^i{}_j = \Phi^a(T_a)^i{}_j$, corresponding to the components of the matrices in the adjoint representation, the propagator turns out to be (with $\operatorname{Tr}(T_aT_b) = \delta_{ab}$)

$$\langle\Phi^i{}_j(x)\Phi^k{}_l(y)\rangle = \left(\delta^i{}_l\delta^k{}_j - \frac{1}{N}\delta^i{}_j\delta^k{}_l\right)\frac{g^2}{4\pi^2(x-y)^2} \approx \delta^i{}_l\delta^k{}_j\frac{g^2}{4\pi^2(x-y)^2}. \tag{1.38}$$

In the limit $N \to \infty$, we can drop the term that's proportional to $1/N$ if we're mainly interested in the leading-order behavior. The Kronecker deltas in the resulting propagator suggest that each row in one matrix only 'talks' to a particular column of the other matrix, and vice versa. Therefore, instead of a single line corresponding to a matrix-valued field, we can also represent Φ using *two* lines in Feynman diagrams, roughly corresponding to its columns and vectors. Pictorially, this gives

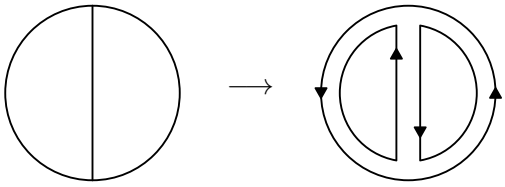
$\Phi^i{}_j(x)$  $\Phi^k{}_l(y)$ (1.39)

This is known as the ('t Hooft) double line notation. Note that the arrows serve to distinguish column and row vectors.

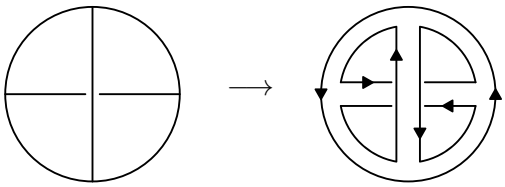
We now want to analyze the coupling constants that enter in a particular diagram in this notation.

- As we can see from Equation (1.38), each propagator comes with a factor of g^2. If we have E edges in a given diagram, the propagators therefore contribute g^{2E}.
- From the Lagrangian, we see that the quadratic and cubic vertices both come with a factor of g^{-2}. For V vertices, a Feynman diagram therefore picks up g^{-2V}.
- Since $\Phi(x)$ is a $N \times N$ matrix, its rows and columns have N entries. As a result, closed loops in the double line notation (which lead to a sum over the i indices) correspond to a factor N. Having L closed loop in a diagram therefore corresponds to a factor N^L.

Let's work out a few examples and rewrite the resulting factors in terms of $\lambda = g^2 N$.



$$\sim \left(\frac{\lambda^3}{N^3}\right)\left(\frac{N^2}{\lambda^2}\right) N^3 = N^2\lambda, \tag{1.40a}$$



$$\sim \left(\frac{\lambda^6}{N^6}\right)\left(\frac{N^4}{\lambda^4}\right) N^2 = N^0\lambda^2, \tag{1.40b}$$

The diagram in Equation (1.40a) has three propagators (so a factor of $g^6 = \lambda^3/N^3$) and two vertices (so a factor of $g^4 = N^2/\lambda^2$). Once we write it in double-line notation, we also see that it has three closed loops, corresponding to a factor of N^3. On the other hand, the diagram in Equation (1.40b) has six propagators and four vertices, but only *two* closed loops. Multiplying all factors, the diagram in Equation (1.40b) is subleading compared to the diagram in Equation (1.40a) in the limit $N \to \infty$.

In general, a diagram with V (interaction) vertices, E edges (propagators) and L loops will come with a factor of

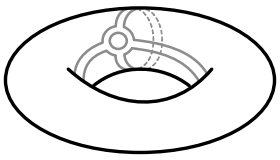
$$g^{-2V} g^{2E} N^L = N^{V-E+L} \lambda^{E-V} \tag{1.41}$$

If we fill in each loop in the above diagrams, we can think of them as polygons, with each loop corresponding to a face. (Note that in this counting, the 'outside' loop also corresponds to a face.) The propagators are then the edges of these polyhedra, while the interaction vertices are its vertices. With this interpretation, we can introduce the *Euler number*

$$\chi = V - E + L = 2 - 2g_t. \tag{1.42}$$

This invariant depends only on the topology of the corresponding polyhedron. It determines the simplest compact surface on which you can draw the associated diagrams without having any of the propagators cross each other. Here, g_t corresponds to the (topological) *genus* of the surface, which roughly counts the number of 'handles' it has.

For example, the double-line diagram in Equation (1.40b)

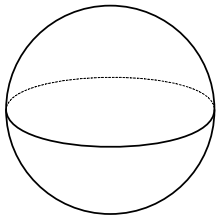


(1.43)

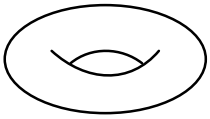
can be drawn without intersecting lines on a torus, but not on a sphere. On the other hand, the diagram in Equation (1.40a) can obviously be drawn without intersections on

a plane, and since we want to 'close' the outside loop to identify it with a face, it's most natural to think of this diagram as living on a sphere.

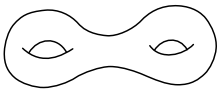
Topologically, we can model the sphere using a tetrahedron with four vertices, six edges and four faces, which gives us $\chi = 2$ and $g_t = 0$. On the other hand, the torus and the two-torus have $\chi = 0$ and $\chi = -2$, respectively, corresponding to $g_t = 1$ (one 'handle') and $g_t = 2$ (two 'handles'), as illustrated below:



$$\chi = 2, \qquad g_t = 0, \tag{1.44a}$$



$$\chi = 0, \qquad g_t = 1, \tag{1.44b}$$



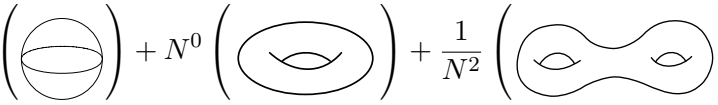
$$\chi = -2, \qquad g_t = 2. \tag{1.44c}$$

With these conventions, the prefactor of each Feynman diagram can then be organized in powers of N^{2-2g_t}, where g_t is the genus of the corresponding surface. Since $g_t \geq 0$, in the large N limit this expansion is dominated by the diagrams with $g_t = 0$ which you can draw on a plane (or sphere) and that are therefore called *planar diagrams*. In contrast, all higher-genus diagrams (with $g_t > 0$) are suppressed in the large N limit.

This greatly simplifies the number of diagrams that one has to consider. Corrections to the large N limit are suppressed in $1/N^2$, so even for relatively small N such as $N = 3$ this approximation is still pretty good. If we can additionally do perturbation theory in λ, we've therefore made our life a lot easier by reducing the number of diagrams we have to consider.

However, since the beta function of λ is still negative, this additional large N expansion is only immediately useful at large energies, so it looks like we're still stuck. At low energies, we're at strong coupling and still cannot trust perturbation theory. However, we can suggestively summarize what we learned in the following picture:

$$N^2 \left(\quad \right) + N^0 \left(\quad \right) + \frac{1}{N^2} \left(\quad \right) + \cdots \tag{1.45}$$



This is strikingly similar to something you might have seen in string theory. There, interactions correspond to the merging and splitting of strings, and we can therefore imagine that the parameter N is somehow related to the coupling constant of a *string* theory. If we then additionally allow λ to be large, we know perturbation theory 'fails' since we would have to take more and more diagrams into account at any given order.

With this 'stringy' expansion picture in mind, on the other hand, this could actually be an advantage: one could think that the diagrams that contribute at each order in N 'fill in' the worldsheet of the string.

While this qualitative observation was already made by 't Hooft in the seventies, it turned out to be very hard to make precise. In 1997, Maldacena found a way to argue that the large N limit of particular *conformal* field theories could be very precisely related to a particular type of string theory. Our next goal is therefore to learn more about conformal symmetry in field theory.

References and further reading

The Schwinger–Dyson equation and Ward identities are discussed in many places, but depending on the circumstances different aspects will be emphasized. The QFT book by Schwarz [2] discusses them in Sections 14.7 and 14.8. Another useful perspective is Section 2.4.4 of what is known as the 'big yellow book' on Conformal Field Theory by Di Francesco, Mathieu and Sénéchal [3]. (The latter works in Euclidean signature, so some factors of i and signs will be different.)

Wilson loops were introduced by Ken Wilson in the paper [4], where they are motivated through their relation to confinement. The same paper [4] is also at the foundation of the lattice approach to QCD. The computation that recovers the Coulomb potential (1.32) in the weak-coupling limit which we skipped can for example be found in Jan Ambjørn's lecture notes [5].

The large N limit was introduced by Gerard 't Hooft in the paper [6]. A famous account of this limit can be found in Chapter 8 of Sidney Coleman's book [7]. Additionally, the lecture notes on gauge theory by David Tong [8] are particularly recommended, which discuss large N in Section 6. (Note that these notes discuss Wilson loops, for example in Section 2.1.3 and Section 2.5, in a *much* more thorough way than we did above. They're also 455 pages long. Still, highly recommended.)

2 Conformal field theory

Quantum field theories typically behave very differently at different energy scales. An infinitesimal change in the energy scale of a theory changes its coupling constants in a manner determined by the beta function,

$$\frac{\partial g}{\partial \log(\mu)} = \beta(g). \tag{2.1}$$

This determines a trajectory in the space of coupling constants, known as the renormalization group flow. Clearly, such a trajectory ends when the beta function vanishes, at which point changing the scale no longer affects the theory. Such 'fixed points' of the RG flow are *scale invariant*.

Such a fixed point of the RG flow will sometimes just be a free, massless theory. For example, at high energies the mass coupling m of a free massive scalar field is suppressed and the theory flows to a UV fixed point described by a free massless scalar. But in other cases non-trivial fixed points can arise, such as the Wilson-Fisher fixed point that can arise in the IR of ϕ^4 theory. In such cases, the original Lagrangian description of the theory is typically not very useful anymore.

Under general circumstances, the scale invariance of such a fixed point is enhanced into *conformal symmetry*. This additional symmetry allows us to place strong additional constraints on the theory. In this section, we will explore conformal symmetry and see what effect it has on correlators. In particular, we will find that the two- and three-point functions can be fixed completely in terms of a small amount of data. For this reason, such *conformal field theories* (CFTs) are often described directly in terms of this data. In fact, only very few non-trivial conformal field theories are known to have a Lagrangian description, although we will see a prominent example later on in the course.

For now, though, our focus will be on conformal symmetries and their consequences for correlation functions. The results we obtain are general in the sense that they hold for *any* conformal field theory. As we will see later on, the fact that our upcoming results can be reconstructed using fields living on Anti-de Sitter spacetime in a similarly general way is a strong indication for the wide range of applicability of the AdS/CFT correspondence.

2.1 Weyl transformations and conformal transformations

Conformal transformations are *angle-preserving* transformations. The angle between two spacetime vectors u^μ and v^μ at a point x^μ in spacetime is determined by

$$\cos\theta_{uv} = \frac{g_{\mu\nu}(x)u^\mu v^\nu}{\sqrt{|g_{\mu\nu}(x)u^\mu u^\nu|}\sqrt{|g_{\mu\nu}(x)v^\mu v^\nu|}} \tag{2.2}$$

This is preserved by transformations that act on the metric as

$$g_{\mu\nu}(x) \to \Omega(x)^2 g_{\mu\nu}(x), \tag{2.3}$$

which is known as a *Weyl transformation* with scale factor determined by $\Omega(x)$.

Such transformations may be symmetries of a scale-invariant action. For example, a two-dimensional free massless scalar field coupled to a general background metric $g_{\mu\nu}$,

$$S[\phi] = -\frac{1}{2}\int d^2x\sqrt{-g}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi, \tag{2.4}$$

is invariant under Weyl transformations, since the square root of the metric determinant transforms with a factor $\Omega(x)^2$ which is cancelled by the $\Omega(x)^{-2}$ coming from the inverse metric. As a result, the theory cannot distinguish between metrics that are related by a Weyl transformation (2.3).

In general, it is useful to characterize these transformations in terms of coordinate transformations. Recall that under a general coordinate transformation $x^\mu \to x'^\mu(x)$, the metric transforms as

$$g_{\mu\nu}(x) \to g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu}g_{\rho\sigma}(x). \tag{2.5}$$

Conformal transformations are then defined as the coordinate transformations such that

$$g'_{\mu\nu}(x') = \Omega(x)^2 g_{\rho\sigma}(x). \tag{2.6}$$

In other words, they act on the metric in the same way as the Weyl transformations (2.3). As a result, even though all coordinate transformations are symmetries of generally-covariant theories, the conformal transformations play a distinguished role in theories that are Weyl invariant. Since the conformal transformations only transform the metric with an overall Weyl factor (2.6), which drops out of the action, they are present even after the background metric is fixed. In particular, this allows us to restrict to flat metrics $g_{\mu\nu} = \eta_{\mu\nu}$ while still keeping invariance under the conformal coordinate transformations.

2.2 Conformal algebra

Let's determine which coordinate transformations satisfy Equation (2.6) with $g_{\mu\nu} = \eta_{\mu\nu}$. For an infinitesimal transformation parametrized by $\epsilon^\mu(x)$ and $\Omega(x)^2 \approx 1 + \omega(x)$,

$$\mathcal{L}_\epsilon\eta_{\mu\nu} = \partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = \omega(x)\eta_{\mu\nu} \quad \Longrightarrow \quad \omega(x) = \frac{2}{d}\partial_\mu\epsilon^\mu(x). \tag{2.7}$$

Here, we have taken a trace to determine $\omega(x)$ in terms of $\epsilon^\mu(x)$. We have to solve

$$\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = \frac{2}{d}\partial_\rho\epsilon^\rho\eta_{\mu\nu}. \tag{2.8}$$

After some algebra, this leads to the equation

$$\partial^\mu\partial_\mu\epsilon_\nu = \frac{1}{d}(2-d)\partial_\nu\partial_\lambda\epsilon^\lambda. \tag{2.9}$$

This shows that $d = 2$ is a special case, which we will consider later. For now, we'll assume $d > 2$, where it turns out that we get

$$\epsilon^\mu(x) = a^\mu + \omega^\mu{}_\nu x^\nu + \lambda x^\mu + b^\mu x^2 - 2x^\mu b\cdot x. \tag{2.10}$$

These are all conformal transformations. They contain translations parametrized by a^μ and rotations/Lorentz boosts parametrized by the antisymmetric matrix $\omega^{\mu\nu}$. That makes sense, since these are the isometries of flat space and therefore satisfy the conformal Killing equation (2.7) with $\omega(x) = 0$.

In addition, the conformal transformations contain infinitesimal scaling transformations generated by λ. Their finite version is just given by

$$x^\mu \to x'^\mu = \lambda x^\mu. \tag{2.11}$$

Finally, they also contains the *special conformal transformations* generated by b^μ, whose finite version is given by

$$x^\mu \to x'^\mu = \frac{x^\mu + b^\mu x^2}{1 + 2b \cdot x + b^2 x^2}, \tag{2.12}$$

where we have abbreviated the inner product coming from the metric $\eta_{\mu\nu}$. They are closely related to the inversions,

$$x^\mu \to x'^\mu = \frac{x^\mu}{x^2}, \tag{2.13}$$

In fact, you'll see that the finite special conformal transformations can be obtained from a combination of translations and inversions.

The coordinate representation of the generators is given by

$$P_\mu = -i\partial_\mu, \tag{2.14a}$$
$$J_{\mu\nu} = i\left(x_\mu \partial_\nu - x_\nu \partial_\mu\right), \tag{2.14b}$$
$$D = -i x^\mu \partial_\mu, \tag{2.14c}$$
$$K_\mu = -i\left(x^2 \partial_\mu - 2x_\mu x^\nu \partial_\nu\right), \tag{2.14d}$$

whose commutators are given by

$$[J_{\mu\nu}, P_\rho] = i\left(\eta_{\mu\rho} P_\nu - \eta_{\nu\rho} P_\mu\right) \tag{2.15a}$$
$$[J_{\mu\nu}, J_{\rho\sigma}] = i\left(\eta_{\mu\rho} J_{\nu\sigma} + \eta_{\nu\sigma} J_{\mu\rho} - \eta_{\nu\rho} J_{\mu\sigma} - \eta_{\mu\sigma} J_{\nu\rho}\right), \tag{2.15b}$$
$$[J_{\mu\nu}, K_\rho] = i\left(\eta_{\mu\rho} K_\nu - \eta_{\nu\rho} K_\mu\right), \tag{2.15c}$$
$$[J_{\mu\nu}, D] = 0, \tag{2.15d}$$
$$[D, P_\mu] = i P_\mu, \tag{2.15e}$$
$$[D, K_\mu] = -i K_\mu, \tag{2.15f}$$
$$[K_\mu, P_\nu] = -2i\left(\eta_{\mu\nu} D - J_{\mu\nu}\right), \tag{2.15g}$$
$$[K_\mu, K_\nu] = 0. \tag{2.15h}$$

As you'll show in the exercises, the conformal algebra in d spacetime dimensions turns out to be isomorphic to $so(d, 2)$. Later on in the course, we will see that this algebra is closely related to Anti-de Sitter spacetime.

2.3 Conformal weight

Earlier, we saw that the action (2.4) for free massless scalar fields in $d = 2$ is invariant under Weyl transformations. At first sight, however, this does not seem to hold for the same action in $d > 2$,

$$S[\phi] = -\frac{1}{2}\int d^d x\sqrt{-g}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi, \tag{2.16}$$

since now the factor of Ω^d coming form the square root of the determinant of the metric no longer cancels against the factor Ω^{-2} from the inverse metric. However, there is a natural solution due to the fact that in $d > 2$, the field ϕ in the action (2.16) is dimensionful. If the overall action should have $[S] = 0$, we find that

$$[\partial_\mu\phi] = \frac{d}{2} \quad \Longrightarrow \quad [\phi] = \frac{d-2}{2}. \tag{2.17}$$

This 'mass dimension' vanishes for $d = 2$, but in general dimensions the field ϕ should therefore be sensitive the (local) scale changes of a Weyl transformations (2.3). For this reason, it is natural to allow fields to transform under the Weyl transformation as well,

$$\phi(x) \to \Omega(x)^{-\Delta_\phi}\phi(x). \tag{2.18}$$

Here, Δ_ϕ denotes the *scaling weight* associated to the field ϕ. This is a scalar number that should be taken as a defining property of a conformal field theory. Now recall that conformal transformations are defined as the coordinate transformations such that

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu}\frac{\partial x^\sigma}{\partial x'^\nu}g_{\rho\sigma}(x) = \Omega(x)^2 g_{\rho\sigma}(x). \tag{2.19}$$

Comparing this to the field transformation (2.18), we can then require that

$$\phi(x) \to \phi'(x') = \left|\frac{\partial x'}{\partial x}\right|^{-\Delta_\phi/d}\phi(x). \tag{2.20}$$

For this reason, Δ_ϕ is also known as the *conformal weight* of ϕ. Here, the factor of d arises since we are taking the determinant of the Jacobian. Also note that the Jacobian is related to the conformal factor $\Omega(x)$ by $|\partial x'/\partial x| = \Omega(x)^{-d}$, which follows from Equation (2.6) above.[1] Scalars that transform in this way under conformal transformations are known as *primary* fields.[2]

Now let's restrict to the scale transformations (2.11), where we have

$$x'^\mu = \lambda x^\mu \quad \Longrightarrow \quad \phi'(x') = \lambda^{-\Delta_\phi}\phi(x). \tag{2.21}$$

[1] Note that the power of Equation (2.20) is then opposite to the power in Equation (2.18). This is a slightly subtle point, but it is related to the fact that the coordinates are transformed by a conformal transformation but not by a Weyl transformation.

[2] As we'll see later on, in $d = 2$ the conformal algebra is much larger, and as a result there are more constraints we would like to impose on a primary field. For that reason, fields that transform as (2.20) are only *quasi-primary* in two-dimensional CFTs.

This coordinate transformation also means that

$$\frac{\partial}{\partial x'^{\mu}} = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \frac{\partial}{\partial x^{\nu}} = \lambda^{-1} \frac{\partial}{\partial x^{\mu}}, \qquad d^d x' = \lambda^d d^d x. \tag{2.22}$$

For the total action (2.16) to be invariant, we must therefore have

$$\Delta_\phi = \frac{d-2}{2}. \tag{2.23}$$

In other words, for a free scalar theory in d dimensions, the conformal weight equals the mass dimension (2.17). This is not true in general: for interacting theories, the conformal weight will receive quantum corrections compared to the mass dimension. Such *anomalous* scaling dimensions of $\mathcal{N} = 4$ super-Yang–Mills play an important role in detailed checks of the AdS/CFT correspondence.

You may have noticed that the action (2.16) is not actually invariant under the Weyl transformations (2.3) and (2.18). For this, we need to couple ϕ to the Ricci scalar of the background metric,

$$S[\phi] = -\frac{1}{2} \int d^d x \sqrt{-g} \left(g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{d-2}{4(d-1)} R \phi^2. \right) \tag{2.24}$$

Note that this does not change our previous analysis of the conformal dimension Δ_ϕ, since R contains two derivatives.

So far, we have only focused on scalar fields, but we can also consider fields that transform in non-trivial representations of the Lorentz algebra. Since the Lorentz algebra is a subalgebra of the conformal algebra (2.15), these representations (such as fermions or vectors) can then be extended into representations of the full conformal algebra. In addition to their spin, these representations are labeled by their conformal weight Δ.

Recall from Equation (2.14c) that the coordinate scaling transformations in Equation (2.21) are generated by

$$D = -i x^\mu \partial_\mu. \tag{2.25}$$

Acting on a primary field with conformal weight Δ, this generators is represented by

$$\mathcal{D} = -i x^\mu \partial_\mu - i\Delta, \tag{2.26a}$$

and the rest of the conformal generators are represented by

$$\mathcal{P} = -i\partial_\mu, \tag{2.26b}$$

$$\mathcal{J}_{\mu\nu} = i(x_\mu \partial_\nu - x_\nu \partial_\mu) - S_{\mu\nu}, \tag{2.26c}$$

$$\mathcal{K}_\mu = i\left(-x^2 \partial_\mu + 2x_\mu x^\rho \partial_\rho + 2x_\mu \Delta\right) - 2x^\nu S_{\mu\nu}, \tag{2.26d}$$

where $S_{\mu\nu}$ is the associated Lorentz representation. These 'curly' generators also satisfy the commutation relations (2.15) and therefore form a representation of the conformal algebra on a particular primary field. This representation incorporates its quantum numbers (conformal dimension Δ and spin s from the Lorentz representation $S_{\mu\nu}$). In

this course, we will mostly set $S_{\mu\nu} = 0$ and focus on spinless fields only. However, due to the scaling generator of the conformal algebra, we will always have access to one additional quantum number Δ.

Finally, due to the commutation relations (2.15), note that acting with the special conformal transformations on a given field lowers the scaling weight by one, while acting with the translation generator raises it. An alternative definition of a primary field is that its special conformal transformations give zero (when evaluated at $x = 0$, which is a fixed point of these transformations as can be seen from Equation (2.12)). As a result, a primary field has 'minimal' conformal dimension, but we can generate fields with higher conformal dimension from it by acting with the 'raising' operator corresponding to translations. Such 'derived' fields are known as *descendant* fields.

2.4 Correlation functions

Earlier in the course, we saw that translation invariance plus Lorentz invariance implies that n-point functions only depend on the distance between the insertion points,

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = f\left(|x_1 - x_2|, |x_2 - x_3|, \ldots, |x_n - x_1|\right). \tag{2.27}$$

Without specifying the theory we're considering, we can typically not say much more about the form of the function f in the above. However, in conformal field theory, it turns out that the form of the two- and three-point functions are *fully determined* by the additional scaling and special conformal symmetries! In particular, we will show that for primary fields $\phi_i(x)$, we have

$$\langle \phi_1(x_1)\phi_2(x_2) \rangle = \frac{\delta_{\Delta_1,\Delta_2}}{(x_1 - x_2)^{2\Delta_1}}, \tag{2.28}$$

for the two-point function, and

$$\langle \phi_1(x_1)\phi_2(x_2)\phi_3(x_3) \rangle = \frac{C_{123}}{x_{12}^{\Delta_1+\Delta_2-\Delta_3} x_{23}^{\Delta_2+\Delta_3-\Delta_1} x_{13}^{\Delta_3+\Delta_1-\Delta_2}}. \tag{2.29}$$

for the three-point function. Here, Δ_i denotes the conformal weight of ϕ_i. The scalar numbers C_{ijk} are determined by the properties of the theory.

Together with the scaling weights Δ_i, these numbers C_{ijk} can be taken as a *defining* property of a conformal field theory. The reason for this is that higher-point functions in CFTs can be decomposed into three-point functions. Furthermore, such decompositions impose very stringent constraints on the initial data (Δ_i, C_{ijk}). Rather than defining a theory by specifying a Lagrangian and computing this data using diagrams, one can therefore define a CFT using such initial data and then check if the consistency conditions coming from conformal symmetry are satisfied. This idea is known as the *conformal bootstrap* and has seen a huge revival over the last ten years. One of the ultimate goals of this research program is to classify the space of all possible conformal field theories.

2.5 Conformal symmetry in $d = 2$

Up until now, we have focused on dimensions $d > 2$, but as we saw in Equation (2.9) something special happens in two spacetime dimensions. For simplicity, we'll go to Euclidean signature, which changes the Minkowski metric $\eta_{\mu\nu}$ into the flat Euclidean metric $\delta_{\mu\nu}$ and the Minkowski coordinates (t, x) into the Cartesian coordinates (x^1, x^2). With $d = 2$, the defining equation (2.8) for the generator $\epsilon^\mu = (\epsilon^1, \epsilon^2)$ of conformal transformations then implies

$$\partial_1\epsilon_2 = -\partial_2\epsilon_1, \quad \partial_2\epsilon_2 = \partial_1\epsilon_1. \tag{2.30}$$

These are the Cauchy–Riemann equations for the components of a holomorphic function! That means we can form the holomorphic and anti-holomorphic functions

$$\epsilon(z) = \epsilon_1 + i\epsilon_2, \quad \bar{\epsilon}(\bar{z}) = \epsilon_1 - i\epsilon_2, \tag{2.31}$$

where $z, \bar{z} = x_1 \pm ix_2$ and the ϵ_i satisfy Equation (2.30). This means that the two-dimensional conformal transformations are parametrized by two *arbitrary* functions. As a result, the algebra of two-dimensional conformal transformations is infinite-dimensional!

A set of coordinate transformations generating this algebra is given by

$$\ell_n = -z^{n+1}\partial_z, \quad \bar{\ell}_n = -\bar{z}^{n+1}\partial_{\bar{z}}. \tag{2.32}$$

They satisfy the following commutation relations,

$$[\ell_m, \ell_n] = (m-n)\ell_{m+n}, \quad [\bar{\ell}_m, \bar{\ell}_n] = (m-n)\bar{\ell}_{m+n}, \quad [\ell_m, \bar{\ell}_n] = 0. \tag{2.33}$$

Most of these generators are singular at $z \to 0$ or at $w = 1/z \to 0$, except for those with $m = -1, 0, +1$. The latter form the globally defined part of the two-dimensional conformal algebra, which is isomorphic to

$$sl_2(\mathbb{R}) \oplus sl_2(\mathbb{R}) \simeq so(2, 2) \tag{2.34}$$

which agrees with the higher-dimensional conformal algebra $so(d, 2)$ for $d = 2$. (The latter identification brings us back to Lorentzian signature.) In addition to these global transformations, a two-dimensional conformal symmetry has an infinite amount of local symmetry generators which place very strong constraints on the theory.

A representation of this algebra in the quantized theory, with generators denoted by L_m, can have a *central extension*,

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}. \tag{2.35}$$

(The factor of $1/12$ is conventional.) This extension is parametrized by a single scalar number c, which is known as the *central charge* and determines important aspects of the quantum theory. A complete classification of unitary $2d$ conformal field theories with $c < 1$ is known. In addition, two-dimensional CFTs have many appealing duality properties when they are placed on compact surfaces such as those in Equation (1.45). These properties are particularly important in the context of string theory, where the two-dimensional worldsheet theory turns out to be described by a conformal field theory.

2.6 Energy-momentum tensor

Now let's return to a general spacetime dimension d. Previously, you've probably seen the energy-momentum tensor defined as the Noether charge associated to translation symmetry. There exists an alternative definition that is particularly useful in the context of CFTs and the gauge/gravity correspondence.

For this, let's consider a general spacetime translation parametrized by a vector ϵ^μ, which acts on the coordinates as

$$x^\mu \to x^\mu + \epsilon^\mu. \tag{2.36}$$

Following our discussion of the Noether current derivation in Section 1.1, we then allow the parameter ϵ^μ to depend on spacetime,

$$x^\mu \to x^\mu + \epsilon^\mu(x). \tag{2.37}$$

By Equation (1.9a), we can then identify the energy-momentum tensor as the Noether current associated to the translation symmetry,

$$\delta S = -\int d^d x T^\mu{}_\nu \partial_\mu \epsilon^\nu. \tag{2.38}$$

Here, we're assuming we are in flat spacetime. If we furthermore assume that the energy-momentum tensor is symmetric, we can write the above as

$$\delta S = -\frac{1}{2}\int d^d x T^{\mu\nu}\left(\partial_\mu \epsilon^\nu + \partial_\nu \epsilon_\mu\right) = -\frac{1}{2}\int d^d x T^{\mu\nu}\delta_\epsilon \eta_{\mu\nu}, \tag{2.39}$$

where the variation $\delta_\epsilon \eta_{\mu\nu} = \mathcal{L}_\epsilon \eta_{\mu\nu}$ corresponds to the Lie derivative of the metric $\eta_{\mu\nu}$ with respect to the vector field $\epsilon^\mu(x)$. We can generalize this expression to non-flat metrics as follows,

$$\delta S = -\frac{1}{2}\int d^d x \sqrt{-g} T^{\mu\nu}\delta_\epsilon g_{\mu\nu}, \tag{2.40}$$

where again $\delta_\epsilon g_{\mu\nu} = \mathcal{L}_\epsilon g_{\mu\nu}$. This expression can be used to define a symmetric energy-momentum tensor $T^{\mu\nu}$ associated to a theory on a spacetime with metric $g_{\mu\nu}$ by setting

$$T^{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S}{\delta g_{\mu\nu}}. \tag{2.41}$$

Conservation and charges Recall that the Lie derivative of a general metric is

$$\delta_\epsilon g_{\mu\nu} = \mathcal{L}_\epsilon g_{\mu\nu} = \nabla_\mu \epsilon_\nu + \nabla_\nu \epsilon_\mu, \tag{2.42}$$

where ∇_μ is the covariant derivative associated to the Levi-Civita connection. Plugging this expression in Equation (2.40), where the left-hand side vanishes since we are just doing a diffeomorphism, we find the conservation law

$$\nabla_\mu T^{\mu\nu} = 0. \tag{2.43}$$

The energy-momentum tensor (2.41) is therefore always conserved in a theory that is invariant under diffeomorphisms. However, it only leads to a conserved Noether current $j^\mu = T^\mu{}_\nu \epsilon^\nu$ and an associated Noether charge if ϵ^μ is a Killing vector field of $g_{\mu\nu}$.

Weyl invariance In a theory with Weyl invariance, the transformation

$$g_\mu\nu(x) \to \Omega(x)^2 g_{\mu\nu}(x) \tag{2.44}$$

is a symmetry of the action. Then the definition (2.41) implies that

$$0 = \delta S = -\frac{1}{2}\int d^d x\sqrt{-g}T^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}\int d^d x\sqrt{-g}T^{\mu}{}_{\nu}\Omega^2 \qquad \Longrightarrow \qquad T^{\mu}{}_{\nu} = 0, \tag{2.45}$$

since $\Omega(x)$ is arbitrary. Classically, this means that the energy-momentum tensor is traceless in conformally-invariant theories. However, just like the classical conservation law $\partial_\mu j^\mu = 0$ gets corrections in the quantum theory, this tracelessness condition turns out to be modified by quantum effects in even spacetime dimensions.

References and further reading

The standard introductions to conformal field theory are the lecture notes [9] by Ginsparg and the 'big yellow book' [3] by Di Francesco, Mathieu and Sénéchal. The former is available on arXiv, for the latter you'll probably need to go to the library. Another nice reference are the lecture notes [10] by Blumenhagen and Plauschinn, which is available online through the KU library, as well as the lecture notes [11] by Qualls on arXiv.

Most of the previous references cover the material on CFTs in $d > 3$ that we discussed, but then focus mainly on the $d = 2$ case afterwards. This is useful from the perspective of string theory, but for more information on higher-dimensional CFTs and in particular the conformal bootstrap, the lectures [12] by Simmons-Duffin are highly recommended, as well as the more advanced lectures [13] by Rychkov.

3 Anti-de Sitter spacetime

We now turn from field theory to gravity. Here, we will take the spacetime dimension to be $d+1$. In particular, we'll focus on *Anti-de Sitter* (AdS) spacetime. This is a solution of Einstein's equation in the presence of cosmological constant Λ,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 0. \tag{3.1}$$

Taking the trace, we see that the sign of the Ricci scalar corresponds to the sign of Λ,

$$R = 2\Lambda\frac{d+1}{d-1}. \tag{3.2}$$

We now want to restrict ourselves to maximally symmetric spacetimes, which have the property that their Riemann tensor is fully determined by the Ricci scalar. This means it must be of the form

$$R_{\mu\nu\rho\sigma} = \frac{R}{d(d+1)}\left(g_{\mu\rho}g_{\nu\sigma} - g_{\nu\rho}g_{\mu\sigma}\right). \tag{3.3}$$

Such geometries can be conveniently described in terms of an embedding in a higher-dimensional geometry. This embedding description also gives us a convenient way of obtaining their isometries. We will be particularly interested in solutions with $\Lambda < 0$, which have negative scalar curvature. Finally, we will discuss several coordinate systems and discuss aspects of the geometry of AdS spacetime that hint towards the AdS/CFT correspondence.

3.1 AdS as a hypersurface and global coordinates

One consequence of a spacetime being maximally symmetric is that it looks the same at every point, and in every direction at every point. In the following, we will mainly be interested in Lorentzian manifolds (meaning that the metric has a single negative eigenvalue) with negative curvature, as in Equation (3.2). However, to set up some of our toolkit it is useful to look at some more familiar examples with Euclidean signature.

Euclidean spheres and hyperboloids A two-dimensional sphere is a good example of a maximally symmetric spacetime: no point and no direction on it is distinct from any other. One convenient way to describe a sphere is using the equation

$$x^2 + y^2 + z^2 = \ell^2. \tag{3.4}$$

This corresponds to a sphere of radius ℓ embedded in a three-dimensional space described by the coordinates x, y and z. A more covariant version of this equation would be

$$\delta_{ab}x^a x^b = \ell^2, \tag{3.5}$$

where δ_{ab} is the flat Euclidean metric on $\mathbb{R}^3$. The two-dimensional sphere is a hypersurface in this three-dimensional flat Euclidean manifold, but it can equally described

directly as a two-dimensional manifold, independently of its embedding in $\mathbb{R}^3$. For this, we can use the parametrization

$$x = \ell \sin\theta \cos\varphi, \tag{3.6a}$$
$$y = \ell \sin\theta \sin\varphi, \tag{3.6b}$$
$$z = \ell \cos\theta. \tag{3.6c}$$

We then obtain the metric of the two-sphere as a restriction of the flat metric on $\mathbb{R}^3$ to the submanifold described by these coordinates,

$$ds^2 = \delta_{ab} dx^a dx^b = \ell^2 \left(d\theta^2 + \sin^2\theta d\varphi^2 \right). \tag{3.7}$$

From this, we can compute its Ricci tensor, which can easily be found to be of the form (3.3) with $R = 2/\ell^2$.

The coordinates $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$ obviously cannot cover the entire manifold, but they get pretty close: they only leave out the north and south pole and a line between them. For this reason, they are sometimes loosely referred to as *global coordinates*.

Obtaining the *isometries* of the two-sphere metric (3.7) by solving the Killing equation is not particularly easy in terms of the two-sphere coordinates (θ, φ). The isometries can be more easily obtained from the embedding description (3.4). First, the isometries of the embedding $\mathbb{R}^3$ are well known: they consist of three-dimensional translations and rotations. In terms of the Euclidean coordinates $x^a = (x, y, z)$, these are given by

$$P_a = -i\partial_a, \qquad J_{ab} = i\left(x_a \partial_b - x_b \partial_a \right). \tag{3.8}$$

The isometries of the two-sphere embedded in $\mathbb{R}^3$ are then given by the subset of the Poincaré generators that leave the two-sphere invariant. This corresponds to just the rotations, which we can then express in terms of the sphere coordinates (θ, φ) as

$$-iJ_{12} = \partial_\varphi, \tag{3.9a}$$
$$-iJ_{13} = -\cos\varphi \partial_\theta + \cot\theta \sin\varphi \partial_\varphi, \tag{3.9b}$$
$$-iJ_{23} = -\sin\varphi \partial_\theta - \cot\theta \cos\varphi \partial_\varphi. \tag{3.9c}$$

One can then check that these vector fields leave the induced metric (3.7) invariant.

Euclidean Anti-de Sitter spacetime Next, we want to look at a maximally symmetric manifold with *negative* scalar curvature. In the three-dimensional setting of our previous example, this would be

$$-x^2 + y^2 + z^2 = -\ell^2 \tag{3.10}$$

leading to a hyperboloid along the z-axis. In a more covariant language, this corresponds to a two-dimensional submanifold in a Euclidean $\mathbb{R}^3$ with a special vector field distinguishing the x-direction. Alternatively, we can see this hyperboloid as a submanifold of a three-dimensional *Minkowski* manifold, where the x-direction plays the role of 'time',

$$\eta_{ab} x^a x^b = -\ell^2. \tag{3.11}$$

Note, however, that the induced metric on the hyperboloid is still of Euclidean signature. We can check this explicitly using the parametrization

$$x = \ell \cosh\rho \cosh\tau, \tag{3.12a}$$
$$y = \ell \cosh\rho \sinh\tau, \tag{3.12b}$$
$$z = \ell \sinh\rho, \tag{3.12c}$$

which results in the metric

$$ds^2 = \eta_{ab} dx^a dx^b = \ell^2 \left(\cosh^2\rho d\tau^2 + d\rho^2\right). \tag{3.13}$$

This is a maximally symmetric metric with scalar curvature $R = -2/\ell^2$. It corresponds to two-dimensional Euclidean Anti-de Sitter space.

We can easily generalize it to higher dimensions by replacing the z-direction with a d-vector $\vec{z}$, which leads to

$$-x^2 + y^2 + \vec{z}^2 = -\ell^2 \tag{3.14}$$

If we now use the parametrization

$$x = \ell \cosh\rho \cosh\tau, \tag{3.15a}$$
$$y = \ell \cosh\rho \sinh\tau, \tag{3.15b}$$
$$\vec{z} = \ell \sinh\rho \vec{\Omega}, \tag{3.15c}$$

where $\vec{\Omega} \in \mathbb{R}^d$ satisfy $\vec{\Omega}^2 = 1$, so that they parametrize a $(d-1)$-sphere, the total embedding metric becomes

$$ds^2 = \ell^2 \left(\cosh^2\rho d\tau^2 + d\rho^2 + \sinh^2\rho d\Omega_{d-1}^2\right). \tag{3.16}$$

where $d\Omega_{d-1}^2$ is the metric of the $(d-1)$-sphere. This results in a $(d+1)$-dimensional Euclidean Anti-de Sitter space. Its isometries are given by the subgroup of the $(d+2)$-dimensional Poincaré group that leave the hyperboloid invariant, corresponding to the $(d+2)$-dimensional Lorentz group $SO(d+1,1)$.

Lorentzian Anti-de Sitter spacetime To obtain a Lorentzian metric on a hyperboloid, we can flip an additional sign in the embedding equation,

$$-x^2 - y^2 + z^2 = -\ell^2. \tag{3.17}$$

One can again see this as a hypersurface in a three-dimensional spacetime, but now this spacetime does not have a Minkowski metric but instead a metric with two minus signs. The embedding coordinates are then given by

$$x = \ell \cosh\rho \cos t, \tag{3.18a}$$
$$y = \ell \cosh\rho \sin t, \tag{3.18b}$$
$$z = \ell \sinh\rho, \tag{3.18c}$$

which results in the following metric with a Lorentzian signature,

$$ds^2 = \ell^2 \left(-\cosh^2 \rho dt^2 + d\rho^2 \right). \tag{3.19}$$

This is two-dimensional Lorentzian Anti-de Sitter spacetime. Note that it is related to the two-dimensional Euclidean Anti-de Sitter metric (3.13) by identifying $t = -i\tau$.

Its higher-dimensional generalization again follows from replacing the z-direction with a d-dimensional vector $\vec{z}$, which leads to

$$ds^2 = \ell^2 \left(-\cosh^2 \rho dt^2 + d\rho^2 + \sinh^2 \rho d\Omega_{d-1}^2 \right). \tag{3.20}$$

This is the metric for $(d+1)$-dimensional Lorentzian Anti-de Sitter spacetime. Its isometries are given by the subset of the $(d+2)$-dimensional symmetries that leave the hypersurface intact, which corresponds to $SO(d,2)$. Again, we can relate it to the Euclidean solution (3.16) by setting $t = -i\tau$.

Note that the time coordinate t as defined by the embedding coordinates above is currently periodic. That is clearly undesirable for a physical spacetime (we don't want closed timelike curves!), but this can be easily addressed by 'rolling out' the time coordinate in the resulting spacetime. The result is the universal cover of the hypersurfaces we studied above, where the time coordinate now can take any real value. This is almost always what people mean when they talk about (Lorentzian) Anti-de Sitter spacetime.

3.2 Conformal boundary and Penrose diagrams

In field theory courses, one often assumes that fields 'fall off sufficiently quickly towards infinity', for example so that one does not have to take total derivatives in integrals into account. However, even in relatively simple spacetimes such as Minkowski or Anti-de Sitter, it is not obvious what the precise meaning of such a statement is.

Part of the problem is that physically interesting spacetimes have infinite volume, so the topology of their boundary can often not be read off from the metric. Instead, one can look for a spacetime whose metric $\tilde{g}_{\mu\nu}$ is related by a Weyl transformation to the metric $g_{\mu\nu}$ we're interested in,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = e^{\Omega(x)} \tilde{g}_{\mu\nu} dx^\mu dx^\nu. \tag{3.21}$$

so that the volume of the spacetime manifold measured using $\tilde{g}_{\mu\nu}$ is finite. Such metrics are said to be conformally related, or to be 'in the same conformal class'. As a result, we can 'draw' the resulting metric on a finite-sized piece of paper, which results in a *Penrose diagram* for the original spacetime.

Of course, geodesics and curvature will typically be different for $\tilde{g}_{\mu\nu}$ than for the original metric. However, as we saw in Equation (2.2), a Weyl transformation preserves angles, so in particular a timelike vector with respect to the original metric will still be timelike with respect to the rescaled metric. Likewise, the notion of null and spacelike vectors remain invariant. As a result, the *causal structure* of both spacetimes are the same, since Weyl transformations leave the light cone structure invariant. This makes conformal compactifications and Penrose diagrams a useful tool for analyzing causal structures of physical spacetimes such as Minkowski and Anti-de Sitter.

Minkowski space Let's start with two-dimensional Minkowski space, with the metric

$$ds^2 = -dt^2 + dx^2 = -du\, dv. \tag{3.22}$$

Here, we have introduced the null coordinates $u = t + x$ and $v = t - x$. Note that both t, x and u, v run over the full real line. Instead, the coordinates

$$U = \arctan u, \qquad V = \arctan v, \tag{3.23}$$

only take values on the interval $(-\pi/2, \pi/2)$. In these coordinates, the metric becomes

$$ds^2 = -\frac{1}{\cos^2 U \cos^2 V} dU\, dV. \tag{3.24}$$

Even though these new coordinates have finite range, the volume is still infinite, since the metric diverges when U, V approach $\pm\pi/2$. However, the metric is now of the form (3.21), so we see that Minkowski space is conformally related to the finite-size metric

$$d\tilde{s}^2 = -dU\, dV \tag{3.25}$$

using the conformal factor $1/\cos^2 U \cos^2 V$. We can draw this as in Figure 1.

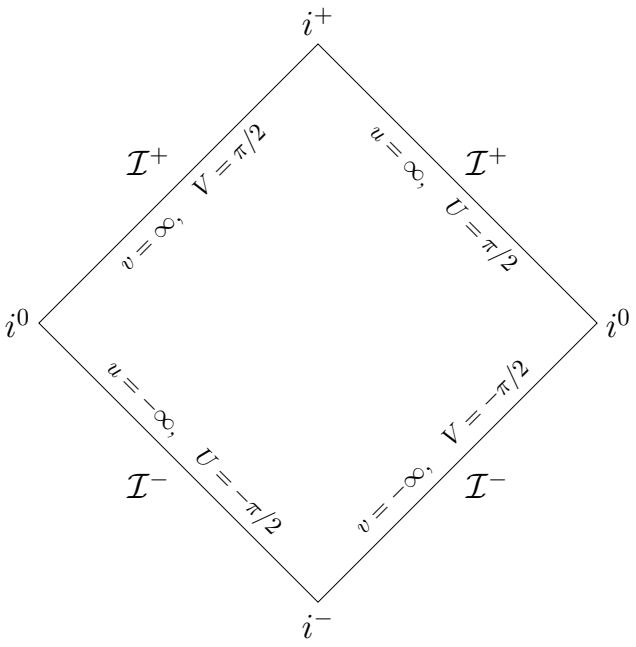


Figure 1: Penrose diagram for two-dimensional Minkowski spacetime

This is the Penrose diagram for two-dimensional Minkowski space. Here, the edges of the diamond correspond to $U, V = \pm\pi$. It turns out that these edges can only be reached by massless observers, which is why they are known as *null infinity*. Instead,

the corners of the diamond correspond to *spatial infinity* and *past and future infinity*, respectively. When one defines boundary conditions for fields on Minkowski spacetime, all these need to be taken into account.

We can easily generalize the above to a $(d+1)$-dimensional Minkowski space if we now interpret x as the norm of a d-dimensional vector $\vec{x}$. The only thing that changes is that x must now be positive, so the diagram is halved, which leads to Figure 2. At

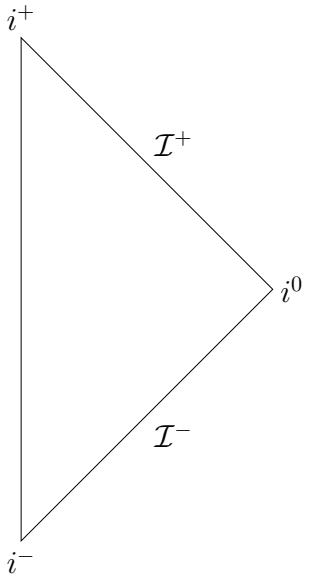


Figure 2: Penrose diagram for $(d+1)$-dimensional Minkowski spacetime, which implicitly contains an S^{d-1} at each point.

each point of this Penrose diagram there is an implicit S^{d-1}.

Anti-de Sitter Now let's do the same thing for Anti-de Sitter in global coordinates. We can compactify the coordinate $\rho \in (0, \infty)$ by defining

$$\tan \chi = \sinh \rho, \tag{3.26}$$

so that $\chi \in (0, \pi/2)$, which results in the metric

$$ds^2 = \frac{\ell^2}{\cos^2 \chi} \left[-dt^2 + d\chi^2 + \sin^2 \chi d\Omega_{d-1}^2 \right]. \tag{3.27}$$

Now $\rho \to \infty$ corresponds to $\chi \to \pi/2$, at which point the conformal factor diverges. The boundary of the conformally related metric (also known as the *conformal boundary* of the original metric) is parametrized by $(t, \vec{\Omega})$ and corresponds to a Lorentzian cylinder. Suppressing the $(d-2)$-sphere, this gives us the Penrose diagram for Anti-de Sitter as in Figure 3.

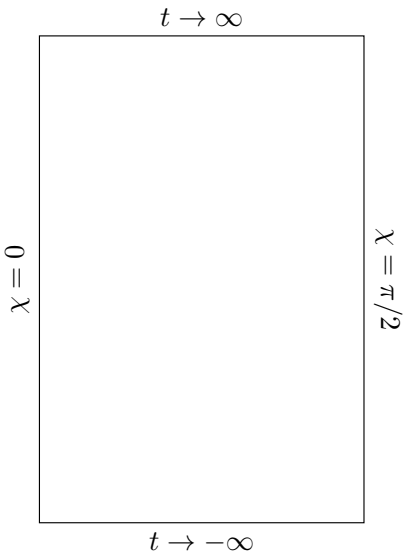


Figure 3: Penrose diagram for Anti-de Sitter spacetime. The top and bottom edges of the rectangle are understood to be infinitely far away. This time coordinate can also be compactified, see for example page 179-180 of [14].

Domains of dependence Recall that the *future domain of dependence* of a spacetime region Σ is defined as all future points that can be reached from this region using causal (timelike or null) geodesics. Penrose diagrams are a particularly useful tool for studying domains of dependence, since they give a very compact representation of the causal structure of a spacetime.

In the AdS/CFT correspondence, we will see that the behavior of fields at the (conformal) boundary of Anti-de Sitter space plays an important role in specifying the initial conditions of a given problem. Its Penrose diagram above gives a hint for why this might make sense. Imagine we insert some perturbation a certain time $t = t_0$ on the conformal boundary of AdS, as in Figure 4. After some time has passed, its future domain of dependence includes the *entire* spatial slice! Thus, it is not unreasonable to think that one can 'cook up' a wide range of phenomena in Anti-de Sitter spacetime using suitable boundary conditions. To be precise, the boundary conditions of fields along the entire conformal boundary of Anti-de Sitter space contribute to the time evolution of the fields.

This does not happen in Minkowski space, where only insertions at $\mathcal{I}^-$ can reach spatial surfaces Σ, and insertions on $\mathcal{I}^+$ cannot. Relatedly, the initial values of a field on a spatial surface Σ in Minkowski space (which must necessarily end at spatial infinity i^0) fully determines the future evolution of the fields to a future spatial surface Σ'. This is an indication that (if one can get it to work at all), the gauge/gravity correspondence would work quite differently in Minkowski compared to Anti-de Sitter.

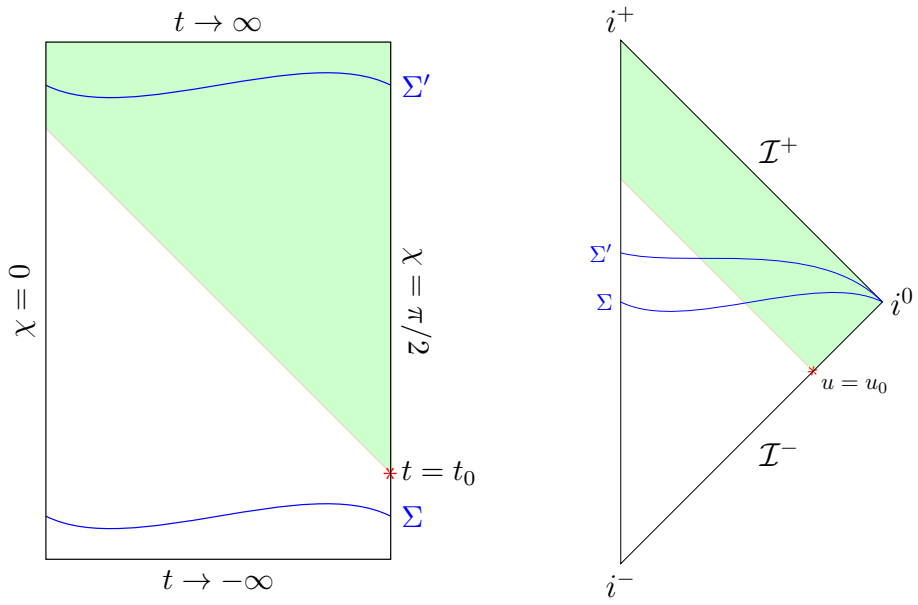


Figure 4: Left: an insertion $*$ on the conformal boundary of AdS at $t = t_0$, and its future domain of dependence, denoted by the green region. Non-trivial boundary behavior in AdS therefore adds information to Σ' that cannot be determined from earlier spatial slices Σ. Right: in Minkowski space, only insertions on $\mathcal{I}^-$ can reach spatial surfaces Σ, and the time evolution of fields from Σ to Σ' is independent of what happens at $\mathcal{I}^+$.

3.3 Poincaré coordinates

Global coordinates are not the only possibility to describe AdS space. Indeed, there are many choices that can be done, each of them having more convenient aspects or applications than others. Starting from the hyperboloid description, we obtain the so-called Poincaré coordinates by using the following parametrization:

$$X^\mu = \ell \frac{x^\mu}{z}\,, \qquad X^d = \frac{\ell}{2}\frac{1-x^2-z^2}{z}\,, \qquad X^{d+1} = \frac{\ell}{2}\frac{1+x^2+z^2}{z}\,, \tag{3.28}$$

with

$$\mu \in \{0,\dots,d-1\}\,, \qquad x^2 = \eta_{\mu\nu}x^\mu x^\nu\,, \qquad z \geq 0\,. \tag{3.29}$$

We immediately notice that these coordinates do not cover the entire AdS_{d+1} spacetime, because $X^d + X^{d+1} \geq 0$. The metric reads

$$ds^2 = \frac{\ell^2}{z^2}\left(\eta_{\mu\nu}dx^\mu dx^\nu + dz^2\right)\,. \tag{3.30}$$

It is evident that the conformal boundary of this metric is given by $d-$dimensional Minkowski space, since we can simply collect the conformal factor

$$e^{\Omega(x)} = \frac{\ell^2}{z^2}\,. \tag{3.31}$$

This allows to find the Penrose diagram for AdS space in Poincaré coordinates as depicted in Fig. 5, where the horizon is located at $z = \infty$ and the boundary at $z = 0$.

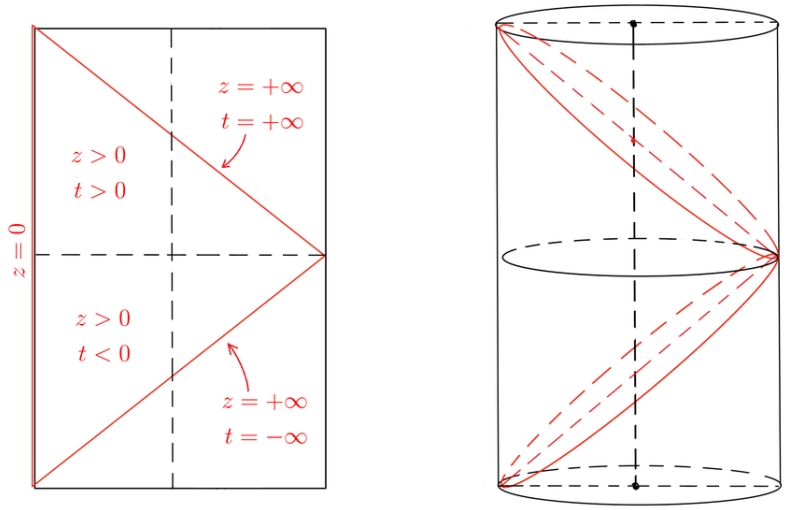


Figure 5: Conformal diagram of anti de Sitter space in Poincaré coordinates (left). On the right, we show the Poincaré patch inside the cylinder which describes the global coordinate system.

Given this structure of the metric, the $d-$dimensional Poincaré invariance is manifest and it is generated by

$$J_{\mu\nu} = i\left(x_\mu \partial_\nu - x_\nu \partial_\mu\right) , \qquad P_\mu = -i\partial_\mu \, . \tag{3.32}$$

The dilatations correspond to a simultaneous rescaling $(z, x_\mu) \to (\lambda z, \lambda x_\mu)$ and the corresponding generator reads

$$D = -i\left(x^\mu \partial_\mu + z\partial_z\right) . \tag{3.33}$$

Putting everything together, the manifest symmetries are $\mathrm{SO}(1,1)\times\mathrm{SO}(3,1)$. In addition, there are also the so-called K_μ generators, which at the boundary correspond to special conformal transformations.

It is possible to get another intuitive feeling of the Penrose diagram in Poincaré coordinates by starting with with the full cylinder for AdS space in global coordinates, see the picture on the right in Fig. 5. The Poincaré patch is obtained by performing two diagonal cuts; the region that we obtain in this way can be unrolled to give Minkowski space! The fact that the topology of the boundary is different in global and Poincaré coordinates corresponds to two different ways to approach infinity. Since we can only constrain the boundary metric only up to a conformal factor, the spherical metric (that we obtain in global coordinates) and the flat one (that we obtain in Poincaré coordinates) are equivalent; only the global structure differs.

There are two other choices which are commonly used in the literature to describe Poincaré coordinates: the first one simpy amounts to define

$$r = \frac{\ell^2}{z}\,, \tag{3.34}$$

and has some a natural interpretation as an energy scale associated to RG flows, as we will briefly discuss later in the course. The other possibility corresponds to further set $r = \ell e^{\rho}$. In this way the metric reads

$$\begin{aligned} ds^2 &= \ell^2 \frac{dr^2}{r^2} + \frac{r^2}{\ell^2}\eta_{\mu\nu}dx^\mu dx^\nu = \\ &= \ell^2 d\rho^2 + e^{2\rho}\eta_{\mu\nu}dx^\mu dx^\nu\,. \end{aligned} \tag{3.35}$$

In these coordinates, the horizon is mapped to $r = 0$ and $\rho = -\infty$, while the boundary is located at $r = \infty$ and $\rho = \infty$.

As it happens in the global case, one can also take the Euclidean continuation $t \to -i\tau$. Using for example the coordinates with radial distance r, this gives

$$ds^2 = \ell^2 \frac{dr^2}{r^2} + \frac{r^2}{\ell^2}\left(d\tau^2 + d\vec{x}^2\right)\,. \tag{3.36}$$

The $r = \infty$ boundary plane $\mathbb{R}^{1,d-1}$ of the Minkowskian version is now replaced by $\mathbb{R}^d$. On the other hand, the $r = 0$ plane now shrinks to a single point. It is sometimes convenient to compactify the boundary of the flat d–dimensional space to $\mathbb{S}^d$ by adding the point $r = 0$ to the boundary of $\mathbb{R}^d$. One can show that the space is diffeomorphic to a $(d+1)$–dimensional ball in $\mathbb{R}^{d+1}$ with metric

$$ds^2 = \frac{dy^2}{(\ell^2 - |y|^2)^2}\,, \qquad \sum_{I=1}^{d+1}(y_I)^2 \leq \ell^2\,. \tag{3.37}$$

References and further reading

You can find the details on many coordinate systems in AdS space in Chapter IX.11 of [15]. Other General Relativity notes or lectures where you can find related discussions are Section 4 of [14], or Section 39 from [16]. More advanced topics are treated in [17].

4 Pre-strings AdS/CFT

In the previous lectures, we have seen two apparently different ways of realizing theories with $\mathrm{O}(d,2)$ symmetry:

- Conformal field theories in d dimensions
- Relativistic gravitational theories in AdS_{d+1}.

Since the two objects live in spacetimes with different number of dimensions, it is difficult to imagine a relation between them. However, holography has always been a favorite principle when dealing with gravity. For several reasons, it is thought that in quantum gravity the number of degrees of freedom of a region of spacetime grows with the area of a region and not the volume. For example, Bekenstein and Hawking discovered that black holes can be understood as thermodynamical systems with an entropy proportional to the area of their horizon.

In this Section, we will see another form of holography, related to the fact that the boundary of AdS space is Minkowski. All the dynamics in AdS_{d+1} can be reformulated as a boundary effect which is captured by a $d-$dimensional local field theory. This will be an example of a *duality*, which is a relation between two different ways to describe the same theory, in the sense that all the physical observables are the same, but they are obtained in different ways from the two sides of the duality.

We start with a general construction of this correspondence without referring to explicit realizations. In particular, we will construct, for every $(d+1)-$dimensional gravitational theory with AdS_{d+1} vacuum, a set of correlation functions for operators which satisfy all constraints required by a $d-$dimensional local field theory with conformal invariance. In this Section we will specifically work at the level of an effective action in 5 dimensions, dealing with a theory that at low energy scales reduce to General Relativity (GR) coupled to gauge and matter fields. We will assume the existence of a suitable UV completion, but this will not be used for the following discussion. We shall enlighten some general aspects of the duality which do not refer to explicit realizations, making clear that all we need to set up such correspondence is a spacetime with the topological structure of AdS_{d+1} and a conformal boundary.

4.1 Formulation of the correspondence

In order to define a correspondence, we need to find a map relating observables from the two sides of the duality, as well as a prescription for computing physical quantities and amplitudes. We refer to the fields living in $(d+1)-$dimensional AdS space as the *bulk* fields, and we assume that their interaction is described by an effective action

$$S_{\mathrm{AdS_{d+1}}}(g_{\mu\nu}, A_\mu, \phi, \psi, \dots) \tag{4.1}$$

with an AdS_{d+1} vacuum. We included in the action the metric, gauge fields, scalars and fermions; in supergravity applications, it will also contain tensor fields.

We refer instead to the fields leaving on the CFT as the *boundary* fields, with an associated $d-$dimensional action $S_{\rm CFT}$. We remind that due to gauge invariance, most of the elementary fields will not be observables in the quantum theory. The spectrum will be specified by a complete set of primary operators in the CFT.

A basic point of the correspondence is the following statement: we will associate a field h in AdS with an operator $\mathcal{O}$ in the CFT with the same quantum numbers. They will be related to each other via the behaviour close to the boundary. We formalize more precisely this idea: from the $d-$dimensional point of view on the boundary, every operator $\mathcal{O}$ can be associated to a source h such that the action reads

$$S_{\rm CFT} + \int d^d x\, h(x)\mathcal{O}(x)\,. \tag{4.2}$$

In this sense, $h(x)$ is simply a standard $d-$dimensional background field that is introduced in order to compute correlation functions for the operator $\mathcal{O}(x)$, *i.e.* we have

$$e^{W(h)} = \left\langle e^{\int h\mathcal{O}} \right\rangle_{\rm QFT}\,, \tag{4.3}$$

where $W(h)$ is the generating functional for connected diagrams which gives

$$\langle \mathcal{O}\dots\mathcal{O}\rangle_{\rm c} = \frac{\delta^n W}{\delta h^n}\Big|_{h=0}\,. \tag{4.4}$$

At this point, we change our point of view. We consider the source $h(x)$ as the boundary value of a $(d+1)-$dimensional bulk field $h(x, x_{d+1})$. This means that for every source configuration $h(x)$ in the field theory, there is a $(d+1)-$dimensional field configuration

$$h(x) \to \hat{h}(x, x_{d+1})\,, \tag{4.5}$$

obtained by requiring that $\hat{h}(x, x_{d+1})$ solves the $(d+1)-$dimensional equations of motion derived from $S_{{\rm AdS}_{d+1}}$ in Eq. (4.1). This extension from the boundary to the bulk is unique if we impose suitable boundary conditions at the horizon of the AdS space.

The statement of the correspondence is

$$e^{W(h)} = \left\langle e^{\int h\mathcal{O}} \right\rangle_{\rm QFT} = \exp\left[-S_{{\rm AdS}_{d+1}}(\hat{h})\right]\,, \tag{4.6}$$

where on the left hand-side we have a functional depending on an arbitrary $d-$dimensional off-shell configuration $h(x)$ and on the right-hand side the on-shell value of the $(d+1)-$dimensional action, evaluated on the solutions of the equations of motion that reduce to $h(x)$ at the boundary. Since the knowledge of $W(h)$ for all possible sources of composite operators determines completely the CFT, the previous formula precisely states the required equivalence between the CFT and the bulk theory.

We comment on some immediate consequences of this statement.

- **Boundary conditions.** The equations of motion in the AdS geometry are second order and then we need to specify two boundary conditions in order to determine

a unique solution. Since typically the solutions of the equations of motion vanish or diverge at the boundary, we cannot simply require a condition like $\hat{h}(x, x_{d+1}) = h(x)$ at the boundary. Rather, we need to impose something of the form

$$\hat{h}(x, x_{d+1}) \sim f(x_{d+1}) h(x) \,, \tag{4.7}$$

with an appropriate choice of $f(x_{d+1})$. The second boundary condition will instead be imposed in the interior of AdS. If we consider the Euclidean version of AdS which is a ball whose center corresponds to a point, we will impose a regularity condition at such point. In Poincaré coordinates, this corresponds to imposing a suitable condition at the horizon.

- **Effective action.** In this general prescription, we are assuming that the CFT is dual to a weekly coupled gravitational system. In case we have a UV completion of the $(d+1)-$dimensional gravity theory, then the right hand-side of last identity in Eq. (4.6) should also be interpreted at the quantum level. Concretely, if the AdS space is embedded in a full string theory background, we need to replace the last equality with the string partition function with boundary condition h. It is only at weak coupling that it reduces to the tree level contribution given by the on-shell effective (supergravity) action.

- **Off-shell and on-shell.** We stress that the $d-$dimensional theory on the boundary is off-shell, while from the bulk side we are imposing the equations of motion coming from the effective action, which means that the gravity side is on-shell. This is a general feature of the AdS/CFT correspondence.

- **Field-operator map.** We will specify only later how to map the CFT operators to fields in the bulk: in fact, this will be specified by the details of the two theories and will be available when we have a constructive way to determine dual pairs. For the moment, we observe that the symmetries are a guiding principle to determine such relations. Some obvious couplings arise in the case of conserved currents; for example we have at the linearized level

$$S_{\text{CFT}} + \int d^d x \, \sqrt{g} \, (g_{\mu\nu} T^{\mu\nu} + A_\mu J^\mu + \ldots) \,, \tag{4.8}$$

which suggests that the operator associated with the graviton is the energy-momentum tensor, the one associated with the gauge field is a conserved current... The conservation of stress-tensor or other currents are associated to the gauge invariance at the level of the sources, in the sense that the generating functional $W(A_\mu, g_{\mu\nu})$ is a gauge-invariant function of $A_\mu, g_{\mu\nu}$. This statement extends by consistency to the bulk theory. Finally, we notice that global symmetries in the CFT correspond to local symmetries in AdS.

We will see in the last part of the course that there is a particularly important example where many details of the AdS/CFT correspondence can be made explicit: it is the relation between

- A particular well-behaved conformal field theory known as $\mathcal{N} = 4$ super Yang-Mills.
- Superstring theory, which is a UV completion of general relativity.

Both of them are theories invariant under a particular exchange between bosons and fermions, which goes under the name of supersymmetry. In such specific example the details of the mapping between operators of the CFT and fields in the gravity side can be given explicitly. Here we will discuss the simpler case (which is part of the total map in the mentioned example) between a scalar field ϕ and the dual primary operator at the boundary. We will specifically work within the realm of AdS_5/CFT_4 duality.

4.2 Field-operator map: physics in the bulk

One of the main points of the AdS/CFT correspondence is the existence of an additional (fifth) radial coordinate in the gravitational picture which is required in order to make sense of an holographic interpretation. In this scenario, it is important to understand that the radial coordinate in AdS space plays the role of an energy scale in the dual field theory picture. This can be seen clearly if we consider the AdS metric in Poincaré coordinates

$$ds^2 = \frac{\ell^2}{z^2}\left(dx_\mu dx^\mu + dz^2\right) . \tag{4.9}$$

A dilatation $x_\mu \to \lambda x_\mu$ at the boundary corresponds from the bulk perspective to the $SO(4,2)$ isometry

$$x_\mu \to \lambda x_\mu\,, \qquad z \to \lambda z\,. \tag{4.10}$$

In this way, we can roughly interpret

$$r = \frac{\ell^2}{z} \sim \mu\,, \tag{4.11}$$

as an energy scale μ similar to a RG flow, where the boundary region $z \ll 1 (\mu \gg 1)$ is associated to the UV regime of the CFT, and the horizon region $z \gg 1 (\mu \ll 1)$ with the IR. This is more than a formal identification: holographic computations of Green functions or Wilson loops associated with a specific reference scale μ are dominated by bulk contributions from the region of coordinates $r \sim \mu$.

Given these premises, let us now consider the dynamics of bulk fields in AdS_5. According to the statement of the correspondence, the boundary values of the fields are arbitrary functions of the four-dimensional coordinates x_μ, while the profile in the fifth directions is determined by imposing the equations of motion. We take a massive scalar field in AdS_5 and we Wick-rotate to Euclidean space. The action reads

$$\begin{aligned} S &= \int d^5x\,\sqrt{g}\left(g^{mn}\partial_m\phi\partial_n\phi + m^2\phi^2\right) = \\ &= \int dz \int d^4x\,\frac{\ell^5}{z^5}\left[\frac{z^2}{\ell^2}(\partial_z\phi)^2 + \frac{z^2}{\ell^2}(\partial_\mu\phi)^2 + m^2\phi^2\right] , \end{aligned} \tag{4.12}$$

with equations of motion

$$\partial_z \left(\frac{1}{z^3} \partial_z \phi \right) + \partial_\mu \left(\frac{1}{z^3} \partial^\mu \phi \right) = \frac{1}{z^5} m^2 \ell^2 \phi \,. \tag{4.13}$$

We start for convenience by looking at a mode independent from the x_μ directions, and we focus on the z dependence. In this way the Eq. (4.13) reduces to

$$z^5 \partial_z \left(z^{-3} \partial_z \phi \right) = m^2 \ell^2 \phi \,, \tag{4.14}$$

which has two independent power-like solutions $\phi \sim z^{\Delta_\pm}$ with the exponents being the roots of the constraint

$$m^2 \ell^2 = \Delta \left(\Delta - 4 \right) \,. \tag{4.15}$$

Since under a dilatation $z \to \lambda z$ and $x_\mu \to \lambda x_\mu$, the quantity Δ corresponds to the scaling dimension of the field, and will be later interpreted as the conformal dimension of the dual operator $\mathcal{O}$. We point out that Eq. (4.15) is the first example of a dictionary between the bulk and the boundary fields, since it relates the mass of a scalar in bulk AdS_5 with the dimension of an operator in the boundary CFT.

This relation can be generalized by similar techniques to the generic $d-$dimensional case, where it reads

$$m^2 \ell^2 = \Delta \left(\Delta - d \right) \,, \tag{4.16}$$

whose roots are given by

$$\Delta_\pm = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2 \ell^2} \,, \qquad \left(\Delta_- \leq \Delta_+ \right) . \tag{4.17}$$

This kind of identity is usually called *mass-conformal weight relation*. Notice that the smaller solution satisfies

$$\Delta_- = d - \Delta_+ \,, \tag{4.18}$$

due to the symmetries of the quadratic equation (4.16). The near-boundary behaviour of the on-shell field is

$$\phi \sim \phi_0 z^{\Delta_-} + \phi_1 z^{\Delta_+} \,, \tag{4.19}$$

where the coefficients correspond to two linearly independent solutions of the second order equation of motion. They can be distinguished by the fact that the solution with coefficient ϕ_0 is *not* normalizable at the boundary, *i.e.* it satisfies

$$\int d^5 x \, \sqrt{g} \, |\phi|^2 = \infty \,, \tag{4.20}$$

while the one corresponding to ϕ_1 is normalizable.

At this point we can include the x_μ dependence in the solution for the scalar field by modifying the solution to

$$\phi(z, x_\mu) \sim z^{\Delta_-} \left(\phi_0(x) + \mathcal{O}(z^2) \right) + z^{\Delta_+} \left(\phi_1(x) + \mathcal{O}(z^2) \right) \,, \tag{4.21}$$

where we still identify the coefficients of the two linearly-independent solutions as $\phi_0(x)$ and $\phi_1(x)$.

We can now discuss the boundary conditions to be imposed on a field near the boundary. Notice that the leading term of the solution in Eq. (4.21) can be singular if $\Delta_- < 0$ or vanish if $\Delta_- > 0$, while it approaches a constant only when $\Delta_- = 0$. In order to have a consistent prescription, we need to impose at the boundary $z = 0$ the condition

$$\phi(z, x_\mu) \to z^{\Delta_-} \phi_0(x)\,, \tag{4.22}$$

where we call $\phi_0(x)$ the boundary value of the field to be identified with the source of the dual operator $\mathcal{O}$. Once the value of $\phi_0(x)$ is specified, we have a unique regular solution which extends to the entire bulk space. In particular, $\phi_1(x)$ will be determined as a functional of $\phi_0(x)$ by imposing the equations of motion and the regularity at the center of Euclidean AdS. By dimensional analysis, $\phi_1(x)$ will be interpreted as the vacuum expectation value for the dual field theory operator $\mathcal{O}(x)$.

We stop here to comment on the mass-dimension relation (4.16) and on the choice of the boundary conditions. We observe that $m^2 > 0$ only for $\Delta = \Delta_+ > d$, which corresponds to an irrelevant operator from the CFT side; we find instead a massless mode when $\Delta_+ = d$, which is associated to a marginal operator. Naively, we would expect that a consistent theory could arise only when $m^2 \geq 0$, but it turns out that in AdS space there exist modes with imaginary mass (corresponding to relevant operators in the dual field theory) without giving rise to a tachyon. In particular, energy is positive as long as the following constraint holds

$$m^2 \ell^2 \geq -\frac{d^2}{4}\,, \tag{4.23}$$

which goes under the name of Breitenlohner-Freedman bound. The reason is that the curvature contribution in AdS gives rise to a positive contribution for the energy of a scalar field propagating in the spacetime, and as a result the bound is changed with respect to flat space.

This bound can be derived starting from the action of a massive scalar field $\phi(z)$ dependent only from the radial coordinate of AdS_{d+1}

$$S \sim \ell^{d-1} \int dz d^d x\, \frac{1}{z^{d+1}} \left(z^2 \partial_z \phi \partial_z \phi + m^2 \ell^2 \phi^2 \right)\,, \tag{4.24}$$

and performing the coordinate transformation $y = \log z$ followed by a rescaling of the scalar as $\phi = z^{d/2} \varphi$. Up to boundary terms, this gives

$$S \sim \ell^{d-1} \int dy d^d x \left[\partial_y \varphi \partial_y \varphi + \left(m^2 \ell^2 + \frac{d^2}{4} \right) \varphi^2 \right]\,. \tag{4.25}$$

Since this can be interpreted as the action of a scalar field φ in flat space with effective mass

$$m_{\text{eff}}^2 \ell^2 = m^2 \ell^2 + \frac{d^2}{4}\,, \tag{4.26}$$

the condition $m^2_{\rm eff} \geq 0$ required to obtain a consistent theory gives indeed the bound (4.23).

Even after applying the Breitenlohner-Freedman bound, we are still left with a puzzle. The unitarity bound would in general allow for the condition

$$\Delta \geq \frac{d-2}{2}\,, \tag{4.27}$$

while Eq. (4.23) only includes solutions dual to field theory operators having $\Delta \geq d/2$. There is still a window of conformal dimensions

$$\frac{d-2}{2} \leq \Delta \leq \frac{d}{2}\,, \tag{4.28}$$

which we do not know how to realize from the bulk point of view.

In order to solve the mistery, we need to remind that the solutions for the conformal dimension that we are considering refer to the largest root Δ_+, but when

$$-\frac{d^2}{4} \leq m^2\ell^2 \leq -\frac{d^2}{4}+1\,, \tag{4.29}$$

the mass-dimension relation (4.16) admits two solutions within the unitarity bound. The consequence is that one can choose both the possibilities, and they amount to different choices of the boundary conditions: they correspond to select either $\phi_0(x)$ or $\phi_1(x)$ as the boundary value of the bulk field.

For a 4D Euclidean CFT: rotation group=SO(4), but the full conformal group is SO(5,1). For a 4D Lorentzian CFT: Lorentz group=SO(3,1), and the full conformal group is SO(4,2). So SO(4) is relevant because j1,j2 label the 4D Euclidean rotation representation, SO(4)≈SU(2)L×SU(2)R, but it is not the full conformal group.

The relation between mass and dimension can be found for fields with arbitrary spin with a similar procedure. We summarize the results in Table 1. The mass is a function of the three quantum numbers (Δ, j_1, j_2) or, more geometrically, of the Casimirs of the conformal group SO(4). The mass is in general an ambiguous concept in AdS due to the coupling with negative curvature. For example, for a scalar field one can add the so-called conformal coupling ϕR^2 to the action. The definition used here is consistent with the dual interpretation in terms of conformal fields. For example, the massless gauge field is associated to an operator of conformal dimension $\Delta = d-1$, which corresponds to a conserved current in the dual field theory; and the massless graviton is associated to $\Delta = d$, which matches with the dimension of the energy-momentum tensor.

ξRϕ^2? the normal mass term is also proportional to m^2ϕ^2. ξ is a coupling constant. dimensions should match the mass term. conformal coupling should be dimensionless

Type of field	Spin (j_1, j_2)	Mass-weight relation
Scalar ϕ	$(0,0)$	$m^2\ell^2 = \Delta(\Delta-d)$
Vector A_μ	$(\frac{1}{2},\frac{1}{2})$	$m^2\ell^2 = (\Delta-1)(\Delta-d+1)$
Spin $\frac{1}{2}$ ψ	$(\frac{1}{2},0)+(0,\frac{1}{2})$	$m^2\ell^2 = \Delta - \frac{d}{2}$
Spin $\frac{3}{2}$ ψ_μ	$(\frac{1}{2},1)+(1,\frac{1}{2})$	$m^2\ell^2 = \Delta - \frac{d}{2}$
Symmetric massless spin 2 tensor $g_{\mu\nu}$	$(1,1)$	$m^2\ell^2 = 0 = \Delta(\Delta-d)$
Antisymmetric 2-form $B_{\mu\nu}$	$(1,0)+(0,1)$	$m^2\ell^2 = (\Delta-2)(\Delta-d+2)$

Table 1: Mass-weight relations for fields with various spin.

4.3 Construction of the correlation functions

We want to sketch the general construction of $n-$point correlation functions for a CFT with gravitational dual given by an effective action in AdS_{d+1}. For all reasonable actions S_{AdS}, we can inductively construct a set of correlation functions which satisfy all the requirements of a local field theory.

We will focus for simplicity on a set of scalar fields ϕ_i with masses m_i interacting through the action $S_{\mathrm{AdS}}(\phi)$. We denote with ϕ_i^0 the boundary value of the fields obtained by imposing Eq. (4.21) with $(\Delta_-)_i = d - \Delta_i$. The field ϕ_i^0 is interpreted as the source for the dual operators $\mathcal{O}_i$ of dimensions Δ_i. The CFT generating functional is given by evaluating the bulk action on the solution of the equations of motion with the prescribed boundary conditions. In particular, as in standard QFT, the correlation functions are obtained by differentiation of the on-shell bulk action with respect to the sources, which we set to vanish afterwards. In mathematical language we have

$$\langle \mathcal{O}_1 \dots \mathcal{O}_n \rangle = \frac{\delta^n S}{\delta\phi_1^0 \dots \delta\phi_n^0}\Big|_{\phi_i^0=0} . \tag{4.30}$$

Since on-shell fields vanish when the sources ϕ_i^0 are turned off, it is obvious that in order to obtain $n-$point functions we need to consider interactions in the bulk action which contain at most n fields, because terms with more of them will otherwise vanish.

We will consider in turn one, two and three-point functions. Up to order $n = 2$, from the above discussion we know that we only need to keep the quadratic terms in the action, so that the equations of motion correspond to Klein-Gordon equations in the bulk. We need a Green function or a *boundary-to-bulk* propagator to extend a field ϕ with mass m and conformal dimension Δ from the boundary of spacetime to the bulk. It reads

$$\phi(z, x_\mu) = \int dx'_\mu \, K(z, x_\mu - x'_\mu)\phi^0(x'_\mu) , \tag{4.31}$$

which satisfies

$$\left(-\Box + m^2\right) K = 0 , \tag{4.32}$$

$$K \to z^{4-\Delta}\, \delta(x_\mu - x'_\mu) \qquad z \to 0 . \tag{4.33}$$

Let's give the solution to this equation. We work in Euclidean AdS_5 and we compactify the boundary to S^4 by adding the point $z = \infty$. We have seen previously that the Euler-Lagrange equations (4.13) have a solution independent from the x_μ coordinates, with behaviour $\phi \sim z^\Delta$. This solution vanishes on most of the boundary, where $z = 0$, and then suddenly diverges when it reaches the point compactified from infinity: then it corresponds to a Dirac delta function for the insertion of a source at $z = \infty$. Since the equations of motion are covariant under the Euclidean conformal group $\mathrm{O}(5, 1)$, we can find the generic Green function (4.33) by mapping $z = \infty$ to a generic point on the boundary by a conformal transformation. In particular, we can send it to the point $(z = 0, x_\mu = 0)$ with the coordinate change

$$z \to \frac{z}{z^2 + x_\mu^2} , \qquad x_\mu \to \frac{x_\mu}{z^2 + x_\mu^2} . \tag{4.34}$$

Applying this transformation to the solution $\phi \sim z^\Delta$, we get the Green function

$$K(z, x_\mu) = c\, \frac{z^\Delta}{(z^2 + x_\mu^2)^\Delta}\,, \tag{4.35}$$

with c an appropriate normalization. Its specific value is not important for the discussion below, but it can be found by computing

$$\begin{aligned} &\int dx\, K(z,x)\phi_0(x) = cz^{4-\Delta} \int dx\, \frac{z^{2\Delta-4}}{(z^2+x^2)^\Delta}\phi_0(x) = \\ &= cz^{4-\Delta} \int dy\, \frac{\phi_0(zy)}{(y^2+1)^\Delta} \xrightarrow[z\to 0]{} c \int \frac{dy}{(1+y^2)^\Delta}\, z^{4-\Delta}\phi_0(0)\,. \end{aligned} \tag{4.36}$$

In going from the second to the third line, we changed variables to $x = yz$. In order to obtain the correct normalization at the boundary, we impose

$$c^{-1} = \int \frac{dy}{(1+y^2)^\Delta}\,. \tag{4.37}$$

At this point we expand the solution in powers of z as in Eq. (4.21) and we plug such expression in Eq. (4.31) to get

$$\phi_1(x) = c \int dx'\, \frac{\phi_0(x')}{|x-x'|^{2\Delta}}\,. \tag{4.38}$$

Now we consider the action, which after integrating by parts reads

$$S_{\text{AdS}} \sim \int_{\partial\mathcal{M}} \sqrt{g}\, \phi \partial^n \phi + \int_{\mathcal{M}} \sqrt{g}\, \phi \left(-\Box + m^2\right) \phi\,, \tag{4.39}$$

where $\mathcal{M}$ is the bulk AdS spacetime with metric $g_{\mu\nu}$, we denote with $\partial\mathcal{M}$ its boundary with induced metric $h_{\mu\nu}$ and ∂^n is the normal derivative as in the divergence theorem. On-shell, the term proportional to the equations of motion vanishes and we obtain

$$S_{\text{AdS}} \sim \frac{\ell^3}{z^3}\, \phi \partial_z \phi \Big|_{\partial\mathcal{M}}\,. \tag{4.40}$$

By inserting Eq. (4.21), we observe that there are some divergent terms in z, all proportional to local functions of ϕ_0. Since these are contact terms, we can always get rid of them with an appropriate choice of local counterterms and for this reason we will disregard them. The terms containing ϕ_1 are instead non-local, but they only contribute to finite expressions in z which read

$$S_{\text{AdS}} \sim \int dx\, \phi_0(x)\phi_1(x) \sim \int dx dx'\, \frac{\phi_0(x)\phi_0(x)}{|x-x'|^{2\Delta}}\,. \tag{4.41}$$

From this expression, we can immediately derive the one-point functions as

$$\langle \mathcal{O}(x) \rangle = \frac{\delta S_{\text{AdS}}}{\delta \phi_0(x)}\Big|_{\phi_0=0} = \phi_1(x)\Big|_{\phi_0=0}\,. \tag{4.42}$$

Indeed, it is clear that the normalizable solution to the equations of motion plays the role of the vacuum expectation value of the dual boundary operator. Since $\phi_1 \propto \phi_0$, after setting the source to zero we obtain a vanishing one-point function, which is consistent with conformal invariance.

From Eq. (4.41) we also obtain the two-point function

$$\langle \mathcal{O}(x)\mathcal{O}(x')\rangle = \frac{\delta^2 S_{\rm AdS}}{\delta\phi_0(x)\delta\phi_0(x')}\Big|_{\phi_0=0} = \frac{1}{(x-x')^{2\Delta}}\,. \tag{4.43}$$

This result agress with the behaviour of two-point functions of primary scalar fields of dimension Δ in a CFT.

In order to compute higher-point functions, we need to consider all the terms up to order n in the fields from the action

$$S_{\rm AdS} = \int d^5x \left(\frac{1}{2}\sum_i (\partial\phi_i)^2 + \frac{m_i^2}{2}\phi_i^2 + \sum_{k=3}^{n} \lambda_{i_1\ldots i_k}\phi_{i_1}\ldots\phi_{i_k}\right)\,. \tag{4.44}$$

For simplicity, we are considering fields with canonical kinetic terms, no higher derivative interactions and we are neglecting other couplings with gauge fields, fermions, gravity and other possibilities. These ingredients can be incorporated into the AdS/CFT correspondence with technical efforts, but they are conceptually treated in the same way.

Since the equations of motion contain higher order terms, now they cannot be solved anymore explicitly and then it is necessary to set up a perturbative expansion. For example, the typical equation

$$\left(-\Box + m^2\right)\phi = \lambda\phi^n \tag{4.45}$$

can be solved in power series of λ by starting from the Green functions for the Klein-Gordon equation in the bulk. Indeed, at order 0 in λ we know that the solution is

$$\phi^{(0)}(z, x_\mu) = \int dx'\, K(z, x_\mu - x'_\mu)\phi^0(x'_\mu)\,, \tag{4.46}$$

where K is the boundary-to-bulk propagator defined in Eq. (4.31). At first order in λ, we need to consider a bulk-to-bulk Green function G defined by solving Eq. (4.45) as

$$\phi^{(1)}(z, x_\mu) = \lambda \int dz' \int dx'\, G(z-z', x_\mu - x'_\mu)\left[\phi^{(0)}(z', x'_\mu)\right]^n\,, \tag{4.47}$$

and satisfying the initial condition problem

$$\left(-\Box + m^2\right) G(z-z', x_\mu - x'_\mu) = \frac{1}{\sqrt{g}}\,\delta(z-z')\delta(x_\mu - x'_\mu)\,. \tag{4.48}$$

The recursive procedure continues by putting the first-order solution $\phi^{(1)}$ in the right hand-side of Eq. (4.45) to determine the second order $\phi^{(2)}$, and one can proceed until the desired order in the expansion parameter is reached.

For the purposes of computing correlation functions, however, we need to remember that all the solutions are function of ϕ_0 (the boundary counterpart of the bulk field), and then it is sufficient to stop after a finite number of steps: all the contributions containing more than n powers of ϕ_0 are discarded after setting the sources to vanish. This procedure resembles a perturbative computation in QFT, the difference being that propagators can connect points at the boundary or in the bulk. These pictorial representations are called Witten diagrams. Some examples of them are depicted in Fig. 6.

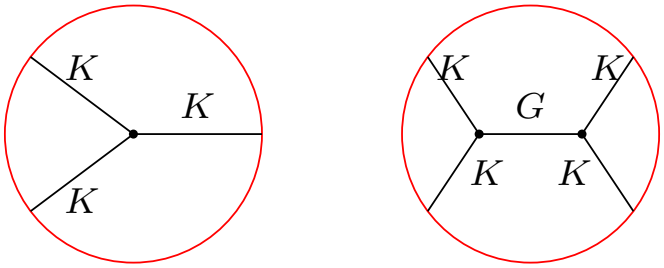


Figure 6: Witten diagrams associated to the computation of three and four-point functions, where K is a boundary-to-bulk propagator and G a bulk-to-bulk one.

One can employ this procedure for the case of three-point functions, where the only non-vanishing contribution comes from three external points (placed at the boundary) connected in the bulk with a cubic vertex. Performing the above-mentioned computations explicitly, for a cubic interaction $\lambda_{ijk}\phi_i\phi_j\phi_k$ one gets

$$\langle \mathcal{O}_i(x_i)\mathcal{O}_j(x_j)\mathcal{O}_k(x_k)\rangle \sim \frac{\lambda_{ijk}}{|x_i-x_j|^{\Delta_i+\Delta_j-\Delta_k}|x_j-x_k|^{\Delta_j+\Delta_k-\Delta_i}|x_k-x_i|^{\Delta_k+\Delta_i-\Delta_j}} \,. \tag{4.49}$$

This behaviour is consistent with the three-point functions for scalar primary operators in a CFT.

We conclude the Section with some general remarks:

- We observed that the computations of the correlation functions involve divergent terms. The local expressions in the generating functional $W(\phi_0)$ can always be eliminated with the choice of appropriate counterterms, which can be performed *e.g.* with the introduction of a UV cutoff in the effective action. We remark that this procedure can be sistematically performed and goes under the name of holographic renormalization.

- Similar results hold for fermions, vectors etc. For example, a massless vector field

will give rise to a two-point function of the form

$$\langle J_\mu(x) J_\nu(x') \rangle = \frac{\delta_{\mu\nu}}{|x-x'|^6} - 2\,\frac{(x-x')_\mu (x-x')_\nu}{|x-x'|^8}\,, \tag{4.50}$$

which corresponds to a field (operator) of dimension $\Delta = 3$ in a four-dimensional QFT, as it happens for conserved currents.

References and further reading

Most part of this Section is inspired from Sections 2.1−2.3 of [18], which explains many physical concepts of the AdS/CFT correspondence before being exposed to a significant introduction to string theory. Similar material can be found in Chapters 5.3−5.4 of [1]; the example with the scalar field is motivated from the dimensional reduction of supergravity fields.

For further reading on more advanced general aspects of the AdS/CFT correspondence, we suggest:

- Holographic renormalization (systematic procedure to regularize divergencies in the context of AdS/CFT duality) [19].
- An application of holography: the computation of the Weyl anomaly using the AdS/CFT correspondence [20].

5 Supersymmetry

5.1 What is supersymmetry?

Symmetry has always been a guiding principle in physics: using the invariances of a system under certain transformations, we can find conserved quantities (*e.g.* via Noether theorem, first integrals of motion or other techniques), build consistent models and get a simpler explanation of many phenomena. One could believe that if a symmetry is possible, Nature should realize it; on the other hand, in recent physics we are facing many beautiful mathematical structures which suggest promising interpretations of General Relativity (GR) and Quantum Mechanics (QM), but it is not obvious how they should be realized.

Among the possible invariances that are not at the basis of the Standard Model, but is a promising scenario to investigate physical phenomena, there is supersymmetry (SUSY): it is a spacetime symmetry relating bosons and fermions, *i.e.* particles with different spin. As it is standard, one expects that there should be a (fermionic) generator, called supercharge, associated to this invariance. This can be schematically represented as

$$Q|\text{boson}\rangle = |\text{fermion}\rangle\,, \qquad Q|\text{fermion}\rangle = |\text{boson}\rangle\,. \tag{5.1}$$

This action has already a couple of important consequences:

- It changes the spin of a particle (from integer to half-integer or viceversa) and hence its spacetime properties: this is the reason why it is not an internal symmetry.
- In a theory where supersymmetry is realized, each single-particle state has at least a superpartner.

In particular, supersymmetric-invariant field theories will contain (super)multiplet of particle states. It is important to get a first feeling of their behaviour from the algebra of the other generators with the supercharge

$$[Q, P_\mu] = 0\,, \qquad [Q, G] = 0\,, \qquad [Q, J_{\mu\nu}] \neq 0\,, \tag{5.2}$$

where P_μ is the generator of translations, G of internal symmetries and $J_{\mu\nu}$ of the Lorentz group. This implies in particular that particles belonging to the same supermultiplet will have the same mass and quantum numbers, even if they will differ by the spin.

The symmetry that we presented should be thought as an invariance that should be respected when building the action of the theory, including the interactions. Imposing such a requirement will imply many constraints on the spectrum of the theory and on the behaviour of the coupling constants under the Renormalization Group flow. This symmetry can also be made local by requiring consistency with general coordinate transformations: in such case, it will give rise to *supergravity*. Since supersymmetry is a spacetime symmetry, and since the algebra schematically contains the anticommutator

$$\{Q, \bar{Q}\} \sim P_\mu\,, \tag{5.3}$$

we conclude that having diffeomorphism invariance is equivalent to performing a gauging of supersymmetry. The gauge mediator will be a spin 3/2 particle, called *gravitino*.

Up to this point we generically discussed of the existence of supersymmetry generators, but it is interesting to remark that there can be more copies of them in a certain model: we usually count such copies as Q^I with $I \in \{1, \ldots, \mathcal{N}\}$. Their number is constrained by the following observation:

Any supermultiplet contains particles with spin at least as large as $\mathcal{N}/4$.

This implies that in four dimensions:

- $\mathcal{N} \leq 4$ to describe a local gauge theory with maximum spin one particles.
- $\mathcal{N} \leq 8$ to describe gravity with a spin two particle.

Similar statements can be made in other dimensions, where the number $\mathcal{N}$ will change but the counting of the number of real components for all the supercharges will be the same.

5.2 Motivations for supersymmetry

After giving a feeling of how supersymmetry works, we would like to present several motivations to consider it.

Theoretical reasons

Guided by the principle to include in the description of Nature all the possible invariances that could be realized, one can ask what are the most general symmetries that the S-matrix could allow. This is established by the Coleman-Mandula theorem (1967), which states that under the assumptions of locality, causality, positivity of energy, the only possible continuous symmetries of the S-matrix are generated by the Poincaré group generators $P_\mu, J_{\mu\nu}$ together with a generator G for an internal symmetry group which commutes with them

$$[G, P_\mu] = [G, J_{\mu\nu}] = 0\,. \tag{5.4}$$

There are some possibilities to evade the Coleman-Mandula theorem, by weakening some of its assumptions. The one of interest for supersymmetry is that we can include anti-commutators in the algebra, as we have seen is necessary to consider supercharges, see Eq. (5.3). It was shown in 1975 by Haag, Lopuszanski and Sonhius that supersymmetry is the only possible such option. In this way, the biggest set of symmetries is enlarged to

$$\text{Poincaré} \times \text{Internal} \;\to\; \text{SuperPoincaré} \times \text{Internal}\,. \tag{5.5}$$

We say in passing that another way out to Coleman-Mandula theorem is to consider conformal invariance, since the invariance under dilatations does not allow for the existence of asymptotic states and then of an S-matrix.

Another reason to consider supersymmetry resides in the search of a theory of unification: since this framework naturally includes both bosons and fermions, and in addition also includes in the same multiplets matter and radiation, it is a serious candidate for a consistent theory containing all the interactions.

Finally, we will see that supersymmetry is required in order to have a consistent UV completion of gravity in the form of String Theory, which naturally admits the existence of a graviton in the spectrum. This is another theoretical reason to expect that supersymmetry should be included in a model unifing GR and QM.

Elementary particle perspective

In addition to more abstract theoretical reasons, there are some hints from particle physics that suggest to take into account supersymmetry. We refer in particular to *naturalness* and the *hierarchy problem*.

We explain it here in detail. All the fundamental forces of Nature, excluded gravity, are comprised in the formulation of the Standard Model (SM) whose typical electroweak scale is

$$M_{\rm ew} \sim 250 \ \ {\rm GeV}\,, \tag{5.6}$$

and it is very well tested experimentally in this regime. On the other hand, we expect that gravity will not be negligeble when increasing the energy, in particular at the Planck scale

$$M_{\rm Pl} \sim 10^{19} \ {\rm GeV}\,. \tag{5.7}$$

Since $M_{\rm ew} \ll M_{\rm Pl}$, we also expect that new physics should arise at intermediate energies. In order to see this, consier that the Higgs potential in the Standard Model reads

$$V(H) \sim \mu^2 |H|^2 + \lambda |H|^4 \qquad \text{where} \qquad \mu^2 < 0\,. \tag{5.8}$$

Careful with the wording
The VEV itself does not alone predict 125 GeV.

Experimentally, the minimum of this potential is around $\langle H \rangle = \sqrt{-\mu^2/2\lambda} \sim 174$ GeV which in turn implies that the Higgs mass is actually measured to be $m_H \sim 125$ GeV. $m_H^2 \sim \lambda v^2$

In perturbation theory, scalar masses are usually subject to quadratic divergences after renormalization; indeed the SM fermion f coupled to the Higgs boson via the Yukawa interaction $-\lambda_f H \bar{f} f$ induces a one-loop correction which reads

$$\Delta m_H^2 \sim -2\lambda_f^2 \Lambda^2\,, \tag{5.9}$$

where Λ is a UV cutoff, see Fig. 7. In order to protect the Higgs mass, the UV cutoff should be in the TeV scale and the SM should be seen as an effective theory valid at the energy scales $E < M_{\rm eff} \sim$ TeV.

Beyond such scale, we expect new physics to appear in such a way to protect the otherwise divergent perturbative contribution to the Higgs mass; in particular, new bosonic and fermionic degrees of freedom, and possibly additional couplings to the Higgs scalar, can appear. In turn, they will give rise to other perturbative corrections of the order $M_{\rm Pl}$, since this will be the corresponding UV cutoff of the theory. This implies that we need a very huge fine tuning in order to keep the Higgs mass around 125 GeV!

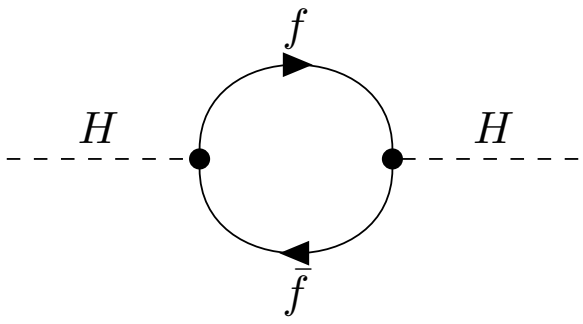


Figure 7: Feynman diagram representing the one-loop self-energy correction to the Higgs mass due to the fermion coupling.

Such phenomenon is called hierarchy problem, because the experimental value of the Higgs mass is unnaturally smaller than the theoretical value that should be expected from these reasonings.

However, it is possible to find a conceptually simple way out to this problem: since fermions contribute to negative signs in loop corrections, we could obtain precisely the cancellation of the UV divergences needed to solve the hierarchy problem by assuming that the model admits another coupling of the form $-\lambda_S|H|^2|S|^2$. Indeed, such vertex would induce an interactions whose contribution, represented pictorially in Fig. 8, reads

$$\Delta m_H^2 \sim \lambda_S \Lambda^2 \,. \tag{5.10}$$

We then see that an appropriate choice of λ_f, λ_S allows to cancel the quadratic divergences in Eqs. (5.9) and (5.10). A natural way to invoke such cancellations without an ad-hoc requirement is to introduce supersymmetry in the model, since it precisely relates bosonic and fermionic degrees of freedom!

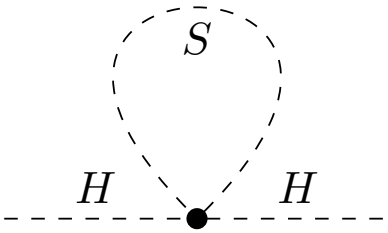


Figure 8: Feynman diagram representing the one-loop self-energy correction to the Higgs mass due to the scalar coupling.

However, we need to remark that the known bosons and fermions cannot be superpartner of each others: apart from the absence of degeneracies in the mass of elementary particles that are classified, they cannot be related to each other by supersymmetry

because gauge bosons transform in the adjoint representation of the SM group, while matter fields transform in the fundamental. We should instead expect to observe new particles which are superpartners of the ones that we know. The problem, up to these days, is that the hierarchy problem requires that the scale where we should observe these new particles should be around 1 TeV, even if we admit that supersymmetry is spontaneously broken. Since these new particles were not observed at LHC, there are ongoing discussions on the possibility to admit or not supersymmetry in the group of invariances of a theory aiming to describe Nature.

From a phenomenological point of view, there is another reason to believe that supersymmetry could provide a solution. The SM is invariant under the symmetry group

$$\mathrm{SU}(3) \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \,, \tag{5.11}$$

which contains three coupling constants running with the energy scale μ along the RG flow as

$$\frac{4\pi}{g_i^2(\mu)} = \frac{b_i}{2\pi} \log\left(\frac{\mu}{\Lambda_i}\right) \,. \tag{5.12}$$

If we suppose that there no particles except the ones contained in the SM would arise at higher energies, the three coupling constants tend to approximately converge at the value

$$M_{\mathrm{GUT}} \sim 10^{15} \text{ GeV} \,. \tag{5.13}$$

This observation naturally leads to the formulation of a Grand Unified Theory (GUT) with gauge group SU(5) which is spontaneously broken to the SM group of invariances at lower energy. Such a process can be implemented by requiring the existence of a scalar, called an heavy Higgs, which exactly like its lighter counterpart should give the correct masses to the particles at lower energies. This scenario suffers however some problems: the heavy Higgs boson would present a similar hierarchy problem as the one discussed above; in addition, there is a proton deacy problem, since the additional gauge bosons in the GUT can mediate baryon number violating transitions, thus allowing for a process like

$$p \to e^+ + \pi_0 \,. \tag{5.14}$$

Then the proton is not fully stable and its expected timelife in the GUT framework is violated by the present experimental measurements.

Suprisingly, it turns out that supersymmetry can solve all these problems at once:

- The three coupling constants of the SM precisely meet (not only approximately) at a common energy scale.
- The GUT scale is slightly raised, allowing compatibility of the lifetime of the proton with current experimental bounds.
- The hierarchy problem is solved as discussed for the light Higgs boson.

Additional comments

In addition to theoretical and elementary particle physics motivations, other reasons can come from certain models of dark matter and cosmology. We should also consider supersymmetry as a useful tool to understand the dynamics of strongly coupled gauge theories. Due to the large amount of symmetries that it brings with it, many non-renormalization theorems can be found and then the running of the coupling constants greatly simplifies. The knowledge and the techniques acquired to study these cases can be hopefully applied also to other scenarios.

Finally, there are some reasons to consider supersymmetry in the context of AdS/CFT correspondence. The construction of interacting conformal field theories that survive quantization is extremely difficult, because we need all the β functions to vanish. One possibility is to consider fixed points of RG flows, but it is difficult to have control over them, since they are very often strongly coupled theories, and in some situations they do not even have a Lagrangian formulation. In these cases, the additional constraints imposed by supersymmetry lead to well-behaved theories with nice UV behaviour. This will be the case of $\mathcal{N} = 4$ super Yang-Mills, which will play a central role in AdS/CFT duality.

5.3 Simplest renormalizable SUSY theory: Wess-Zumino model

In order to concretely understand the features of supersymmetry that we listed above, we consider the simplest SUSY-invariant theory: the so-called *Wess-Zumino model*. We start with a free theory composed by a complex scalar and a complex (left-handed) Weyl fermion

$$\mathcal{L} = -|\partial\phi|^2 - i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{m}{2}\left(\psi\psi + \bar{\psi}\bar{\psi}\right) - m^2|\phi|^2 \,. \tag{5.15}$$

A minimal requirement that we should satisfy in order to map bosonic fields into fermionic ones and viceversa is that their degrees of freedom match. On-shell, *i.e.* at the level of the EOM, it can be easily verified that the theory propagates two (real) bosonic and two spinorial degrees of freedom, the latter corresponding to the two possible spin polarizations of ψ.

We can also perform the counting off-shell. The Weyl fermion contains two complex components, which correspond to four real degrees of freedom. Since the Dirac action is first order in the derivatives, the corresponding EOMs an be used as constraints which allow to get rid of the canonical momenta (which are not anymore independent variables). This is responsible for reducing the original four real degrees of freedom for the Weyl field to only two. In the action (5.15) we did not solve yet for the constraints (*i.e.* we did not integrate out the non-physical degrees of freedom), and for this reason there is a mismatch between the bosonic and fermionic degrees of freedom. This is not a fundamental problem, but rather a nuisance: we always need to remember that the constraint holds, but it would be easier to get rid of it. It is possible to do that by introducing an auxiliary complex field F, which has no dynamics but its role is simply to match the off-shell degrees of freedom between scalars and fermions. For these reasons,

one can use the Lagrangian

$$\mathcal{L} = -|\partial\phi|^2 - i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi + |F|^2 - m\left(\frac{1}{2}\psi\psi + \frac{1}{2}\bar{\psi}\bar{\psi} + F\phi + \bar{F}\bar{\phi}\right) . \tag{5.16}$$

By inspection of either Eq. (5.15) or (5.16), we notice that the symmetries of the action are

- Poincaré invariance.
- Global U(1) phase invariance.

Less obviously, the action is supersymmetric, *i.e.* it is invariant under the following spacetime transformations relating bosons and fermions:

$$\delta_\varepsilon\phi = \sqrt{2}\varepsilon^\alpha\psi_\alpha \,, \tag{5.17}$$

$$\delta_\varepsilon\psi_\alpha = \sqrt{2}\varepsilon_\alpha F + \sqrt{2}i(\sigma^\mu)_{\alpha\dot{\beta}}\bar{\varepsilon}^{\dot{\beta}}\partial_\mu\phi \,, \tag{5.18}$$

$$\delta_\varepsilon F = \sqrt{2}i\bar{\varepsilon}_{\dot{\beta}}(\bar{\sigma}^\mu)^{\dot{\beta}\alpha}\partial_\mu\psi_\alpha \,. \tag{5.19}$$

We formally define the transformations of the fields under supersymmetry as coming from the action of a fermionic generator acting on them

$$\delta_\varepsilon\phi = i\left[\varepsilon Q + \bar{\varepsilon}\bar{Q}, \phi\right] , \qquad \delta_\varepsilon\psi = i\left[\varepsilon Q + \bar{\varepsilon}\bar{Q}, \psi\right] , \tag{5.20}$$

where $\varepsilon^\alpha, \bar{\varepsilon}_{\bar{\alpha}}$ are spinorial parameters with $\alpha, \dot{\alpha} \in \{1, 2\}$, and the contraction is performed as

$$\varepsilon Q \equiv \varepsilon^\alpha Q_\alpha = \epsilon^{\alpha\beta}\varepsilon_\beta Q^\alpha \,, \qquad \bar{\varepsilon}\bar{Q} \equiv \bar{\varepsilon}_{\dot{\alpha}}\bar{Q}^{\dot{\alpha}} = (\varepsilon Q)^* \,. \tag{5.21}$$

The SUSY algebra can be explicitly derived by computing two variations of the fields, and then commuting them. We consider for example the scalar field: an explicit computation gives

$$[\delta_\varepsilon, \delta_\eta]\phi = -2i\left(\eta\sigma^\mu\bar{\varepsilon} - \varepsilon\sigma^\mu\bar{\eta}\right)\partial_\mu\phi \quad \Rightarrow \quad [\delta_\varepsilon, \delta_\eta] = -2i\left(\eta\sigma^\mu\bar{\varepsilon} - \varepsilon\sigma^\mu\bar{\eta}\right)\partial_\mu \,. \tag{5.22}$$

In this way we find

$$[\varepsilon Q, \bar{\eta}\bar{Q}] = 2\varepsilon\sigma^\mu\bar{\eta}P_\mu \,, \qquad [\varepsilon Q, \eta Q] = [\bar{\varepsilon}\bar{Q}, \bar{\eta}\bar{Q}] = 0 \,. \tag{5.23}$$

Stripping off the dependence on the spinorial parameters, we get

$$\{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \,, \tag{5.24}$$

$$\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2(\sigma^\mu)_{\alpha\dot{\alpha}}P_\mu \,. \tag{5.25}$$

This is the precise relation between the anticommutator of the supercharges and the momentum operator, which is the generator of translations.

The couple $Q, \bar{Q}$ forms a Majorana spinor in four dimensions, and then the commutation rules with the Lorentz generators are

$$[Q_\alpha, J^{\mu\nu}] = (\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta \,, \tag{5.26}$$

where the generator of Lorentz transformations for left-handed spinors is

$$(\sigma^{\mu\nu})_\alpha{}^\beta = \frac{i}{4}\left[\sigma^\mu_{\alpha\dot\alpha}(\bar\sigma^\nu)^{\dot\alpha\beta} - \sigma^\nu_{\alpha\dot\alpha}(\bar\sigma^\mu)^{\dot\alpha\beta}\right] . \tag{5.27}$$

Using the Jacobi identity, one can also obtain the commutation rule with the momentum

$$[Q_\alpha, P^\mu] = 0 . \tag{5.28}$$

Analogous relations hold for $\bar Q$.

We call the algebra that we obtained a $\mathcal{N} = 1$ SUSY algebra. In this case, arbitrary phase rotations of the supercharges leave the algebra invariant

$$Q_\alpha \to e^{i\lambda} Q_\alpha , \qquad \bar Q_{\dot\alpha} \to e^{-i\lambda}\bar Q_{\dot\alpha} . \tag{5.29}$$

This is called U(1) R-symmetry and it has an associated bosonic generator R such that

$$[Q_\alpha, R] = Q_\alpha , \qquad [\bar Q_{\dot\alpha}, R] = -\bar Q_{\dot\alpha} . \tag{5.30}$$

The R-symmetry plays an important role in spontaneously broken SUSY, or in the determination of non-perturbative properties of SUSY-invariant theories. Collecting all the (anti)commutation relations determined above, we obtain the $\mathcal{N} = 1$ algebra

$$\boxed{\begin{aligned} &\{Q_\alpha, Q_\beta\} = \{\bar Q_{\dot\alpha}, \bar Q_{\dot\beta}\} = 0 , && \{Q_\alpha, \bar Q_{\dot\alpha}\} = 2(\sigma^\mu)_{\alpha\dot\alpha} P_\mu , \\ &[Q_\alpha, J^{\mu\nu}] = (\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta , && [Q_\alpha, P^\mu] = 0 . \end{aligned}} \tag{5.31}$$

When the theory contains more than one supercharge, we label them as Q^A with $A \in \{1, \dots, \mathcal{N}\}$ and we talk about *extended supersymmetry*. In this case R-symmetry gets promoted to U($\mathcal{N}$) global symmetry; in particular, when $\mathcal{N} > 1$ the algebra can be *centrally extended*. This means that the anticommutator $\{Q^A, Q^B\}$ and its hermitian conjugate do not vanish, but close into a central charge, *i.e.* a generator which commutes with all the other elements in the algebra. The entire SUSY algebra gets generalized to

$$\boxed{\begin{aligned} &\{Q^A_\alpha, Q^B_\beta\} = \color{red}{\epsilon_{\alpha\beta} Z^{AB}} , && \{\bar Q_{\dot\alpha A}, \bar Q_{\dot\beta B}\} = \color{red}{\epsilon_{\dot\alpha\dot\beta} (Z^{AB})^*} , \\ &\{Q^A_\alpha, \bar Q_{\dot\alpha B}\} = 2(\sigma^\mu)_{\alpha\dot\alpha} P_\mu \delta^A_B , \\ &[Q^A_\alpha, J^{\mu\nu}] = (\sigma^{\mu\nu})_\alpha{}^\beta Q^A_\beta , && [Q^A_\alpha, P^\mu] = 0 . \end{aligned}} \tag{5.32}$$

In the previous expression, $Z^{AB} = -Z^{BA}$ is the central charge and we put in red the new terms with respect to the $\mathcal{N} = 1$ case. The internal global symmetry, which is promoted to a U($\mathcal{N}$) symmetry with generators T_a, reads

$$[Q^A_\alpha, R] = \color{red}{(T_a)^A{}_B} Q^B_\alpha , \qquad [\bar Q_{\dot\alpha A}, R] = -\bar Q_{\dot\alpha B} \color{red}{(T_a)^B{}_A} . \tag{5.33}$$

5.4 Representations of the supersymmetry algebra

Given the set of brackets (5.32), it is natural to investigate the representations of the SUSY algebra. To this aim, it is convenient to recall the working procedure in the simpler case of the Poincaré group, which has two Casimir (*i.e.* operators which commute with all the generators)

$$P^2 = P^\mu P_\mu \,, \qquad W^2 = W_\mu W^\mu \,, \tag{5.34}$$

with the Pauli-Lubanski vector given by

$$W^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu J_{\rho\sigma} \,. \tag{5.35}$$

The Casimir operators are a useful mean to classify the irreducible representations of a group; in the case of Poincaré, they correspond to particles. In order to proceed, a convenient tool is Wigner's little group: one goes to the rest frame, and studies which are the transformations which leave the momentum invariant. $\hat{P}_\mu|p\rangle = p_\mu|p\rangle$

In the case of a massive particle, the rest frame is characterized by $P_\mu = (m,0,0,0)$ and the Casimir operators become $P^2 = m^2$ and $W^2 = -m^2 j(j+1)$ with j being the spin of the particle. This second equality easily follows from the fact that $W_\mu P^\mu = 0$, which implies that in the rest frame $W_0 = 0$. This shows that the massive particles are characterized by their mass and spin. & $\vec{W} = m\vec{J}$ & $\vec{J}^2 = j(j+1)$

In the case of massless particles, the rest frame becomes instead $P_\mu = (E,0,0,E)$ and the Casimir operators become $P^2 = W^2 = 0$. In particular $W^\mu = J_{12} P^\mu$, which means that the two operators composing the Casimirs are proportional to each other with a constant of proportionality given by $J_{12} = \pm j$, the helicity of the particle. This implies that the spin is fixed and different states are characterized by the energy and the sign of the helicity. boosts?

The same method can be applied to the supersymmetric case, which is an extension of the Poincaré group. Since Poincaré is a subalgebra of the SUSY one, it follows that any irreducible representation of the Poincaré algebra will be in general reducible. This implies that a superparticle will correspond to a collection of particles, the latter being related to each others by the action of the supercharges. The total object collecting all of these particles will be called a *supermultiplet*.

an irreducible SUSY representation is generally reducible as a Poincare´ representation?

Before diving into the explicit representation of super-Poincaré algebra, we stress some generic properties that all of them enjoy:

- P^2 will still be a Casimir of the algebra, but the square of the Pauli-Lubanski vector will not: this is clear from the fact that the supercharges carry spin 1/2, and then modify the spin of the particles they act on. Therefore the particles in the same supermultiplet will have the same mass but different spin. In the case of the SUSY algebra, we have a new Casimir element $\tilde{W}^2 = \tilde{C}_{\mu\nu}\tilde{C}^{\mu\nu}$ arising from

$$\tilde{C}_{\mu\nu} = \tilde{W}_\mu P_\nu - \tilde{W}_\nu P_\mu \,, \qquad \tilde{W}_\mu = W_\mu - \frac{1}{4}\bar{Q}_{\dot{\alpha}A}(\sigma_\mu)^{\dot{\alpha}\alpha} Q^A_\alpha \,, \tag{5.36}$$

  where W_μ is the Pauli-Lubanski vector defined in Eq. (5.35).

- (Anti)-commutation relations in Eq. (5.32) imply that the Hamiltonian is positive definite. Given an arbitrary state, we get indeed

$$\langle\phi|\{Q^A_\alpha, \bar{Q}_{\dot{\alpha}A}\}|\phi\rangle = 2(\sigma^\mu)_{\alpha\dot{\alpha}}\langle\phi|P_\mu|\phi\rangle\delta^A_A = \langle\phi|\left(Q^A_\alpha(Q^A_\alpha)^\dagger + (Q^A_\alpha)^\dagger Q^A_\alpha\right)|\phi\rangle = \\ = ||(Q^A_\alpha)^\dagger|\phi\rangle||^2 + ||Q^A_\alpha|\phi\rangle||^2 \geq 0\,. \tag{5.37}$$

  If we sum the previous expression over $\alpha, \dot{\alpha} \in \{1, 2\}$ and we use the property $\mathrm{Tr}\,(\sigma^\mu) = 2\delta^{\mu 0}$, we get

$$4\langle\phi|P_0|\phi\rangle \geq 0\,. \tag{5.38}$$

- A supermultiplet contains the same number of bosonic and fermionic degrees of freedom.

Massless supermultiplets

We assume that all the central charges vanish[3], *i.e.* $Z^{AB} = 0$. Looking at the algebra (5.32), this implies that the supercharges Q will commute among themselves. We explain in detail the procedure to find the irreducible representations for the massless case:

1. We go to the rest frame $P_\mu = (E, 0, 0, E)$, where

$$\sigma^\mu P_\mu = \begin{pmatrix} 0 & 0 \\ 0 & 2E \end{pmatrix}\,. \tag{5.39}$$

   Plugging this expression into the anticommutator of the supercharges, we get

$$\{Q^A_\alpha, \bar{Q}^B_{\dot{\alpha}}\} = \begin{pmatrix} 0 & 0 \\ 0 & 4E \end{pmatrix}_{\alpha\dot{\alpha}} \delta^{AB} \Rightarrow \{Q^A_1, \bar{Q}^B_{\dot{1}}\} = 0\,. \tag{5.40}$$

   Due to the positiveness of the Hilbert space, this implies that both Q^A_1 and $\bar{Q}^B_{\dot{1}}$ are trivially realized, *i.e.* they vanish since we obtain

$$0 = \langle\phi|\{Q^A_1, \bar{Q}^B_{\dot{1}}\}|\phi\rangle = ||Q^A_1|\phi\rangle||^2 + ||\bar{Q}^A_{\dot{1}}|\phi\rangle\,. \tag{5.41}$$

   This means that we are only left with half of the generators, *i.e.* the supercharges Q^A_2 and $\bar{Q}^B_{\dot{2}}$.

2. We introduce a different normalization and notation for the non-trivial generators, in such a way to resemble the case of the harmonic oscillator:

$$a_A \equiv \frac{1}{\sqrt{4E}} Q^A_2\,, \qquad a^\dagger_A \equiv \frac{1}{\sqrt{4E}} \bar{Q}^A_{\dot{2}}\,. \tag{5.42}$$

   Indeed, one can show that they satisfy the commutation rules

$$\{a_A, a^\dagger_B\} = \delta_{AB}\,, \qquad \{a_A, a_B\} = \{a^\dagger_A, a^\dagger_B\} = 0\,. \tag{5.43}$$

[3] We will see that this is the only relevant case for massless representations.

In the following manipulations, we also need to observe that the operators Q_2^A and $\bar{Q}_{\dot{2}}^B$ lower (raise) the helicity by 1/2, since

$$[J_{12}, Q_2^A] = -\frac{1}{2} Q_2^A \,, \qquad [J_{12}, \bar{Q}_{\dot{2}}^A] = \frac{1}{2} \bar{Q}_{\dot{2}}^A \,. \tag{5.44}$$

3. In analogy to the harmonic oscillator, we build a representation by starting with a Clifford vacuum (*i.e.* a state annihilated by all the a_A). This state has $m = 0$ and it carries a certain helicity λ_0, so that we call the vacuum using this label:

$$a_A |\lambda_0\rangle = 0 \,. \tag{5.45}$$

   In principle this state can be either bosonic or fermionic, and it is not the vacuum state of the theory (which instead refers to the state of minimal energy).

4. The full representation, which corresponds to a supermultiplet, is obtained by acting on $|\lambda_0\rangle$ with the raising operators in the following way:

$$|\lambda_0\rangle \,, \qquad a_A^\dagger |\lambda_0\rangle \equiv \left|\lambda_0 + \tfrac{1}{2}\right\rangle_A \,, \tag{5.46}$$

$$a_A^\dagger a_B^\dagger |\lambda_0\rangle \equiv |\lambda_0 + 1\rangle_{AB} \,, \quad \ldots \quad a_1^\dagger a_2^\dagger \ldots a_N^\dagger |\lambda_0\rangle \equiv \left|\lambda_0 + \tfrac{\mathcal{N}}{2}\right\rangle . \tag{5.47}$$

   This shows that the state with highest helicity has $\lambda = \lambda_0 + \mathcal{N}/2$. Due to the antisymmetry under exchanging of $A \leftrightarrow B$, the total number of states in the irreducible representation is

$$\sum_{k=0}^{\mathcal{N}} \binom{\mathcal{N}}{k} = 2^{\mathcal{N}} \,, \tag{5.48}$$

   half of them bosonic and half of them fermionic.

5. Finally, there is a last important observation coming from CPT invariance. Since CPT flips the sign of helicity, a supermultiplet can be invariant under this discrete transformation only if the helicity is distributed symmetrically around 0. When this does not happen, we need to double the supermultiplet built in the previous steps by adding its CPT conjugate.

Now we concretely apply the above-mentioned technique to specific cases, focusing on four-dimensional Minkowski space.

$\mathcal{N} = 1$ SUSY. In this case there is only one supercharge to act on the states. This will always gives rise to the states

$$|\lambda_0\rangle \,, \quad \bar{Q}|\lambda_0\rangle \,. \tag{5.49}$$

Depending from the helicity of the vacuum state, various possibilities are allowed:

- Chiral (matter) multiplet:

$$\lambda_0 = 0\,, \quad \Rightarrow \quad \left(0, \frac{1}{2}\right) \underset{\text{CPT}}{\oplus} \left(-\frac{1}{2}, 0\right)\,. \tag{5.50}$$

  The degrees of freedom correspond to one scalar and one Weyl fermion (on shell: in fact, remember that we are constructing representations of the SUSY algebra on the states). In $\mathcal{N} = 1$ SUSY theory, this is the representation where the matter sits. This field content is exactly the same entering the Wess-Zumino model.

- Vector (gauge) multiplet:

$$\lambda_0 = \frac{1}{2}\,, \quad \Rightarrow \quad \left(\frac{1}{2}, 1\right) \underset{\text{CPT}}{\oplus} \left(-1, -\frac{1}{2}\right)\,. \tag{5.51}$$

  The degrees of freedom are those of one vector and one Weyl fermion. This is the representation where the gauge fields sit in a SUSY-invariant theory. Since internal symmetries (like R-symmetry) commute with the supercharges, this implies that the representation of the Weyl fermion must be the same as the gauge field, *i.e.* the adjoint. This implies that the SM matter (both quarks and leptons) cannot be accomodated in these multiplets, since they transform instead in the fundamental.

- Spin 3/2 multiplet:

$$\lambda_0 = 1\,, \quad \Rightarrow \quad \left(1, \frac{3}{2}\right) \underset{\text{CPT}}{\oplus} \left(-\frac{3}{2}, -1\right)\,. \tag{5.52}$$

  helicity?

  It contains a spin 1 particle and a spin 3/2 one, and such multiplet can only appear in a $\mathcal{N} = 1$ SUSY theory if also a graviton (a spin 2 particle) is present.

- Graviton multiplet:

$$\lambda_0 = \frac{3}{2}\,, \quad \Rightarrow \quad \left(\frac{3}{2}, 2\right) \underset{\text{CPT}}{\oplus} \left(-2, -\frac{3}{2}\right)\,. \tag{5.53}$$

  It contains the graviton (a spin 2 particle) and the SUSY partner which is called the gravitino (a spin 3/2 particle).

The last two supermultiplets can appear only in a theory of gravity, *e.g.* supergravity. The case with $\lambda_0 \geq 2$ gives rise to higher spin particles, which we cannot include inside interacting local field theories.

$\mathcal{N} = 2$ SUSY. In this case we have at disposal two supercharges to play with, and then starting from a vacuum with given helicity we can create a total of four states (plus the CPT conjugates). We list the most relevant ones:

- Hypermultiplet:

$$\lambda_0 = -\frac{1}{2}\,, \quad \Rightarrow \quad \left(-\frac{1}{2}, 0, 0, \frac{1}{2}\right) \underset{\text{CPT}}{\oplus} \left(-\frac{1}{2}, 0, 0, \frac{1}{2}\right) . \tag{5.54}$$

  The degrees of freedom correspond to two Weyl fermions and two complex scalars. This is the supermultiplet where $\mathcal{N} = 2$ matter sits, and corresponds in $\mathcal{N} = 1$ language to two chiral multiplets with opposite chirality (which is flipped by CPT conjugation).

- Vector multiplet:

$$\lambda_0 = 0\,, \quad \Rightarrow \quad \left(0, \frac{1}{2}, \frac{1}{2}, 1\right) \underset{\text{CPT}}{\oplus} \left(-1, -\frac{1}{2}, -\frac{1}{2}, 0\right) . \tag{5.55}$$

  The degrees of freedom correspond to one vector, two Weyl fermions and one complex scalar. In $\mathcal{N} = 1$ language, it splits into a vector and a chiral multiplet, both transforming in the same representation.

- Spin 3/2 multiplet:

$$\lambda_0 = -\frac{3}{2}\,, \quad \Rightarrow \quad \left(-\frac{3}{2}, -1, -1, -\frac{1}{2}\right) \underset{\text{CPT}}{\oplus} \left(\frac{1}{2}, 1, 1, \frac{3}{2}\right) . \tag{5.56}$$

  It contains a spin 3/2 particle, two vectors and one Weyl fermion.

- Graviton multiplet:

$$\lambda_0 = -2\,, \quad \Rightarrow \quad \left(-2, -\frac{3}{2}, -\frac{3}{2}, -1\right) \underset{\text{CPT}}{\oplus} \left(1, \frac{3}{2}, \frac{3}{2}, 2\right) . \tag{5.57}$$

  It contains a graviton, two gravitinos and a vector (usually called graviphoton in the supergravity literature).

$\mathcal{N} = 4$ SUSY. There exists only one supermultiplet in this case, which is the vector one and is given by

$$\lambda_0 = -1\,, \quad \Rightarrow \quad \left(-1, \mathbf{4} \times -\frac{1}{2}, \mathbf{6} \times 0, \mathbf{4} \times \frac{1}{2}, 1\right) . \tag{5.58}$$

In total, it contains the degrees of freedom of one gauge field, four Weyl fermions and three complex scalars (or six real ones). In $\mathcal{N} = 1$ language, this corresponds to one vector superfield and three chiral ones, all transforming in the adjoint representation. The R-symmetry group is given by SU(4), and under this invariance the vector is a singlet, the fermions transform in the fundamental **4** and the scalars in the double antisymmetric representation **6**, which is the fundamental of SO(6). This will be precisely the field content of $\mathcal{N} = 4$ super Yang-Mills, one of the main protagonists in the AdS/CFT duality that we will introduce later.

We observe that the $\mathcal{N} = 4$ supermultiplet is CPT self-conjugate. One may wonder why we did not consider $\mathcal{N} = 3$ representations: it turns out that after adding the CPT conjugate, the multiplet that we obtain is the same as the $\mathcal{N} = 4$ vector.

$\mathcal{N} > 4$ SUSY. As we anticipated above, a theory without gravity can accomodate a maximum of $\mathcal{N} = 4$ extended supersymmetry in four dimensions. This can be justified as follows. When $\mathcal{N} > 4$, we have at least five non-trivial creation operators; starting from a vacuum with λ_0, we increase the helicity by $1/2$ each time. Since the massless representations also contain the CPT conjugate, this implies that we will always generate at least one state with helicity $3/2$ or higher, which is not admitted in a local field theory without gravity.

It is also interesting to observe that the maximal amount of supersymmetry admitted in a theory with gravity would be instead $\mathcal{N} = 8$, since in that case the highest value of helicity that we can allow is 2 (corresponding to the graviton).

The previous statements are specifically referred to the four-dimensional case. However, it is possible to perform a counting of the amount of supersymmetry which is independent from the dimension considered, if we enumerate the number of real components instead of the copies of different supercharges. In four dimensions, the supercharges are Majorana spinors, which means that they have four real components. Then an amount of $\mathcal{N} = 4$ SUSY corresponds to 16 real components: this number is independent from the dimension of the spacetime where the theory lives.

If we want to write the restrictions on the maximal amount of supersymmetry in terms of the copies $\mathcal{N}$ of supercharges in dimensions different than four, all that we need to do is to count the number of real components of a Clifford spinor in the given dimension, and take combine various copies of the supercharges until we obtain 16 real components. For example, in 10 dimensions the spinors turn out to be both Majorana and Weyl, and then they have 16 real components. Thus $\mathcal{N} = 1$ SUSY in 10 dimensions corresponds exactly to the maximal amount of SUSY. Since in 11 dimensions the spinors can only be Majorana and contain 32 real components, we conclude that no SUSY-invariant and local field theories exist!

In the case with gravity, $\mathcal{N} = 8$ in four dimensions corresponds to 32 real components, which corresponds to $\mathcal{N} = 2$ in 10 dimensions and $\mathcal{N} = 1$ in 11 dimensions. So, we find that it is possible to define a consistent theory of supergravity at most in 11 dimensions, but not in higher ones.

Massive supermultiplets

The logical steps that we need to follow for massive representations are similar to the massless case, but there is a main difference that becomes clear once we go to the rest frame $P_\mu = (m, 0, 0, 0)$ to evaluate the anticommutator of supercharges: now we still have the full set of $2\mathcal{N}$ creation and $2\mathcal{N}$ annihilation operators. Indeed we find that Eq. (5.32) becomes

$$\{Q^A_\alpha, \bar{Q}^B_{\dot{\alpha}}\} = 2m\delta_{\alpha\dot{\alpha}}\delta^{AB}\,, \tag{5.59}$$

so that no supersymmetric generators are trivially realized. Loosely speaking, this means that massive representations are *longer* than the massless ones: since we can act with more generators, starting from the Clifford vacuum we can create more states belonging to the same family. In addition, we remark that in this case we need to talk about the spin of the particles instead of their helicity; hence, the Clifford vacuum will have mass m, spin j and a corresponding degeneracy $2j+1$.

$\mathcal{N}=1$ SUSY. We define the following creation and annihilation operators

$$a_{1,2} \equiv \frac{1}{\sqrt{2m}} Q_{1,2}\,, \qquad a^\dagger_{1,2} \equiv \frac{1}{\sqrt{2m}} \bar{Q}_{\dot{1},\dot{2}}\,, \tag{5.60}$$

normalized in such a way to satisfy the usual commutation relations for the harmonic oscillators. By looking at the quantum numbers carried by the supercharges, it is clear that $a_1^\dagger$ lowers the spin by half unit, while $a_2^\dagger$ raises it by the same amount. The Clifford vacuum $|m,j\rangle$ is annihilated by

$$a_1|m,j\rangle = a_2|m,j\rangle = 0\,, \tag{5.61}$$

and we act with the two creation operators to construct the corresponding massive representations.

Now we apply the previous technique to find the explicit multiplets:

- Matter multiplet:

$$j=0 \quad \Rightarrow \quad \left(-\frac{1}{2}, 0, 0, \frac{1}{2}\right)\,. \tag{5.62}$$

  The number of degrees of freedom is the same as in the massless case; the multiplet is composed by a massive complex scalar and a massive Majorana fermion.

- Vector multiplet:

$$j=\frac{1}{2} \quad \Rightarrow \quad \left(-1, \mathbf{2}\times -\frac{1}{2}, \mathbf{2}\times 0, \mathbf{2}\times \frac{1}{2}, 1\right)\,. \tag{5.63}$$

  The degrees of freedom turn out to correspond to a massive gauge field, a massive Dirac fermion and a massive real scalar. Notice that they are the same as those of a massless vector multiplet plus a massless chiral one. The reason is that in a renormalizable supersymmetric theory, one can generate massive vector multiplets with a generalization of the Higgs mechanism.

Extended SUSY. We now consider the case of extended SUSY, *i.e.* $\mathcal{N}>1$, which allows for non-trivial central charges. This will also allow us to discuss why it was not restrictive to consider $Z^{AB}=0$ for massless representations.

Due to the antisimmetry of the central charges, by performing a U($\mathcal{N}$) rotation we can always put them into the standard block-diagonal form

$$Z^{AB} = \begin{pmatrix} 0 & Z_1 & \dots & 0 & 0 \\ -Z_1 & 0 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & Z_{\mathcal{N}/2} \\ 0 & 0 & \dots & -Z_{\mathcal{N}/2} & 0 \end{pmatrix}, \tag{5.64}$$

where we are assuming for simplicity that $\mathcal{N}$ is even. In this context, we define the creation and annihilation operators as

$$a^1_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^1_\alpha + \epsilon_{\alpha\beta} (Q^2_\beta)^\dagger \right), \qquad b^1_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^1_\alpha - \epsilon_{\alpha\beta} (Q^2_\beta)^\dagger \right), \tag{5.65}$$

$$a^2_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^3_\alpha + \epsilon_{\alpha\beta} (Q^4_\beta)^\dagger \right), \qquad b^2_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^3_\alpha - \epsilon_{\alpha\beta} (Q^4_\beta)^\dagger \right), \tag{5.66}$$

$$\dots$$

$$a^{\mathcal{N}/2}_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^{\mathcal{N}-1}_\alpha + \epsilon_{\alpha\beta} (Q^{\mathcal{N}}_\beta)^\dagger \right), \qquad b^{\mathcal{N}/2}_\alpha = \tfrac{1}{\sqrt{2}} \left(Q^{\mathcal{N}-1}_\alpha - \epsilon_{\alpha\beta} (Q^{\mathcal{N}}_\beta)^\dagger \right). \tag{5.67}$$

They satisfy the algebra

$$\{a^r_\alpha, (a^s_\beta)^\dagger\} = (2m + Z_r)\, \delta_{rs} \delta_{\alpha\beta}\,, \qquad \{b^r_\alpha, (b^s_\beta)^\dagger\} = (2m - Z_r)\, \delta_{rs} \delta_{\alpha\beta}\,, \tag{5.68}$$

$$\{a^r_\alpha, (b^s_\beta)^\dagger\} = \{a^r_\alpha, a^s_\beta)\} = \{b^r_\alpha, b^s_\beta)\} = 0\,, \tag{5.69}$$

with $r, s \in \{1, 2, \dots, \mathcal{N}/2\}$ and $\alpha \in \{1, 2\}$.

Notice that the positiveness of the Hilbert space implies that the oscillator algebra should satisfy

$$2m \geq |Z_r|\,. \tag{5.70}$$

This is called a BPS (Bogomolny, Prasad, Sommerfield) bound. The first important consequence is that $m = 0$ implies that the central charges must be trivially realized, *i.e.* $Z^{AB} = 0$ which is the choice that we performed to obtain the massless representations.

If none of the BPS bounds (5.70) are saturated, *i.e.* we have

$$\forall r : \qquad 2m > Z_r\,, \tag{5.71}$$

then we create a total of $2^{2\mathcal{N}}$ states starting from the Clifford vacuum.

If instead some of the bounds (5.70) are saturated, from the oscillator algebra we notice that the corresponding $b-$type oscillators become trivial, and the dimension of the representation accordingly becomes smaller. If $k < \mathcal{N}/2$ BPS bounds are saturated, we only create $2^{2(\mathcal{N}-k)}$ states, and we speak about *short multiplets*. The extreme case is $k = \mathcal{N}/2$, where half of the creation operators become trivial and the multiplet is called *ultrashort*, with dimension $2^{\mathcal{N}}$.

One can make use of these observations to find the explicit massive multiplets. Since we will not need to explicitly use them, here we just comment that they combine into matter and vector multiplets, similarly to the massless case.

5.5 SUSY-invariant actions and non-renormalization theorems

So far, we have discussed supersymmetry representations on states, but we would like to also discuss representations on fields in order to build QFT SUSY-invariant actions. Here we will not dive into the details of this procedure, but we highlight the most important results.

- A convenient way to treat covariantly SUSY-invariant theories is to introduce *superspace*, an extension of ordinary spacetime with Grassmann coordinates $(\theta^\alpha, \bar{\theta}_{\dot{\alpha}})$ which are associated to the action of supercharges. Similarly to Minkowski space, which can be seen as the coset

$$\text{Minkowski} = \frac{\text{Poincaré}}{\text{Lorentz}}\,, \tag{5.72}$$

  superspace is given by

$$\text{superspace} = \frac{\text{super-Poincaré}}{\text{Lorentz}}\,. \tag{5.73}$$

  SUSY transformations correspond to translations in superspace.

- Fields living in superspace are called *superfields*. The simplest example is the so-called chiral superfield, which contains the field content of the Wess-Zumino model:

$$\Phi(x,\theta) = \phi(x) + \sqrt{2}\theta\psi(x) - \theta^2 F(x)\,. \tag{5.74}$$

  This object is precisely the representation on fields of the $\mathcal{N} = 1$ (massive or massless) chiral multiplet, that we found when studying the representations on states. One can find similar expressions for the vector superfield, corresponding to the vector multiplet derived above.

- Superfields are a convenient way to build SUSY-invariant actions, since they can be easily combined in order to manifestly satisfy this symmetry.

- Feynman diagrams can be built using directly superfields (supergraph techniques).

- Superfields are a powerful tool to prove non-renormalization theorems (either perturbatively and non-perturbatively).

5.6 $\mathcal{N} = 4$ super-Yang Mills theory

Symmetries usually introduce more constraints in a theory, since they give rise to more conserved quantities and to invariances that need to be satisfied by physical quantities. We have mentioned that supersymmetry gives rise to non-renormalization theorems: the correlation functions are better behaved and the interacting terms in the action are not renormalized at any order both in perturbation theory and at the non-perturbative level. If we increase the amount of supersymmetry of the model, further constraints arise:

one can show that $\mathcal{N} = 2$ super Yang-Mills[4] is one-loop exact, *i.e.* the wavefunction renormalization of the kinetic term encodes all the renormalization properties of the theory, and it runs only at one-loop order.

The situation becomes even better when we consider $\mathcal{N} = 4$ SUSY: the system contains so many constraints that the theory is perturbatively finite, *i.e.* there is no renormalization of both the wavefunction and the coupling constants, and no UV divergences arise in the computation of correlation functions! In addition, $\mathcal{N} = 4$ super Yang-Mills enjoys the property to be superconformal invariant at the full quantum level: this means that both supersymmetry and conformal invariance are preserved at all loops and in particular all the β functions of the theory vanish!

The task of this subsection is to introduce the action and the properties of $\mathcal{N} = 4$ super Yang-Mills (SYM) in four dimensions. This model plays an important role in AdS/CFT correspondence because the big amount of symmetries that it enjoys allows to formulate explicit statements on the duality that can be matched from the gravity side.

Action

The field content of the theory corresponds to the $\mathcal{N} = 4$ massless vector multiplet discussed in Section 5.4: it contains six real scalar fields ϕ_i transforming in the anti-symmetric **6** representation of the R-symmetry group $\mathrm{SO}(6) \simeq \mathrm{SU}(4)$, which can be combined into three complex scalar fields as $\Phi_a = \frac{1}{\sqrt{2}}(\phi_{2a-1} + i\phi_{2a})$ with $a \in \{1, 2, 3\}$; four Weyl fermions ψ^A with $A \in \{1, 2, 3, 4\}$ transforming in the fundamental representation **4** of SU(4); and one gauge field A_μ which is a singlet under the R-symmetry group. The action is unique and there are two main techniques to derive it:

- Since $\mathcal{N} = 4$ SYM theory is also $\mathcal{N} = 1$ invariant by default, we can use the corresponding formalism to build the action. One starts from the chiral and vector superfields in $\mathcal{N} = 1$ language. Standard superspace techniques (which we do not describe here) allow to easily build an action invariant under $\mathcal{N} = 1$ SUSY; we obtain the full $\mathcal{N} = 4$ invariance by combining them in such a way to preserve certain constraints. We simply list the result in component field formalism, *i.e.* after expanding the superfields with respect to the Grassmannian coordinates in order to get regular fields. The Lagrangian reads

$$\mathcal{L} = \mathrm{Tr}\left\{ -\frac{1}{2g^2} F_{\mu\nu}^2 + \frac{\theta}{16\pi^2} F_{\mu\nu}\tilde{F}^{\mu\nu} - (D_\mu \phi_i)^2 - i\psi_A^\dagger \bar{\sigma}^\mu D_\mu \psi^A \right. \\ \left. + g\, C_{AB}^i \psi^A [\phi_i, \psi^B] + g\, \bar{C}^{iAB} \psi_A^\dagger [\phi_i, \psi_B^\dagger] + \frac{g^2}{2} \left([\phi_i, \phi_j]\right)^2 \right\}, \quad (5.75)$$

  where the field strengths are defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]\,, \quad (5.76)$$

[4] As the name suggests, this is the supersymmetric version of a non-abelian gauge theory invariant under the colour group $\mathrm{SU}(N)$.

the covariant derivatives D_μ as

$$D_\mu \phi_i = \partial_\mu \phi_i + i[A_\mu, \phi_i]\,, \tag{5.77}$$
$$D_\mu \psi^A = \partial_\mu \psi^A + i[A_\mu, \psi^A]\,, \tag{5.78}$$

and the dual field strength as

$$\tilde{F}^{\mu\nu} \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}\,. \tag{5.79}$$

The C^i_{AB} are Clebsch-Gordan coefficients coupling two **4** representations and one **6** representation of the R-symmetry group SU(4). The fields in the action transform in the adjoint representation of the gauge group SU(N).

- The other possibility is to start from $\mathcal{N} = 1$ SYM in $d = 10$

$$S = \int d^{10}x \,\mathrm{Tr}\left(-\frac{1}{2}F_{MN}F^{MN} + \frac{i}{2}\bar{\Psi}\Gamma^M D_M \Psi\right)\,, \tag{5.80}$$

and then perform a dimensional reduction along a torus $\mathbb{T}^6$. Here Ψ are Majorana-Weyl spinors in 10 dimensions with 16 real independent components transforming in the adjoint representation of SU(N), and Γ^M are 10-dimensional Dirac matrices. In this case, we rescaled the gauge fields in such a way that the gauge coupling does not enter the free action explicitly, but via the covariant derivative and the definition of field strength

$$F_{MN} = \partial_M A_N - \partial_N A_M + i g_{10}[A_M, A_N]\,, \quad D_M \Psi = \partial_M \Psi + i g_{10}[A_M, \Psi]\,. \tag{5.81}$$

We also record the mass dimension of the fields and coupling constant:

$$[A] = 4\,, \qquad [\Psi] = \frac{9}{2}\,, \qquad [g_{10}] = -3\,. \tag{5.82}$$

The action is invariant under the SUSY transformations

$$\delta_\varepsilon A_M = i\bar{\varepsilon}\Gamma_M \Psi\,, \qquad \delta_\varepsilon \Psi = \Gamma_{MN}F^{MN}\,, \qquad \Gamma_{MN} = \Gamma_{[M}\Gamma_{N]}\,. \tag{5.83}$$

In order to go from a 10-dimensional theory to a 4-dimensional one, we compactify along a torus $\mathbb{T}^6$ in the limit of zero size. For this reason, we split the spacetime indices as

$$M \equiv (\mu, i)\,, \qquad \mu \in \{0,1,2,3\}\,, \qquad i \in \{4,\ldots,9\}\,. \tag{5.84}$$

In this process, we require that all the fields only depend from the coordinates of the lower-dimensional space x^μ and are independent from the internal ones given by x^i. For example, a scalar field is decomposed as

$$\phi(x^M) = \frac{1}{\sqrt{V_6}}\sum_{n\in\mathbb{Z}} e^{i\vec{k}_n\cdot\vec{x}}\phi_n(x^\mu)\,, \tag{5.85}$$

where we defined R to be the radius of the circles in the torus, V_6 its volume and the momentum $k_n = n/R$ is quantized along the circles. Similarly, the 10 dimensional gauge fields split into

$$A_M \to \left(A_\mu(x^\nu), \phi^i(x^\nu)\right) , \tag{5.86}$$

where the first part is interpreted as a lower-dimensional gauge field and the second one as a collection of six real scalars transforming trivially under the Lorentz group.

In this way we work out the field strength components

$$F_{\mu i} = D_\mu \phi_i , \qquad F_{ij} = i g_{10}[\phi_i, \phi_j] , \tag{5.87}$$

and the corresponding term in the Lagrangian becomes

$$\mathrm{Tr}\left(-\frac{1}{2}F_{MN}F^{MN}\right) = \mathrm{Tr}\left(-\frac{1}{2}F_{\mu\nu}F^{\mu\nu} - D_\mu\phi_i D^\mu \phi^i + \frac{1}{2}g_{10}^2[\phi_i,\phi_j]^2\right) . \tag{5.88}$$

When performing the Kaluza-Klein (KK) reduction, we get a factor of R^6 coming from the six-dimensional integration along the internal space, and a factor of R^{-3} due to the normalization like in Eq. (5.85) for each bosonic field. The result is that the last term becomes

$$\int d^{10}x \sqrt{-\det(g_{\mu\nu})}\, \mathrm{Tr}\left(-\frac{1}{2}F_{MN}F^{MN}\right) \supset \int d^4x\, \mathrm{Tr}\left(\frac{g_{10}^2}{2R^6}\,[\phi_i,\phi_j]^2\right) . \tag{5.89}$$

Then we define a four-dimensional coupling constant as

$$g \equiv R^{-3} g_{10} , \tag{5.90}$$

and we notice that $[g] = 0$. This is exactly the SYM coupling constant in four dimensions, and the terms obtained in Eq. (5.88) are part of the total $\mathcal{N} = 4$ SYM action in Eq. (5.75).

In order to perform the reduction of the fermionic kinetic term, we need to decompose the 10-dimensional Dirac matrices into four-dimensional ones γ_μ and six-dimensional ones $\hat{\gamma}^i$ as

$$\Gamma_\mu = \gamma_\mu \otimes \mathbf{1}_6 , \qquad \Gamma_i = \mathbf{1}_4 \otimes \hat{\gamma}_i . \tag{5.91}$$

The final result, combining all the terms, is given by the Lagrangian (5.75).

gamma matrices are defined to encode the geometry of spacetime through the Clifford algebra Since they are different perpendicular spacetime directions,η _μi=0.

γ_5 ?

$\{\Gamma_M, \Gamma_N\} = 2\eta_{MN} = 0$

$(\gamma_\mu \otimes 1)(1 \otimes \hat{\gamma}_i) = (\gamma_i \otimes 1)(\gamma_\mu \otimes 1)$

commute

$\{\gamma_\mu, \gamma_5\} = 0$

Properties

Let's analyze the relevant properties of the $\mathcal{N} = 4$ SYM action (5.75). The symmetries are the following:

- Poincaré invariance.
- $\mathcal{N} = 4$ supersymmetry.

- SU(4) global R-symmetry.
- Conformal invariance.

In particular, at the classical level we can determine the engineering mass dimensions of the fields, which is

$$[A_\mu] = 1\,, \qquad [\psi] = \frac{3}{2}\,, \qquad [\phi^i] = 1\,, \qquad [g] = 0\,. \tag{5.92}$$

As we anticipated, conformal symmetry survives quantization and is responsible for the vanishing of all the β functions. The symmetries of the theory can be unified into the superconformal group PSU$(2,2|4)$, as we are going to explain.

5.7 Superconformal symmetry

We discuss the consequences of supersymmetry in a conformal-invariant field theory, referring in particular to the superalgebra of interest for $\mathcal{N} = 4$ SYM.

We start with a brief review of the non-supersymmetric case. The CFT operators are usually classified by quantum numbers under the non-compact subgroup SO$(1,2) \times$ SO$(1,3) \subset$ SO$(2,4) \simeq$ SU$(2,2)$, which means that we use the scaling dimensions of the fields and their spin. As can be seen from the conformal algebra, P_μ and K_μ act as raising and lowering operators for the dilatation generator D. The lowest state of a representation is annihilated by the K_μ, while the full tower composing the associated irreducible and infinite-dimensional representation is obtained by acting with the raising operator P_μ. In terms of operators evaluated at the origin of space, the stabilizer[5] is composed by the subalgebra $K_\mu, D, J_{\mu\nu}$. A primary conformal field with spins (j_L, j_R) and scaling dimension Δ satisfies

$$[D, \mathcal{O}_{(j_L,j_R)}(0)] = i\Delta\mathcal{O}_{(j_L,j_R)}(0)\,, \qquad [K_\mu, \mathcal{O}_{(j_L,j_R)}] = 0\,. \tag{5.93}$$

The descendants are built by acting with derivatives $\partial \ldots \partial\mathcal{O}_{(j_L,j_R)}(0)$ and fill an infinite-dimensional and irreducible representation.

There are other possibilities to classify states and operators in the conformal algebra, in particular if we want to avoid the creation of non-normalizable states when acting on the vacuum with the operators derived above[6]. One way is to use a compact subgroup SO$(2) \times$ SO$(4) \subset$ SO$(2,4)$, where states are classified according to the eigenvalues of an Hamiltonian

$$H = \frac{P_0 + K_0}{2}\,, \tag{5.94}$$

and two other quantum numbers from the decomposition SU$(2) \times$ SU$(2) \subset$ SO(4). This choice is useful when, upon Euclidean continuation, we quantize the theory on $S^3 \times \mathbb{R}$ (radial quantization).

[5] The stabilizer of an algebra acting on a state is the set of generators which leave the state invariant.

[6] It is necessary that the states classified according to the subgroup SO$(1,2) \times$ SO$(1,3)$ are non-normalizable, otherwise they would form a finite dimensional unitary representation of a non-compact group (the stabilizer of the group at the origin), which is forbidden by group theory.

Superconformal algebra

The superconformal group $\mathrm{SU}(2,2|\mathcal{N})$ corresponds to a theory with $\mathcal{N}$ copies of supersymmetry and it is obtained by adding to the conformal algebra $\mathrm{SO}(4,2)$ the following objects:

- $\mathcal{N}$ copies of supercharges Q.
- $\mathcal{N}$ copies of the superpartners of the supercharges, called S.
- The generators R of a $\mathrm{U}(\mathcal{N})$ global symmetry.

We report the relevant brackets of the superlalgebra including the fermionic operators:

$$[D, Q^A_\alpha] = \tfrac{i}{2} Q^A_\alpha \,, \qquad [D, S^\alpha_A] = -\tfrac{i}{2} S^\alpha_A \,, \tag{5.95}$$

$$[K_\mu, Q^A_\alpha] = -(\sigma_\mu)_{\alpha\dot\alpha} \bar S^{\dot\alpha A} \,, \qquad [P_\mu, S^\alpha_A] = (\bar\sigma_\mu)^{\dot\alpha\alpha} \bar Q_{\dot\alpha A} \,, \tag{5.96}$$

$$\{Q^A_\alpha, \bar Q_{\dot\alpha B}\} = 2\delta^A_B (\sigma^\mu)_{\alpha\dot\alpha} P_\mu \,, \qquad \{S^A_\alpha, \bar S_{\dot\alpha B}\} = 2\delta^A_B (\sigma^\mu)_{\alpha\dot\alpha} K_\mu \,, \tag{5.97}$$

$$\{Q^A_\alpha, S_{\beta B}\} = \epsilon_{\alpha\beta} \left(\delta^A_B \, D + R^A_B \right) + \tfrac{1}{2} \delta^A_B \, J_{\mu\nu} (\sigma^{\mu\nu})_{\alpha\beta} \,. \tag{5.98}$$

The first line gives the scaling dimension of the charges, and shows that Q acts as a raising operator and S as a lowering operator for the dilatation generator. The second line shows that the introduction of the superpartner S is necessary to close the algebra; it can be thought as a fermionic extension of K_μ, which generates the special conformal transformations. The third line contains standard commutation relations for the supersymmetry algebra (which we notice to be satisfied in the same way also by S), and finally the last line contains the bracket between Q and S.

There are other commutation relations which specify the R-symmetry quantum number associated to the fermionic generators

$$[R^A_B, Q^C_\alpha] = \delta^C_B Q^A_\alpha - \frac{1}{4} \delta^A_B Q^C_\alpha \,, \qquad [R^A_B, S^\alpha_C] = -\delta^A_C S^\alpha_B + \frac{1}{4} \delta^A_B S^\alpha_C \,. \tag{5.99}$$

Here R^A_B are generators of the $\mathrm{U}(\mathcal{N})$ internal group; in the specific case of $\mathcal{N} = 4$ SYM, the trace R^A_A decouples from the algebra and gives rise to an outer automorphism. For this reason, the R-symmetry group of the theory is given by SU(4) instead of U(4).

Representations of the superconformal algebra

In order to build representations of the superconformal group, we consider non-vanishing local operators $\mathcal{O}(x)$ constructed from elementary fields which in addition satisfy the requirement of being gauge invariant, if the theory admits such symmetry. As explained above, they can be characterized by the conformal dimension and the spins. Now that we have enlarged the conformal group adding supersymmetry, we can further require that there exists an operator with lowest scaling dimension Δ and spins (j_L, j_R) such that the following conditions hold:

$$[S^A_\alpha, \mathcal{O}_{(j_L, j_R)}(0)\} = 0 \,, \qquad [\bar S_{\dot\alpha A}, \mathcal{O}_{(j_L, j_R)}(0)\} = 0 \,, \tag{5.100}$$

for all $A \in \{1, \dots, \mathcal{N}\}, \alpha \in \{1, 2\}$ and where we denoted with $[\cdot, \cdot\}$ the brackets to remark that they are commutators or anticommutators depending from the statistics of the fields. This requirement is a consequence of the fact that the scaling dimension is bounded from below due to the existence of unitarity bounds, and we notice from the commutation rules (5.95) that the S generators act as lowering operators on the dimension.

Operators satisfying Eqs. (5.93) and (5.100) are called *superprimaries*; such class is also primary, because of the commutation rules $\{S, \bar{S}\} \sim K$. Starting from a superconformal primary operator, we can build superdescendants by acting an infinite number of times with the momentum operator P_μ and the supercharges Q^A_α. Since the latter are fermionic operators, we can act with them only a finite number of times. The supermultiplet obtained by acting with P, Q is called a *Verma module*, and corresponds to an irreducible representation of the superconformal algebra. Different Verma modules are related to each others by supersymmetry transformations.

I would not think of different Verma modules as being related by SUSY. Usually one Verma module is already the full representation generated from one primary by P and Q Doesnt different primaries generally give different representations?

o rmaybe should be: Q connects the different members of the same supermultiplet?

One can consider another subset of superconformal primaries, which correspond to *chiral primary operators*. In addition to the requirements in Eqs. (5.93) and (5.100), they are also annihilated by half of the supercharges

$$[Q^A_\alpha, \mathcal{O}(0)\} = 0\,, \tag{5.101}$$

for at least one value of $A \in \{1, \dots, \mathcal{N}\}$ and $\alpha \in \{1, 2\}$. By the discussion on representations of extended SUSY in Section 5.4, they are BPS operators: this means that the multiplet obtained starting from them is smaller (but it is still infinite-dimensional).

They play a particularly important role because their dimension is protected at quantum level. This can be seen schematically from the commutation relations defining the superalgebra; using the Jacobi identity, we get

$$0 = [\{S, Q\}, \mathcal{O}(0)] = [D + \hat{R} + \hat{J}, \mathcal{O}(0)] \sim (\Delta + R + J)\,\mathcal{O}(0)\,, \tag{5.102}$$

where we defined $\hat{R}, \hat{J}$ as the operators associated to the internal symmetry and the rotations, and with R, J their eigenvalues. Then there is a relation between the scaling dimension, the R-charge and the spins. The quantum numbers R, J are quantized, but the scaling dimension Δ is instead continuous. This implies that the latter cannot change upon quantization, otherwise also the previous charges would be continous.

Superconformal operators in $\mathcal{N} = 4$ SYM

For concreteness, we show an example of chiral primary operators in the context of $\mathcal{N} = 4$ SYM. The elemental fields of the system are given by scalars, fermions and gauge fields and we need to build gauge-invariant operators out of them. It is possible to achieve this task by combining products of scalars, fermions, field strengths, covariant derivatives, and finally taking their trace. One possibility is given by the single trace operator

$$\mathcal{O}(x) = \mathrm{Tr}\left(\phi^i(x) \dots \phi^j(x)\right)\,. \tag{5.103}$$

When we consider instead products of various traces, we talk about double or multiple trace operators.

One particularly important object is given by the symmetrized trace combination

$$\mathcal{O}(x) = \mathrm{Str}\left(\phi^{(i_1}(x)\dots\phi^{i_n)}(x)\right) , \tag{5.104}$$

where

$$\mathrm{Str}\left(T_{a_1}\dots T_{a_n}\right) = \sum_{\text{perm. }\sigma} \mathrm{Tr}\left(T_{\sigma(a_1)}\dots T_{\sigma(a_n)}\right) . \tag{5.105}$$

This turns out to form a completely symmetric object which comprises an irreducible representation of the superconformal algebra, and it is in addition a chiral primary operator, also called a 1/2 BPS state. The last terminology refers to the fact that half of the supercharges annihilate the object, and this corresponds to preserving the maximum possible amount of supersymmetry from the original theory, *i.e.* half of it.

Concretely, the case $n = 2$ reads

$$\mathrm{Str}\left(\phi^{(i}\phi^{j)}\right) = \mathrm{Tr}\left(\phi^i\phi^j\right) - \frac{1}{6}\delta^{ij}\,\mathrm{Tr}\left(\phi^k\phi^k\right) . \tag{5.106}$$

When the product contains n scalars, we get an operator with total dimension $\Delta = n$ and its dimension is protected at quantum level.

On the other hand, there are many non-protected operators in the theory. One example of them is the *Konishi operator*, which reads

$$K = \mathrm{Tr}\left(\phi^k\phi^k\right) , \tag{5.107}$$

and in fact its dimension is affected by loop corrections.

References and further reading

The material for this Section was mainly inspired from the lecture notes [21], see Sec. 1−3. They give a nice and complete explanation of many aspects of supersymmetric field theories, and can also be used as a starting point for further reading on the subject. You can also find part of the material used in this Lecture from Chapter 3.3 of [1], in particular concerning $\mathcal{N} = 4$ SYM. The superconformal algebra is also explained in Appendix A of [18].

There are many lecture notes available online to study supersymmetry. Some of the best ones are [22–25]. For a more advanced reading, we suggest a couple of papers on non-renormalization theorems: the perturbative proof is given in [26], while the non-perturbative result was derived in [27].

6 String theory in a nutshell

6.1 Holography derived from ST (overview)

We start by providing a quick overview of the ingredients and the logic.

ST provides a concrete way to derive a holographic correspondence known as the AdS/CFT correspondence. There are by now various avatars of the original correspondence, but we will focus in the last part of this course on the main one, namely: the duality between type IIB string theory on $AdS_5 \times S^5$ and $\mathcal{N} = 4$ supersymmetric Yang-Mills theory. As we will see, this is a strong-weak coupling duality, meaning that the strong coupling limit of one side is related to the weak coupling limit of the other side (and vice versa).

The heuristic derivation of this builds on a crucial object in string theory, namely D-branes (and in particular the so-called D3-brane). It also uses an essential feature in string theory that it contains both gravity as well as gauge theory, and the fact that a D-brane in string theory has two incarnations: one as a heavy object that curves the spacetime around it in a particular manner (as we know from general relativity) and another as a lower-dimensional hypersurface in spacetime on which a gauge theory lives. Moreover, this is considered in the setting of superstring theory, so that both the gravity as well as the gauge theory side includes a certain amount of supersymmetry (which we have encountered in earlier lectures). These two dual descriptions lie at the heart of the AdS/CFT correspondence.

It is of course impossible to give a complete account of string theory (which would be an entire course, at the least), so our aim in the coming lectures is to present the main elements of string theory and in some sense the shortest route towards understanding the basic ingredients mentioned above, namely the two faces of D-branes.

Let us give a brief overview of what the ingredients are: ST is (as the name suggests) a theory of strings. Instead of particles one assumes the elementary objects are strings. A particular type of excitation of a string will describe a particle. Thus one string describes an infinite tower of particles. The general spectrum takes the form of a set of massless particles (they are very important since they move with the speed of light and thus describe the dynamics at low energies), and then massive particles which are governed by a particular scale, the string length ($M_n \sim \frac{n}{l_s}$, with n being positive integers). Since the string scale is very low (of order of Planck scale) the massive modes are extremely heavy and will not propagate at low energies. Moreover, there can be (unwanted) tachyons in the spectrum for which $M^2 < 0$.

Strings can be either closed or open, and depending on that describe different types of excitations. As mentioned above, of particular importance is the massless modes in the spectrum. For closed strings one of these is the graviton field $G_{\mu\nu}$. This is one of the amazing things about string theory that it naturally includes gravity. In addition there are other massless fields, more about that later. On the other hand for open strings one finds that the massless fiels include (possibly non-abelian) gauge fields A_μ. Thus string theory naturally gives rise to both gravity and gauge theory, and other stuff, though tightly constrained). The basic string theory is bosonic string theory, but this suffers

from having a tachyon. This is solved by considering superstring theory.

Interactions in string theory are encoded in the the splitting and joining of (closed/open) strings, and such interactions are governed by another parameter in string theory (beyond the string length mentioned above), which is the string coupling g_s.

D-branes are particularly important objects in string theory. The follow from considering open strings and realizing that the endpoints of open strings can have either Neumann or Dirichlet boundary conditions, meaning they are either free to move in spacetime or fixed at a particular value (in a given direction in spacetime). They can be characterized by hypersurfaces in spacetime on which open strings can end. It turns out that these are dynamical objects by themselves. They have a (lower-dimensional) gauge theory living on them and then transverse scalar fields describing their fluctuations in the ambient spacetime.

If one considers a stack of N D-branes on top of each other, something special happens, namely one gets a non-abelian gauge theory (with $U(N)$ gauge symmetry) living on the stack of D-branes. For reasons that will become clear later, this is called the weak-coupling description of D-branes, or the open string picture. It turns out that D-branes in bosonic string theory also have a tachon so they are not stable.

On the other hand, D-branes have a tension (just like strings themselves have tension) and are thus massive objects. As such, when you have many D-branes on top of each other they should backreact on the spacetime that they are sitting in. Another way of saying this is that they backreact, i.e. curve the spacetime around them. This is called the strong-coupling picture of D-branes. As a consequence, we have (in a different regime than the hypersurface picture at weak coupling) a dual description of D-branes as solutions of a the low-energy effective action of string theory (which is described by General Relativity plus other massless fields). This is thus the closed string picture of D-branes.

So far in this story we did not mention the ingredient of supersymmetry. This turns out to be very important in string theory in order to a) get spacetime fermions in the spectrum (which one would like as we know these occur in nature) and b) have a theory without tachyons, since these are unphysical particles that lead to bad instabilities.

Therefore one should really consider superstring theory, which is technically a bit more complicated by essentially has the same features as those of bosonic theory described above, with the addition that everything is "supersymmetrized". For the spectrum of closed strings this means that we get for the massless modes (GR + string massless fields) augmented by SUSY partners of the former. Such theories are known as supergravity theories. For the open string sector (D-branes), this means we get now YM + SUSY partners, i.e. SYM (supersymmetric Yang-Mills theory). Importantly, D-branes with SUSY are stable objects (since they are BPS objects)

Now, the central object in AdS/CFT is the so-called D3-brane. It has two equivalent descriptions valid in different regimes. In the open string picture it is (approximately) described by $\mathcal{N} = 4$ SYM with gauge group $U(N)$. In the closed string picture it is a particular curved background solution of the low-energy limit of superstring theory, known as type IIB supergravity. The AdS/CFT correspondence then follows from taking

a 'decoupling' limit of these two pictures, as we will see.

6.2 Basic starting point of ST: Nambu-Goto and Polyakov action

The starting point of string theory is to consider "fundamental" strings as the constituents of nature. Thus instead of elementary particles, we are going to look at strings. The simplest such strings are bosonic strings. Just like a particle is described by a (one-dimensional) worldline $x^\mu(\tau)$ ($\mu = 0, \ldots D-1$) moving thru an ambient spacetime of dimension D, a string is described by a (two-dimensional) worldsheet $X^\mu(\tau, \sigma)$ (See figure (9)). This is thus an embedding map from the world-sheet parameterized by (τ, σ) to the *target spacetime*.

$$(\tau, \sigma) \to X^\mu(\tau, \sigma) \tag{6.1}$$

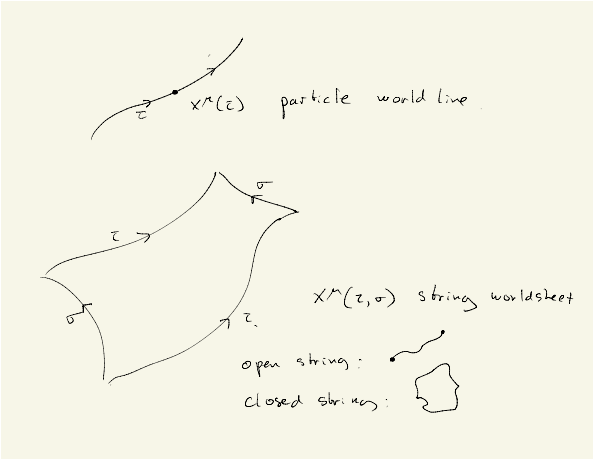


Figure 9: Worldine of a particle and worldsheet of a string

We first need to write down an action. Recall that for a relativistic particle the action is given by

$$S_{\text{particle}} = -m \int_{\text{worldline}} d\tau \tag{6.2}$$

measuring the Lorentz invariant length of the path. The corresponding equation of motion that follows from $\delta S_{\text{particle}} = 0$ is the geodesic equation. The action for a string is the natural generalization of this, namely the area swept out by the string.

$$S = -T \int_{\text{WS}} dA \tag{6.3}$$

Consider first for simplicity the target space to be Minkowskian with metric $\eta^{\mu\nu}$ and let $d^2\xi = d\tau d\sigma$. We have

$$dX^\mu = \frac{\partial X^\mu}{\partial \tau} d\tau + \frac{\partial X^\mu}{\partial \sigma} d\sigma = \frac{\partial X^\mu}{\partial \xi^\alpha} d\xi^\alpha \tag{6.4}$$

Hence the relativistic line element on the worldsheet can be written as

$$\eta_{\mu\nu} dX^\mu dX^\nu = \eta_{\mu\nu} \frac{\partial X^\mu}{\partial \xi^\alpha} \frac{\partial X^\nu}{\partial \xi^\beta} d\xi^\alpha d\xi^\beta = \gamma_{\alpha\beta} d\xi^\alpha d\xi^\beta \tag{6.5}$$

where we define the induced metric on the world-sheet

$$\gamma_{\alpha\beta} = \eta_{\mu\nu} \frac{\partial X^\mu}{\partial \xi^\alpha} \frac{\partial X^\nu}{\partial \xi^\beta} \tag{6.6}$$

and $\alpha, \beta = 0, 1$. An infinitesimal square piece of the worldsheet with corners (ξ^0, ξ^1), $(\xi^0 + d\xi^0, \xi^1)$, $(\xi^0 + d\xi^0, \xi^1 + d\xi^1)$ and $(\xi^0, \xi^1 + d\xi^1)$ has the area

$$dA = d\xi^0 d\xi^1 \sqrt{-\det \gamma} \tag{6.7}$$

Here $\det \gamma$ is the determinant of $\gamma_{\alpha\beta}$. This is the natural stringy generalization of the world-line action of a relativistic point particle, as it is proportional to the area swept out by the string.

Thus the Nambu-Goto string action takes the form

$$S = -T \int d^2\xi \sqrt{-\det \gamma} = -T \int d^2\xi \sqrt{(\dot{X} \cdot X')^2 - \dot{X}^2 X'^2} \tag{6.8}$$

It is customary to introduce $\alpha' = l_s^2$ (with l_s is the string length) and then the tension of the string is $T = \frac{1}{2\pi\alpha'}$.

When writing down an action, it is important to consider its symmetries. The NG action has global Poincare symmetry (transforming $X^\mu \to \Lambda^\mu{}_\nu X^\nu + c^\mu$) and reparametrization invariance of the world-sheet coordinates (which is a local symmetry). The latter implies that only the transverse $D - 2$ coordinates are physical fields (this is explicit in the lightcone gauge).

The NG action is difficult to quantize since it has a square-root in the action, making it non-lineaer. One can find an equivalent form that is simpler, which is the Polyakov action

$$S_{\rm pol}[g, X] = -\frac{T}{2} \int d^2\xi \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} \tag{6.9}$$

which features now also a dynamical world-sheet metric $g_{\alpha\beta}$. The Polyakov action has an important extra symmetry, as compared to the NG action, namely it is also invariant under Weyl transformations

$$g_{\alpha\beta}(\xi) \to \tilde{g}_{\alpha\beta}(\xi) = \Omega(\xi)^2 g_{\alpha\beta}(\xi) \tag{6.10}$$

Once can solve equation of motion for the dynamical 2D metric (which imply the vanishing of the WS energy momentum tensor $T_{\alpha\beta} = 0$) and plug back in and then get the NG action. So the two actions are classicaly equivalent. However, one has to take into account a set of constraints (called the Virasoro constraints) that follow from the the vanishing of the world-sheet energy momentum tensor, which are important at the quantum level. We will not delve in the details of all this in this course (as it requires a full fledged course in string theory).

Instead, to give the main idea, we note here that one can gauge fix the local symmetries of the Polyakov action. Going to the flat gauge $g_{\alpha\beta} = \eta_{\alpha\beta}$ one finds the action

$$S_{\text{Pol,gaugefixed}} = -\frac{T}{2}\int d^2\xi\, \eta^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} \tag{6.11}$$

This is simply an action for D two-dimensional scalar fields. But one should remember that there are non-trivial constraints on them and that there is a residual gauge symmetry (even though we have gauge-fixed), which is a conformal symmetry. The equations of motion of the above action are nothing but the two-dimensional wave equation

$$\partial_\alpha \partial^\alpha X^\mu = 0 \tag{6.12}$$

This shows that for each $\mu = 0, 1, ..., D-1$ the field $X^\mu(\xi)$ obeys the free scalar wave equation in two dimensions. So it is easy to write down classical solutions (see exercise).

Their form depends on the boundary conditions. We have one sector of closed (periodic) strings and another of open strings. The closed string boundary conditions are

$$X^\mu(\tau, \sigma + 2\pi) = X^\mu(\tau, \sigma) \tag{6.13}$$

For open strings the endpoints are conventionally taken to be $\sigma = 0$ and $\sigma = \pi$. Then, for the end point $\sigma = 0$ and for a given $\mu = 0, 1, ..., D-1$ the two possible boundary conditions are

$$\begin{array}{ll} \text{Neumann:} & X'^\mu(\tau, 0) = 0 \\ \text{Dirichlet:} & \dot{X}^\mu(\tau, 0) = 0 \end{array} \tag{6.14}$$

and the same for $\sigma = \pi$. For simplicity we will consider here only NN or DD boundary conditions for a particular μ, i.e. same type of boundary conditions at each end. Note also that for the time coordinate X^0 one always has NN.

These boundary conditions should be familiar from classical mechanics, i.e. one can have Neumann or Dirichlet boundary conditions at the ends, meaning they are either free to move (with momentum conserved at the endpoints) or fixed to a "wall", and momentum non-conserved at the end points, in the direction perpendicular to the wall. More specifcally the walls are hypersurfaces on which open strings can end. They dimensionality of these hypersurfaces (embedded in spacetime) depends on the choice of open string boundary conditions. If you take NN conditions in $p+1$ directions (which include the time direction) and DD conditions in the remaining $D-(p+1)$ dimensions. (See figure (10)).

One can now quantize the string. Note that the string is a two-dimensional quantum field theory consisting of d^D scalar fields. Taking into account the Virasoro constraints makes this a bit more difficult. One there are two routes: i) light-cone quantization that first solves the constraints and then quantize the space of physical configurations or ii) covariant quantization on the full configuration space and implement the constraints as conditions on the Hilbert space. These give the same results. In this entire quantization procedure a special role is played by the fact that two-dimensional field world-sheet theory describing strings is in fact a *conformal field theory*, but we omit the details. In

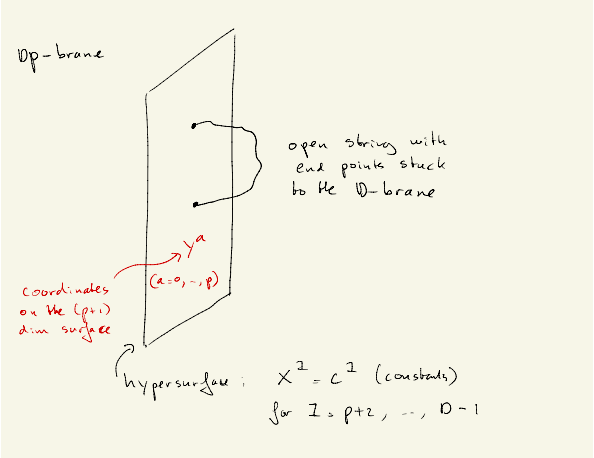


Figure 10: Hypersurface (D-brane) on which endpoints of open strings are fixed.

either way of doing it, one important results is that the number of spacetime dimensions gets fixed. In particular we need $D = 26$, which is the critical dimension of the bosonic string.

Spectrum of the closed bosonic string

The main result is that one finds a quantized spectrum for the bosonic string. For the closed string sector this consists of a tachyon with $M^2 = -\frac{4}{\alpha'}$, and, importantly the following massless fields

$$G_{\mu\nu}(x), B_{\mu\nu}(x), \Phi(x) \tag{6.15}$$

Here $G_{\mu\nu}$ is the graviton field (massless spin 2 field), which is one of the major achievements of string theory, i.e. it naturally includes gravity. The field $B_{\mu\nu}$ is a two-form gauge field (called the NSNS (or Kalb-Ramond) B-field) generalizing the notion of a one-form (or vector) gauge field A_μ that we know from QED and Φ is a scalar field called the dilaton. On top of this there is an entire tower of massive modes in the spectrum, with $M^2 \sim \frac{n}{\alpha'}$. These are conceptually important in that string theory predicts a specific set of stringy particles, but their particular form will not be important for our purposes.

Higher form gauge fields

To understand the field B better we present here a short intermezzo on higher form gauge fields. It is useful (also for the understanding of D-branes which will be discussed below), to consider a general n-form gauge field

$$A_{\mu_0\mu_1\cdots\mu_{n-1}} \tag{6.16}$$

By definition such a field is totally anti-symmetric in all its indices. Just as for the EM field we can introduce the field strength of this field

$$(dA)_{\mu_0\mu_1\ldots\mu_n} = F_{\mu_0\mu_1\ldots\mu_n} = (n+1)\partial_{[\mu_0} A_{\ mu_1\mu_2\ldots\mu_n]} \tag{6.17}$$

where the square brackets denots the completely antisymmetrized sum weighted by the total number of terms. The equation above defines the exterior derivative d acting on an n-form. We can see the the field strenght is gauge invariant under the gauge transformations

$$\delta A_n = d\chi_{n-1} \tag{6.18}$$

where χ is any $(n-1)$ form. We recognize for $n=1$ the gauge transformation of the Maxwell field in electromagnetism. The analogue of the Maxwell equations (in vacuum) for such higher forms take the form

$$\partial_{[\mu_0} F_{\mu_1\cdots\mu_{n+1}]} = 0 \quad , \quad \partial^\mu F_{\mu\nu_1\cdots\nu_n} = 0 \tag{6.19}$$

Note that in so-called form notation these take the compact form

$$dF = 0 \quad , \quad d\star F = 0 \tag{6.20}$$

where the operation $\star$ is Hodge duality. Here, the Hodge dual of an n-form A is the $(D-n)$-form given by (in Minkowski spacetime)

$$\star\, A^{\mu_{n+1}\mu_{n+2}\cdots\mu_D} = \frac{1}{n!} A_{\mu_1\mu_2\cdots\mu_n} \epsilon^{\mu_1\mu_2\cdots\mu_D} \tag{6.21}$$

where the indices in $\star A$ are raised with the Minkowski metric and where $\epsilon^{\mu_1\mu_2\cdots\mu_D}$ is defined as the completely antisymmetric object with $\epsilon^{01\cdots(D-1)} = 1$.

We will return to all this in more detail as it turns out that extended objects, called p-branes (which include strings for $p=0$ can be charged under such $(p+1)$-form potentials. Thus a particle is naturally charged under a one-form potential (we know this from EM), a string is naturally charged under a two-form potential etc. Thus we will later discuss i) how to include such couplings in the action of extended objects and ii) how the dynamics of such higher form potentials arises from string theory, i.e. the generalization of the Maxwell action $S \sim \int d^d x F_{\mu\nu} F^{\mu\nu}$.

a string is p=1, not p=0 (particle) ?

6.3 More on string theory

We now summarize some important results of (bosonic) string theory, focussing on the massless modes in the spectrum. We will first describe how the string propagates in the background that it generates (itself). We will discuss the main elements of string perturbation theory. Then we give the low energy effective action (containing Einstein gravity) which describes the dynamics of the background fields following from quantum consistency conditions. As an example, we discuss a particularly nice background solution, namely that of many fundamental strings on top of each other. Finally, we present the (massless) spectrum of the open bosonic sector (which will later be important when we consider D-branes).

Closed Strings in background fields

We have seen which particular massless fields the closed string generates (when quantizing in flat space). One can ask the question what the action for strings is when they propagate not in flat space but in a curved space generated by the strings themselves, i.e. the fields $G_{\mu\nu}$, $B_{\mu\nu}$ and Φ (in anticipation of the superstring we ignore here for now the tachyon). This action is given by a generalization of the Polyakov action in (6.9), and takes the form

$$S = S_{GB} + S_{\Phi}$$

$$S_{GB} = -\tfrac{1}{4\pi l_s^2}\int d^2\xi \left[\sqrt{g}g^{\alpha\beta}G_{\mu\nu}(X)\partial_\alpha X^\mu \partial_\beta X^\nu + \epsilon^{\alpha\beta}B_{\mu\nu}(X)\partial_\alpha X^\mu \partial_\beta X^\nu\right] \tag{6.22}$$

$$S_{\Phi} = -\tfrac{1}{4\pi}\int d^2\xi \sqrt{g}R^{(2)}\,\Phi(X)$$

where we recall that $\mu,\nu = 0,1,...,D-1$, with D the dimension of the target space. Such actions are also know as non-linear sigma models. This is the natural covariant generalization that one would write down. One can in fact derive this action using string theory (conformal field theory) methods. Let us motivate the terms from a more general perspective. The first term is just like the flat space Polyakov action but with $\eta \to g$. The second term is the unique gauge-invariant coupling of a two-form field strength to a world-sheet (we will return to this). The third term is the simplest term one can write involving the dilaton scalar Φ.

String perturbation theory

Consider the path integral in ST

$$Z = \int_M \mathcal{D}g \int \mathcal{D}X\, e^{-S[g,X]} \tag{6.23}$$

This involves an integral over all possible 2D metrics g and string configurations X. M denotes the space of all two-dimensional Riemannian manifolds.

Let us focus on the dilaton coupling in the action. If we denote Φ_0 as the expectation value of the dilaton $\langle\Phi\rangle = \Phi_0$ then effectively this part of the action takes the form

2(1-g)
else (6.26) doesnt make sense

$$S_\Phi = \Phi_0\chi \quad , \quad \chi = \frac{1}{4\pi}\int d^2\xi\sqrt{g}R^{(2)} \tag{6.24}$$

with $\chi = 2(g-1)$ the Euler number of the 2D Riemann surface. The number g is the *genus* of the manifold, found as the number of holes. Thus $g=0$ is a sphere $g=1$ a torus etc.

Defining the string coupling constant g_s as

$$g_s = e^{\Phi_0} \tag{6.25}$$

we can rewrite the path integral (6.23) as

$$Z = \sum_{g=0}^{\infty} g_s^{2(g-1)} \int_{M_g} \mathcal{D}g \int \mathcal{D}X\, e^{-S_{GB}[g,X]} \tag{6.26}$$

where M_g denotes the space of all two-dimensional Riemannian manifolds with genus g. Thus for small string coupling this defines a perturbative expansion of string theory, with genus zero (sphere) dominating and the next correction arising from the torus, etc. We recall here that the same expansion in topology was seen in the planar 't Hooft limit of large N gauge theory.

Effective action for strings in curved backgrounds

Coming back to the Polyakov action for strings in curved backgrounds, one can ask the question: in what curved backgrounds can a string *consistently* propogate. Consistenly means here in a way that conformal invariance is still maintained, which is an essential feature of the two-dimensional word-sheet theory. One has to ensure that this is maintained also at the quantum level, which requires computing the so called beta-functions and setting them equal to zero. The upshot of this is that one can do this in world-sheet perturbation theory (where the coupling is the string tension). The first non-trivial requirement comes from a one-loop computation on the world-sheet giving rise to the one-loop beta functions. Setting these to zero leads to a set of equations on the closed string background fields, since each of these act as a set of couplings in the sigma-model. The equations on finds to leading order in α' are that $D = 26$ and

$$\begin{aligned} R_{\mu\nu} &= \tfrac{1}{4} H_{\mu\rho\sigma} H_\nu{}^{\rho\sigma} - 2\nabla_\mu \nabla_\nu \Phi \\ \nabla_\rho (e^{-2\Phi} H^{\rho\mu\nu}) &= 0 \\ 4(\nabla\Phi)^2 - 4\nabla_\rho \nabla^\rho \Phi &= R - \tfrac{1}{12} H^2 \end{aligned} \tag{6.27}$$

which can thus be interpreted as the equations of motion of the background fields. $R_{\mu\nu}$ is the Ricci tensor and R the Ricci scalar computed from the D-dimensional metric $G_{\mu\nu}(X)$. Furthermore ∇_ρ is the target space covariant derivative with respect to the metric $G_{\mu\nu}(X)$ and we have introduced the field strength H of the two-form B

$$H_{\nu\rho\sigma} = 3\partial_{[\nu} B_{\rho\sigma]} = \partial_\nu B_{\rho\sigma} + \partial_\rho B_{\sigma\nu} + \partial_\sigma B_{\nu\rho} \tag{6.28}$$

Also, $H^2 = H_{\nu\rho\sigma} H^{\nu\rho\sigma}$.

It furthermore turns out that these equations can be integrated into a spacetime action, called the low energy effective action of the closed bosonic string. This reads

$$S_{\text{bos}} = \frac{1}{16\pi G_N} \int d^{26}x \sqrt{-G} e^{-2\Phi} \left[R + 4(\nabla\Phi)^2 - \frac{1}{12} H^2 \right] \tag{6.29}$$

where G_N is the gravitational coupling. We can write the full dilaton as $\Phi_{\text{full}} = \Phi_0 + \Phi$, so that now Φ contains the fluctuations of the dilaton. So that we see that $G_N \propto g_s^2$. This means that the perturbative (topological) expansion of string theory corresponds to the perturbation theory in gravity in terms of the gravitational coupling constant.

The beautiful thing is that the action above includes Einstein gravity and then string theory predicts two further terms in the action which are the kinetic terms of the B-field and the dilaton. The kinetic term of the B-field is a special case of the gauge-invariant kinetic term of a higher form gauge field.

Fundamental string solution

As one illustration of a particular interesting solution of the equations of motion, let us discuss the fundamental string solution. A fundamental string has a tension (i.e. mass per unit area) and having many fundamental strings on top of each other will thus be a heavy object, which will curve the spacetime around it. We can find this a solution of the equations of motion of the action in (6.29), by making an appropriate ansatz based on symmetries. Consider an infinite static string extending along the z diretion. There are then $D-2$ ($D=26$ for the criticial bosonic string, but we keep it general here) transverse space dimensions. The symmmetry is thus $SO(1,1)\times SO(D-2)$ with the first factor the Lorentz symmetry on the 2D dimensional worldsheet of the string and the second factor rotational symmetry in the transverse space, for which it is thus natural to take spherical coordinates. The ansatz for the metric is then

$$ds^2 = F(r)[-dt^2+dz^2] + G(r)[dr^2 + r^2 d\Omega^2_{D-3}] \tag{6.30}$$

where $d\Omega^2_{D-3}$ is the line element of a unit $(D-3)$-sphere. We also know from the Polyakov action that the string is charged under the Kalb-Ramond field B, for which the corresponding ansatz is

$$B_{tz} = H(r) \tag{6.31}$$

Finally, the dilaton, should just depend on r, i.e.

$$\Phi = I(r) \tag{6.32}$$

Inserting this into the EOMs of the action, one finds the solution

$$F(r) = f(r)^{-1}, G(r) = 1, H(r) = (f(r)^{-1}-1), I(r) = f(r)^{-1} \tag{6.33}$$

where $f(r)$ is a harmonic function on the transverse space,

$$f(r) = 1 + \frac{g_s^2 N l_s^{D-4}}{r^{D-4}} \tag{6.34}$$

By computing the charge of this solution under the B-field one can show that N corresponds to the number of fundamental strings. Furthermore this solution satisifies (in appropriate units) that the tension is equal to the charge.

Spectrum of the open bosonic string

Now, let us present the main features of the spectrum of the open sector of the bosonic string. Open strings have two endpoints. Each endpoint can have either Neumann (N) or Dirichlet (D) boundary conditions. As mentioned above we consider the case in which there are NN conditions in $(p+1)$ directions (which include the time direction) and DD conditions in the remaining $D-(p+1)$ dimensions. We denote the longitudinal directions by $a = 0 \dots p-1$ and the transverse by $I = p+1 \dots (D-1)$. The massless spectrum in that case can be computed to be

$$A^a(y), \Phi^I(y) \tag{6.35}$$

where y^a are the coordinates on a $(p+1)$-dimensional subspace of the full d-dimensional target space (with coordinates x^μ) thus a are Lorentz indices on this subspace. These fields are thus only supported on the subspace spanned by the p-brane, which is thus a hyperplane in the ambient space time. We see that on the brane we have a massless gauge field A, while the other massless degrees of freedom Φ are the transverse fluctuations of this brane. (show lol a picture). Just as for the closed string, on top of this there is an infinite tower of massive open string modes. There is also a tachyon in the spectrum with $M^2 = -\frac{1}{\alpha'}$. We will see later that when one has more than one brane on top of each other, there is an enhancement of massless string modes leading to a non-abelian theory on the stack of branes.

6.4 Superstring theory

Bosonic string theory suffers from the appearance of a closed string tachyon as well as an open string tachyon. Moreover, there are only bosonic fields in the spectrum, and nature also has fermions. This motivates considering superstring theory. The starting point is to consider the (world-sheet) supersymmetric version of the Polyakov action in flat spactime

$$S = -\frac{T}{2}\int d^2\xi\Big(\eta^{\alpha\beta}\partial_\alpha X^\mu \partial_\beta X_\mu - i\bar{\psi}^\mu \rho^\alpha \partial_\alpha \psi_\mu\Big) \tag{6.36}$$

where ρ_α are two-dimensional gamma matrices satisfying the Clifford algebra $\{\rho^\alpha, \rho^\beta\} = -2\eta^{\alpha\beta}$. Now you can basically go thru all the steps of the bosonic string, which is a long and technical story. The end result is that one finds a critical dimension $D = 10$ and that the spacetime fields now also include fermions (both in the closed and open string sector). Moreover the spactime theory is **also** supersymmetric. In particular the low-energy spacetime effective action is a supergravity theory in $D = 10$. In fact there are two possible versions, type IIA and type IIB supergravity, having to do with how you combine right and left moving modes on the string.

One finds that beyond the universal NSNS sector (with graviton, B-field an dilaton), type II (A and B) superstring theory naturally comes with further higher form potentials, called Ramond-Ramond (RR) fields. In particular, we have n-form field gauge potentials $C_{(n)}$ from the R-R sector, with n odd for type IIA and even for type IIB (the corresponding field strengths are thus even for IIA and odd for type IIB). It is technically involved to see how these arise from string theory (and thus not discuss here). We will come back below to these RR fields, as there are important objects in string theory that are charged under them (supersymmetric Dp-branes).

The fermionic fields include two spinors and two gravitinos (from the R-NS and NS-R sectors), and one can show that there are equal bosonic and fermionic degrees of freedom. In all, this corresponds to the field content of ten-dimensional type IIA (IIB) supergravity for the type IIA (IIB) superstring.

Type IIA supergravity has action

$$\begin{aligned} S_{\rm IIA} &= \frac{1}{16\pi G_N}\int d^{10}x\sqrt{-G}\left[e^{-2\Phi}\Big(R+4(\nabla\Phi)^2-\frac{1}{12}H^2\Big)-\frac{1}{4}F_{(2)}^2-\frac{1}{2\cdot 4!}F_{(4)}^2\right] \\ &\quad -\frac{1}{32\pi G_N}\int B_{(2)}\wedge dC_{(3)}\wedge dC_{(3)} + \text{fermionic terms} \end{aligned} \tag{6.37}$$

with the field strengths

$$H_{(3)}=dB_{(2)} \quad , \quad F_{(2)}=dC_{(1)} \quad , \quad F_{(4)}=dC_{(3)}+H_{(3)}\wedge C_{(1)} \tag{6.38}$$

and the supergravity coupling constant

$$16\pi G_N=(2\pi)^7 l_s^8 g_s^2 \tag{6.39}$$

We note that the last bosonic term with $B_{(2)}\wedge dC_{(3)}\wedge dC_{(3)}$ is written using the so-called wedge product of forms. This term is topological, meaning that it does not depend on the metric and we will not consider it in these lectures.

Type IIB supergravity has action

$$\begin{aligned} S_{\rm IIB} &= \frac{1}{16\pi G_N}\int d^{10}x\sqrt{-G}\left[e^{-2\Phi}\Big(R+4(\nabla\Phi)^2-\frac{1}{12}H^2\Big)-\frac{1}{2}F_{(1)}^2-\frac{1}{2\cdot 3!}F_{(3)}^2-\frac{1}{4\cdot 5!}F_{(5)}^2\right] \\ &\quad -\frac{1}{32\pi G_N}\int C_{(4)}\wedge H_{(3)}\wedge F_{(3)} + \text{fermionic terms} \end{aligned} \tag{6.40}$$

with the field strengths

$$H_{(3)}=dB_{(2)} \quad , \quad F_{(1)}=dC_{(0)} \quad , \quad F_{(3)}=dC_{(2)}-C_{(0)}\wedge H_{(3)} \quad , \quad F_{(5)}=dC_{(4)}-C_{(2)}\wedge H_{(3)} \tag{6.41}$$

and the supergravity coupling constant is the same as above.

6.5 Branes in superstring theory

Besides strings, (super) string theory has other extended objects, called D-branes. These are, especially the D-branes in superstring theory, essential ingredients of the AdS/CFT correspondence.

Intermezzo on relativistic branes

We have already introduced the notion of branes, being generalizations of particles (0-branes) and strings (1-branes). Here we discuss some important notions for general p-branes such as their actions and in what way the can be charged. A relativistic p-brane is described by a map

$$(\xi^0,\xi^1,...,\xi^p)\to X^\mu(\xi^0,\xi^1,...,\xi^p) \tag{6.42}$$

from the $(p+1)$-dimensional worldvolume into the target spacetime (taken to be D-dimensional Minkowski spacetime in the following). For a structureless (infinitely thin)

p-brane the simplest action we can think of is the generalization of the worldline (particle) or wordlsheet (string) action see before. Define the induced metric on the brane as

$$\gamma_{ab} = \eta_{\mu\nu}\partial_a X^\mu \partial_b X^\mu \tag{6.43}$$

for $a, b = 0, 1, ..., p$ with $\partial_a = \partial/\partial\xi^a$. Then the action can be taken again to be proportional to the volume swept out by the brane

$$S = -T_p \int d^{p+1}\xi \sqrt{-\det \gamma} \tag{6.44}$$

where T_p is the tension of the p-brane (in units of inverse length to the power $p+1$). This is known as the *Dirac action*. The EOMS (obtained from varying the embedding coordinates) are then

$$\partial_a(\sqrt{-\det\gamma}\,\gamma^{ab}\partial_b X^\mu) = 0 \tag{6.45}$$

Charged branes

Branes can be charged, just like a particle can be charged under a vector field. We consider here electric charge couplings. Recall that the coupling of a charged particle to the Maxwell field has the form

$$S = q\int d\tau\, A_\mu \dot{X}^\mu \tag{6.46}$$

and varying this with respect to X allows one to derive the Lorentz force $qF^\mu{}_\nu \dot{X}^\nu$ which enters the EOM of a massive charged particle.

Using the n-form potentials introduced earlier, we can find the natural generalization of the coupling above to p-branes. This action takes the form

$$S = Q_p \int d^{p+1}\xi \frac{1}{(p+1)!}\epsilon^{a_1\cdots a_{p+1}} A_{\mu_1\cdots\mu_{p+1}}\partial_{a_1}X^{\mu_1}\cdots\partial_{a_{p+1}}X^{\mu_{p+1}} \tag{6.47}$$

Essentially, since a p-brane sweeps out a $(p+1)$-dimensional worldvolume, it couples naturally to a $(p+1)$-form potential. The total action for a charged relativistic p-brane is thus the sum of the Dirac action (6.44) and (6.47). One can derive the equations of motion to be

$$T_p\partial_a(\sqrt{-\det\gamma}\,\gamma^{ab}\partial_b X^\mu) = -\frac{Q_p}{(p+1)!}\epsilon^{a_1\cdots a_{p+1}}F^\mu{}_{\nu_1\cdots\nu_{p+1}}\partial_{a_1}X^{\nu_1}\cdots\partial_{a_{p+1}}X^{\nu_{p+1}} \tag{6.48}$$

where the right-hand side is the generalized Lorentz force, involving the $(p+2)$-form field strength. The action of the latter is

$$S = -\frac{1}{g^2}\int d^D x \frac{1}{(p+1)!}F^{\mu_0\mu_1\cdots\mu_{p+1}}F_{\mu_0\mu_1\cdots\mu_{p+1}} \tag{6.49}$$

Thus, in the presence of an electrically charged p-brane this field strength gets sourced, so that the corresponding Maxwell equation is We also remark that, as far as the dynamical

equations of the gauge field is concerned. $\partial^\mu F_{\mu\nu_1\cdots\nu_{p+1}} = J_{\nu_1\cdots\nu_{p+1}}$ where $J_{\nu_1\cdots\nu_{p+1}}$ is the $(p+1)$-form current that corresponds to the distribution of the charged branes.

A p-brane can also couple magnetically to a $(D-p-3)$-form gauge potential A. This can be seen using Hodge duality. The field strenght of $F_{D-p-2} = dA_{D-p-3}$ is a $(D-p-2)$-form. The latter can be related to $\tilde{F}_{p+2} = \star F_{D-p-2}$ by Hodge duality. The dual gauge field strength can be written as $\tilde{F}_{p+2} = d\tilde{A}_{p+1}$, so that in terms of the dual gauge field $\tilde{A}$ the form of the coupling is as in (6.47), now with Q_p being the magnetic charge. Note that for $D = 4$ and $p = 0$, a particle can couple both electrically and magnetically to a 1-form, as we know from EM.

We also briefly discuss here the way one can compute the charge of a particular field configuration, generalizing what you know from EM. The "electric charge" μ_p is given by

$$\mu_p = \int_{S^{D-p-2}} *F_{p+2},$$

where the integral is over S^{D-p-2}, which surrounds a p-brane, and $*F_{p+2}$ is the Hodge dual of the $p+2$ form field strength of the $p+1$ form gauge field A. The "magnetic dual" charge is

$$\mu_{D-p-4} = \int_{S^{p+2}} F_{p+2},$$

where the integral is over S^{p+2}, which surrounds a $(D-p-4)$-brane. So, the magnetic coupling of the $p+1$ form field is with a $(D-p-4)$-brane, as already remarked above. By a direct analogy with the 4D $U(1)$ case, in which both electric and magnetic charges are point particles (electric and magnetic monopoles), one can show the Dirac quantization condition holds:

$$\mu_p \mu_{6-p} \in 2\pi Z$$

(See Exercise 6.4 of Becker-Becker-Schwawrz)

Finally, note that we can also see from the discussion above that for the $p = 1$ (string) case the action for a charged 1-brane is precisely of the form of the Polyakov action (modulo the dilaton term). Indeed, the term involving the B-field shows that the string is charged under that 2-form, and in particular, the charge turns out to have a very specific value, namely the tension, for the fundamental string.

Dp-branes

We finally focus on (for the purpose of the AdS/CFT correspondence) the most important objects in superstring theory, namely Dp-branes (in fact we will later mainly be concerned with the special case of the D3-brane)

We already showed how D-branes arise from the open string sector in bosonic string theory. When we consider superstring theory such D-branes acquire a number of interesting and crucial properties, namely i) they will be charged under the RR gauge fields of type II supergravity and ii) they will be supersymmetric objects, saturing BPS bounds.

We already discussed (see figure (10)) D-branes in bosonic string theory and mentioned that the massless degrees of freedom are given by a gauge field (living on the

brane) and a set of transverse scalars. In type II superstring theory these bosonic degrees of freedom will get augmented by their supersymmetric partners (namely gauginos and dilatinos).

One can ask the question what the dynamics of these fields is. Just as for the spacetime effective action there are consistency conditions in string theory that allow to compute such an action (we restrict to the bosonic fields). We have seen above the action for an an infinitely thin relativistic p-brane, which only features the embedding map as the dynamical field. In string theory there is further structure. First of all, there is another dynamical massless field on the brane (beyond the embedding map), namely the gauge field A^a. Moreover, D-branes are made out of strings and thus have a thickness of order of the string length. But we can focus on the low energy dynamics (just as for the spacetime action) $E \ll l_s^{-1}$ so that we can approximate the D-brane as infinitely thin.

It can be shown that the action for Dp-brane is the so-called Dirac-Born-Infeld (DBI) action

$$S_{\text{DBI}}(A^a, \Phi^I) = -T_{\text{Dp}} \int d^{p+1}\xi \sqrt{-\det(\gamma_{ab} + 2\pi l_s^2 F_{ab})} \tag{6.50}$$

where γ_{ab} is the induced metric (6.43) and T_{Dp} is the tension of the Dp-brane that can be computed to be

$$T_{\text{Dp}} = \frac{1}{(2\pi)^p g_s l_s^{p+1}} \tag{6.51}$$

The action (6.50) is is a functional of the (massless) open string fluctuating fields in (6.35) given the closed bosonic background fields g, B and ϕ (in the same vein as the Polyakov action describing the dynamics of the string fields X in a closed string background). The dependence on the worldvolume gauge field A^a enters through its field strength F_{ab}. The dependence on the transverse scalars X^I enters through the induced metric γ_{ab} and the embedding coordinates

$$\begin{aligned} X^a(\xi) &= \xi^a \quad \text{for} \quad a = 0, 1, ..., p \\ X^I(\xi) &= c^I + 2\pi l_s^2 \Phi_I(\xi) \quad \text{for} \quad I = p+1, ..., D-1 \end{aligned} \tag{6.52}$$

where c_I are constants denoting the location of the brane in the transverse space.

To get a better idea what this action entails, it is useful to expand the square root in terms of a low energy expansion in powers of the string length l_s. The leading term becomes a constant proportional to the infinite volume of the brane. The next non-zero contribution gives the action

$$S_{\text{A}\Phi} = -\frac{1}{g^2} \int d^{p+1}\xi \left(\frac{1}{4} F_{ab} F^{ab} + \frac{1}{2} \sum_{I=p+1}^{D-1} \partial_a \Phi_I \partial^a \Phi_I \right) \tag{6.53}$$

as expected, exhibiting the kinetic term of the gauge field (electrodynamics in $p+1$ dimensions) along with that of the scalar fields. The normalization is such that $1/g^2 = (2\pi)^2 l_s^4 T_{\text{Dp}}$. Higher derivative terms are suppressed by higher orders in the string length.

We emphasize that we see here that whereas we started with open strings being stuck to a hypersurface, string theory tells us that this hypersurface is not rigid but becomes a dynamical object, with the massless fields Φ^i describing its transverse fluctuations. Since we are consider very small string coupling, it follows from (6.51) that the D-brane tension is very large, so that they can be described classically to a good approximation.

When we consider D-branes in superstring theory, it turns out that they are charged under the RR gauge potentials (recall these area also massless closed string fields, see the gauge potentials C_{p+1} in (6.38) and (6.38)) arising from the quantization of the superstring action). More precisely a Dp-brane is charged under the $(p+1)$-form gauge potential C_{p+1}. This coupling is as we described earlier and takes the form

$$S = S_{\rm DBI} + T_{\rm Dp} \int d^{p+1}\xi \frac{1}{(p+1)!} \epsilon^{a_1 \cdots a_{p+1}} C_{\mu_1 \cdots \mu_{p+1}} \partial_{a_1} X^{\mu_1} \cdots \partial_{a_{p+1}} X^{\mu_{p+1}} \tag{6.54}$$

so that they carry a very particular charge, namely it is equal to the Dp-brane tension $Q_p = T_{\rm Dp}$. This is actually a consequence of supersymmetry and the fact that Dp-branes are BPS objects preserving half of the supersymmetries. The total action also contains fermionic terms which we have omitted.

Note also that it follows from (6.38), (6.38)) that type IIA has Dp-branes with p is even while type IIB with p is odd. Thus, in type IIA string theory one has the D0-brane (also called the D-particle) electrically charged under C_1l; the D2-brane (elec. charged under C_3), D4-brane (magn. charged under C_3); D6-brane (magn charged under C_1);

Likewise in type IIB we have a D1-brane (also called the D-string) electrically charged under C_2, D3-brane (both elec and magn charged under C_4), D5-brane, (mang. charged under C_2). Note the field C_4 is special since it is self-dual.

There are also more exotic branes, such as D8-brane in type IIA and D(-1)-branes (instantons), D7-, and D9-branes in type IIB.

We summarize below: Type IIA:

Form	Electric	Magnetic
C_1	$D0$	$D6$
C_3	$D2$	$D4$.

It turns out D8 can also sometimes exist.

Type IIB:

Form	Electric	Magnetic
C_0	$D(-1)$	$D7$
C_2	$D1$	$D5$
C_4	$D3$	$D3$.

The $D(-1)$ is a D-instanton and only makes sense in Euclidean signature. The D1 is the D-string, which is interchanged with the fundamental string or F-string under S-duality. The D3 brane couples both electrically and magnetically to C_4 and since C_4 is self-dual, the charge is self-dual (i.e., there is no distinction between electric and magnetic in this case).

Stack of D-branes and YM

We need to introduce one more important generalization, namely a so-called stack of D-branes. It is possible to have many (say N) D-branes and put them on top of each other. One can analyze again the spectrum of fluctuations of these, and what one finds is that, while naively one would expect the massless degrees of freedom to be that of a (supersymmetric) $U(1)^N$ gauge theory, there are new massless degrees of freedom (coming from strings connecting two distinct D-branes) such that the gauge group becomes $U(N)$, i.e. non-abelian. This is most easily seen in a picture. If you consider the case $N = 2$ there are now not only strings going from 1-1 or 2-2, but also 1-2 and 2-1 strings. Each of these includes a massless field. Thus while one would have naively expected two independent gauge fields A^a_{11}, A^a_{22} (and thus $U(1)^2$ symmetry), we get an additional two of them, thus four in total. Not only that but one can show that their dynamics describes a non-abelian gauge theory with gauge group $U(2)$. (See figure (11)).

In general this means that on a stack of N Dp-branes in superstring theory we get a $U(N)$ (with dimension N^2) supersymmetric YM theory. The massless fields (see (6.35)) thus carry now also an adjoint index ij $(1 \leq i, j \leq N)$ of $U(N)$, i.e.

$$A^a_{ij}(y), \Phi^I_{ij}(y) \tag{6.55}$$

This means that the DBI action now also involves a trace over the adjoint indices, so that the low-energy limit leads to the action of SYM. This will be very important in the AdS/CFT correspondence.

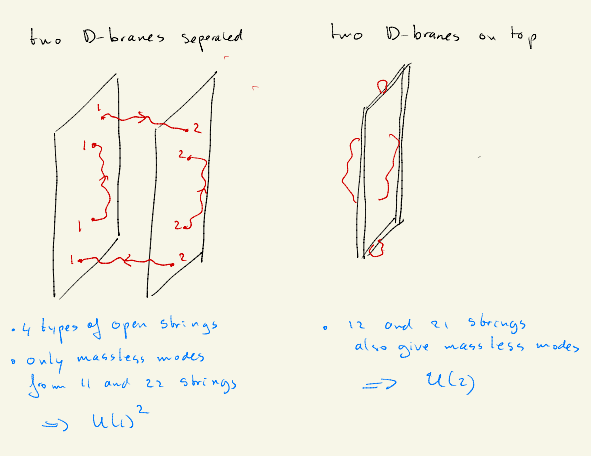


Figure 11: Extra massless degrees of freedom on two stack D-branes stacked on top of each other.

Finally, let us focus on one particular D-brane, namely the D3-brane, which will be the main actor when we heuristically derive the AdS/CFT correspondence. This brane is particular interesting since its worldvolume is 4-dimensional, thus the D3-brane action describes a 4D field theory. In fact the leading order (low energy) DBI action is nothing but 4D $\mathcal{N} = 4$ SYM and if one takes a stack of N D3-branes the YM group is $U(N)$.

D-branes as classical backgrounds in supergravity

D-branes have a tension so they are massive objects (just like the fundamental string discussed earlier). If one puts many such branes on top of each other there is regime in which they will start to curve the spacetime around them, i.e. generate a classical backround in sting theory. Such a classical background is a solution of the equations of motion of the low-energy effective actions (6.37) and (6.40) of type IIA and IIB respectively. To find this solution, we can make again an ansatz for the background fields and solve the EOMs. The symmetry for a Dp-branes is $SO(1,p)\times SO(D-p-1)$ (we keep D general, but $D=10$ is the critical dimension of type II string theory). Furthermore, they are charged under the RR gauge potential C_{p+1}. This motivates the ansatz

$$ds^2 = F(r)\left[-dt^2+\sum_{i=1}^{p} dz_i^2\right] + G(r)[dr^2 + r^2 d\Omega^2_{D-p-2}] \tag{6.56}$$

$$C_{tz_1\ldots z_p} = C(r) \tag{6.57}$$

$$e^{\Phi} = I(r) \tag{6.58}$$

Inserting this into the EOMs of the action, one can find the solution

$$F(r) = H(r)^{\frac{p-7}{8}} \quad , \quad G(r) = H(r)^{\frac{p+1}{8}} \quad , \quad C(r) = 1 - H(r)^{-1} \quad , \quad I(r) = H(r)^{\frac{3-p}{4}} \tag{6.59}$$

with $H(r)$ a harmonic function on the transverse space

$$H(r) = 1 + \frac{\alpha}{r^{7-p}} \quad , \quad \alpha = (4\pi)^{\frac{5-p}{2}}\Gamma(\frac{7-p}{2}) l_s^{7-p} g_s N \tag{6.60}$$

Let us write this explicitly for $D=10$, consider $p=3$, so the solution of a stack of D3-branes. We can also fix all constants by computing the charge (in appropriate conventions) and using the fact that this is equal to the total tension. The resulting solution is

$$ds^2 = H(r)^{-1/2}[-dt^2 + \sum_{i=1}^{3} dz_i^2] + H(r)^{1/2}[dr^2 + r^2 d\Omega^2_{D-p-2}] \tag{6.61}$$

$$C_{tz_1\ldots z_3} = 1 - H(r)^{-1} \tag{6.62}$$

$$\Phi = 0 \tag{6.63}$$

with

$$H(r) = 1 + \frac{\alpha}{r^4} \quad , \quad \alpha = 4\pi l_s^4 g_s N \tag{6.64}$$

References and further reading

There are numerous excellent books on string theory as well as reviews and lectures that are accessible online. Standard books are Green, Schwarz & Witten (two volumes) and the (more modern) volumes by Polchinski. A nice lower level book is by Zwiebach (used in a course for undergraduates). See also the String Theory Wiki

It goes without saying that one can easily fill an entire course (if not more) on an introduction to string theory. There are nice lecture notes at NBI, written by Troels Harmark for his course on this. Other lectures note that can be recommended are those by David Tong.

Further reading in the AE book connected to this section of the notes are sections 4.1.1, 4.1.3, 4.1.4, 4.2.1 (cursory), 4.2.3, 4.4.1, 4.4.2.

7 The AdS/CFT correspondence

In this section we turn to the final aim of the course namely a heuristic derivation of the AdS/CFT correspondence, combining many of the techniques and concepts we have seen so far. This is also very nicely explained in the AE book, so that we will be relatively brief in the text and ask you to also read the relevant sections in the book.

These are specifically Sections 5.1 and 5.2.

7.1 The AdS/CFT correspondence: a first glance

We start with a summary of the most important points of the duality:

- Strongest form of the correspondence:

$\mathcal{N}=4$ SYM	type IIB superstring theory
$SU(N)$ gauge group	string length $\ell_s = \sqrt{\alpha'}$ coupling constant g_s
coupling constant $g_{\rm YM}$	$AdS_5 \times S^5$ background; radius L; N units of F_5 flux on S^5
$g^2_{\rm YM} = 2\pi g_s$	$2g^2_{\rm YM} N = (L/\ell_s)^4$

- The S^5 part of the string theory side is arrived at by Kaluza-Klein dimensional reduction of five of the 10 dimensions.

more precisely,
10D IIB on AdS5×S5→KK reduce on S5→ 5D gravity on AdS5.
We are not really creating the S5 by dimensional reduction?

- Strong form of the correspondence:

$g_{\rm YM} \ll 1$ and $\lambda = N g^2_{\rm YM}$ finite constant	$g_s \ll 1$ and L/ℓ_s finite constant
$N \to \infty$ so planar diagrams	tree-level (genus-0) diagrams
$1/N$ corrections are non-planar diagrams	g_s corrections are higher-genus diagrams

- See Figures 1.4, 1.5 an 1.6 on pp.43-44.

- Weak form of the correspondence:

$g_{\rm YM} \ll 1$ and $\lambda = N g_{\rm YM}^2 \to \infty$	$g_s \ll 1$ and $L/\ell_s \to \infty$
strong coupling	weak coupling
$\mathcal{N} = 4$ $SU(N)$ SYM on $R^{3,1}$	type IIB SUGRA on weakly curved $AdS_5 \times S^5$

- Note that we get SUGRA on the string side because $\ell_s/L \to 0$, namely the string becomes point-like. We are thus insensitive to all but the lowest lying string excitations, which are the massless ones. SUGRA consists of just the massless states of type II superstring theory.

- On the string side, the fact that $g_s \ll 1$ tells us that the string theory is weakly coupled. The fact that $L/\ell_s \to \infty$ and the massive excitations of type IIB superstring theory go like $1/\ell_s^2$ tells us that we're really only sensitive to the massless modes at low energies, $E \ll 1/\ell_s$. The massless modes of type IIB superstring theory is typed IIB SUGRA.

- Note that while $g_s \ll 1$, the combination $N g_s \to \infty$ because $(L/\ell_s)^4 = 4\pi N g_s$. Intuitively, g_s controls the closed string sector, but $N g_s$ controls the open string sector on the D3-branes. In fact, that final statement is actually not quite correct. If $N g_s \ll 1$, then the D-branes are well described as basically rigid undeformable objects on which open strings end and the open strings have six vibrational degrees of freedom orthogonal to the brane. If $N g_s \gg 1$, this picture is no longer a good one and the appropriate picture is really that the branes themselves are vibrating and they have six orthogonal vibrational degrees of freedom. In other words, there are no open strings anymore. There is thus in some sense a transition from an open to closed string picture, and the situation in between is highly complicated. To complete the picture on the gravity side, the fact that $N g_s \gg 1$ tells us that we don't have open strings anymore, just dynamical vibrating D-branes. However, because the D-brane mass scales like $1/\ell_s$, they are also extremely massive in this limit. So, they don't actually vibrate because to make them vibrate would require an enormous amount of energy! In other words, they are just sitting pretty (or ugly, depends on your disposition) deforming spacetime, but are otherwise non-dynamical.

7.2 D3-branes and their two faces

The AdS/CFT correspondence in its original form is obtained by consider the specific case of D3-branes and focussing on the open string perspective on the one hand and the closed string perspective on the other hand. A decoupling limit is then considered which enables to get a non-trivial relation between the two. (See figure (12)

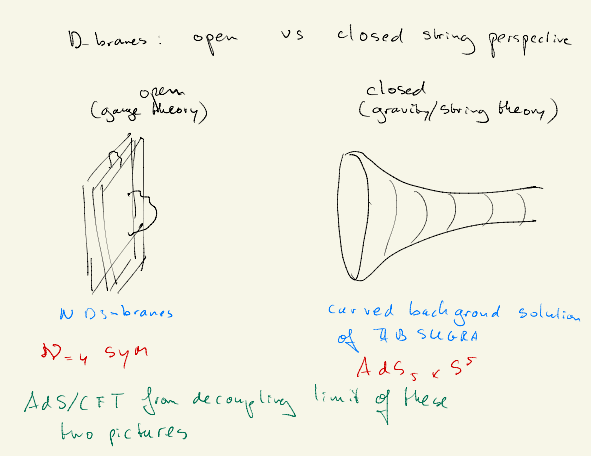


Figure 12: Open and closed perspective of a stack of D3-branes.

Open string perspective of D-branes

- Why consider a stack of N D3-branes? Because the coupling of open string endpoints to this stack can be described by an $SU(N)$ gauge theory in the worldvolume of the D3-branes, which is 4-dimensional. Hence the possibility of YM.
- The motion of the string endpoint in the D3 worldvolume can be described by a vector gauge field A_μ, which is valued in $SU(N)$ depending on the D3-brane in the stack to which the string is attached.
- The vibrations of the string in the 6 orthogonal directions are scalars ϕ^i from the D3-brane worldvolume perspective. This at least covers the bosonic part of $\mathcal{N} = 4$ SYM. There are also fermionic degrees of freedom on the superstring and they will fill in the $\mathcal{N} = 4$ gauge multiplet.
- How to exactly describe the aforementioned states explicitly in terms of the open string states (e.g. using lightcone gauge), is the topic of a string theory course.
- To be more precise, one should consider the worldvolume action of the D3-brane given by the DBI action.
- Note: $[F] = L^{-2}$ and so it must be multiplied by $\alpha' = \ell_s^2$ to get something dimensionless in the determinant. The factor of 2π is just convention. Since we want ϕ^i to be just like A_μ, but orthogonal to the D3-branes, we want it to have the same dimension, which is L^{-1}. On the other hand, we want to identify it with the transverse coordinates, which have dimension L. So, we must multiply by α' again: $x^{i+3} = 2\pi\alpha'\phi^i$. The pullback of the full metric onto the D3-brane worldvolume is

$$\mathcal{P}[g]_{\mu\nu} = g_{MN}\frac{\partial x^M}{\partial x^\mu}\frac{\partial x^N}{\partial x^\nu} = \text{Eqn. 5.5.}$$

- Expand $g = \eta + \kappa h$ and

$$e^{-\phi} \to e^{-\phi_0 + \kappa\phi} = \frac{1}{g_s} e^{\kappa\phi} \approx \frac{1 + \kappa\phi}{g_s}.$$

Using $\ln\det = \mathrm{tr}\ln$, one can derive

$$\sqrt{\det(g_{\mu\nu} + h_{\mu\nu})} = (\det g_{\mu\nu})\left(1 + \frac{1}{2}h + \frac{1}{8}(h^2 - h_{\mu\nu}h^{\mu\nu}) + \cdots\right),$$

where the indices on $h_{\mu\nu}$ are raised using $g^{\mu\nu}$ and $h = g^{\mu\nu}h_{\mu\nu}$.

Recalling that F is anti-symmetric and $h_{\mu\nu}$ is traceless, you should get AE Eqns. (5.6) and (5.7) except (5.7) is multiplied by κ. The text says this is so actually at the end of p.185 and beginning of p.186.

- This is for a single D3-brane. For a stack of N D3-branes, the fields are $SU(N)$-valued. So, there will be matrix traces, derivatives become covariant derivatives, and the ϕ^i's do not commute anymore and pick up an interaction at lowest order of the form expected from SYM: $\propto \mathrm{tr}[\phi^i, \phi^j]^2$.

- Aside: In principle this should all be derived from a non-abelian generalization of the DBI action for on D3-brane in AE Eqn. (5.4) to a stack. This generalization is not known, but there is proposal (to all orders in α') involving the symmetrized trace, and computations to order α'^3. For our purposes we will only need the α'^2 non-abelian part, which is known to give the trace.

- Another aside: One way to think about how the scalars arise is as follows. Consider N space-filling D9-branes. The open superstring spectrum in this case is the gauge multiplet of $\mathcal{N} = 1$ $SU(N)$ SYM. This consists of the 8-vector gauge field in the NS sector and the 8-spinor gaugini in the R sector. Compactify six spatial directions on circles of very small radius (i.e. on a small six-torus) and then T-dualize along those directions. That will leave a very large six-torus (so, basically flat 6D space) and a stack of N D3-branes. You get $\mathcal{N} = 4$ SYM now in 4D as the Kaluza-Klein reduction of $\mathcal{N} = 1$ SYM in 10D (see AE pp.137-138). The $[\phi^i, \phi^j]^2$ term arises from the dimensional reduction of F^2 since $F^{ij} \propto [\phi^i, \phi^j]$.

- Finally, since $\kappa \to 0$ as $\alpha' \to 0$, we also know that the closed string sector, whose action is the SUGRA action in Eqn. (5.3) (the bosonic part) is free in this limit.

- Therefore, when $g_s N \ll 1$, to lowest order in α' (i.e., the low-energy limit), we have $\mathcal{N} = 4$ SYM with $SU(N)$ gauge group living on a stack of N D3-branes and free SUGRA in the closed string sector. These two sectors decouple.

- A final remark is in order: Suppose we start with $N+1$ D3-branes in the flat (9+1)-dimensional spacetime, and we seperate one of them from the stack of N branes, say in the x^9 direction. For definiteness, he stack is at $x^9 = 0$ and the separated brane at $x^9 = r$. This will break the $U(N+1)$ symmetry to $U(N) \times U(1)$, which is

like a Higgs phase due to one of the scalars acquiring a vacuum expectation value $\langle\phi^0\rangle = \frac{r}{2\pi\alpha'}$. In the decoupling limit that we considered, we should keep all the field theory quantities fixed, which means that while we send $\alpha' \to 0$, the quantity

$$u = \frac{r}{\alpha'} = \text{fixed} \tag{7.1}$$

Closed string perspective of D-branes

- We now consider the closed string perspective of the D-brane as a solitonic solution (with a non-trivial background, including a curved spacetime). This is the limit in which $g_s N \gg 1$ (the gravity side).

- First, from the metric of the D3-brane we can read off the Newton gravitational potential for a D3-brane with tension T_3,

$$V_{\text{Newton}} \sim -\frac{\kappa^2 T_3}{4\pi r^4},$$

where κ^2 is the gravitational constant and r is the distance away from the brane in the 6 transverse spatial directions. This goes like r^{d-2} in general, with d transverse spatial directions, which reproduces the usual gravitational potential for a point mass in 3 spatial dimensions $\sim -1/r$.

- Mnemonic for T_p: The weaker g_s is, the more taut the brane appears to open strings. So, we expect $T_3 \sim 1/g_s$. You can also see this from the DBI Lagrangian which has a factor of $e^{\phi} \sim 1/g_s$ multiplying everything. Dimensional analysis:

$$[T_p] = \frac{1}{L^{p+1}} \qquad \Longrightarrow \qquad T_p \sim \frac{1}{\ell_s^{p+1} g_s}.$$

The standard convention (given earlier) is that there is also a factor of $(2\pi)^p$ in the denominator.

- Mnemonic for g_{YM}: g_{YM} measures the strength of "photon" emission. The photon is in the open string sector and emission of open strings scales as $\sqrt{g_s}$. So, $g_{\text{YM}}^2 \sim g_s$. Dimensional analysis:

$$\frac{1}{2g_{\text{YM}}^2}\int \underbrace{d^{p+1}x}_{L^{p+1}}\underbrace{e^{-\phi}}_{L^0}\underbrace{(\partial A)^2}_{L^{-4}} + \cdots \qquad \Longrightarrow \qquad [g_{\text{YM}}^2] = L^{p-3} \qquad \Longrightarrow \qquad g_{\text{YM}}^2 \sim g_s \ell_s^{p-3}.$$

- Mnemonic for κ: κ measures the strength of graviton interaction and, in particular, its emission. The graviton is in the closed string sector and emission of closed strings scales as g_s. So, $\kappa \sim g_s$. Dimensional analysis:

$$\frac{1}{2\kappa^2}\int \underbrace{d^d x}_{L^d}\underbrace{\sqrt{-g}\, e^{-2\phi}}_{L^0}\underbrace{R}_{L^{-2}} \qquad \Longrightarrow \qquad [\kappa^2] = L^{d-2} \qquad \Longrightarrow \qquad \kappa^2 \sim g_s^2 \ell_s^{d-2}.$$

- So, in $d = 10$,

$$V_{\text{Newton}} \sim -\frac{N g_s \ell_s^4}{r^4}.$$

For $N g_s \ll 1$ and $\ell_s/L \to 0$, the region where this potential exists is practically confined to one point. But, for $N g_s \gg 1$, this is no longer the case. The cartoon picture is Figure 5.1 of AE (p.183). It turns out that these geometries generically have singularities and horizons, so they are often called "black-brane" solutions.

- The actual form of the background solution requires a somewhat lengthy derivation. We have given the result in the previous section. Horowitz and Strominger noted that one could reduce the problem of finding black-brane solutions to finding black hole solutions in the transverse dimensions of the black brane [28]. This had already been done by Gibbons and Maeda [29]. The upshot is AE Eqns. (4.115-118). The relationship defining L in terms of N, g_s and ℓ_s (Eqn. 5.13) is derived from setting the flux through the S^5 equal to the number N of D3-branes.

- Exercise: Show that the solution given solves at least the dilaton equation of motion.

$$\delta\mathcal{S} = \frac{1}{2\kappa^2}\int d^{10}x\,\sqrt{-g}\left[e^{-2\phi}(-2\delta\phi)\left(R + 4(\nabla\phi)^2 - \frac{1}{2}|H_3|^2\right) - 8\nabla(e^{-2\phi}\nabla\phi)\delta\phi\right]$$

So, the equation of motion is

$$R - 4(\nabla\phi)^2 + 4\Box\phi - \frac{1}{2}|H_3|^2 = 0.$$

If ϕ is constant and $B_2 = 0$, then this reduces to $R = 0$. This works in general: the Ricci scalar is

$$R = -\frac{5H' + rH''}{2rH^{3/2}} = 0 \qquad \text{when} \qquad H(r) = 1 + \left(\frac{L}{r}\right)^4.$$

You can also see this in the throat region, where we drop the 1 in $H(r)$. Then, you get $AdS_5 \times S^5$ with equal radii L. The sphere has curvature $R_{S^5} = 20/L^2$ and AdS has curvature $R_{AdS_5} = -20/L^2$ and the total curvature is 0.

- The two regions of the D3-brane solution. We now want to study the geometry given in (6.61), (6.62), (6.64), and consider in particular the asymptotic region and the throat (or near-horizon) region. Let us first note again that the length scale of this solution is determined by the Newtonian potential discussed above, i.e by the harmonic function

$$H(r) = 1 + \frac{L^4}{r^4} \tag{7.2}$$

where

$$L^4 = 4\pi g_s N \alpha^2 \tag{7.3}$$

inconsistency
l_s/L —>0 would correspond to g_sN —> inf
not g_sN <<1

Clearly, in the asymptotic region, $r \gg L$ this geometry is flat 10D Minkowski space. On the other hand in the near-horizon region, $r \ll L$, the metric takes a non-trivial form. To see this note that in this region the constant 1 in the harmonic function can be neglected so that the metric becomes

$$ds^2 = \frac{r^2}{L^2}\eta_{\mu nu}dx^\mu dx^\nu + \frac{L^2}{r^2}\delta_{ij}dx^i dx^j + \tag{7.4}$$

where $\mu = 0, 1, 2, 3$ are the brane directions and $i = 4 \dots 9$ the transverse directions. We can rewrite $\delta_{ij}dx^i dx^j = dr^2 + r^2 d\Omega_5^2$, by going to spherical coordinates. Then defining a new coordinate $z = L^2/r$ it is not difficult to see that the metric takes the form

$$ds^2 = \frac{L^2}{z^2}\left[\eta_{\mu nu}dx^\mu dx^\nu + dz^2\right] + L^2 d\Omega_5^2 \tag{7.5}$$

We recognize this metric as that of $\mathrm{AdS}_5 \times S^5$ both with radius L.

- Closed strings can propagate in the near-horizon region or throat ($r \ll L$) or in the asymptotically flat region ($r \gg L$). The throat has the geometry of $AdS_5 \times S^5$ (when you drop the 1 in H). These still matter from the low-energy perspective of an observer far away form the D3-branes because of the gravitational red-shift.

  To see this consider a string excitation with energy $l_s E_r$ (in string theory units) at some fixed radial position r. The energy measured by an observer at infinity is

  $$l_s E_\infty = l_s\sqrt{-g_{00}}E_r = \frac{r}{L}l_s E_r \tag{7.6}$$

  Thus since in the throat region $r \ll L$, even though E_r might be large, the energy measured at infinity is very small.

  Thus, an observer at infinity sees two different low-energy modes, the supergravity modes propagating in flat 10D Minkowski space and the string excitations in the throat region.

- Intuitively, the strings in the throat region are isolated from those outside and they decouple. This also aligns with the decoupling limit in the field theory description described above. There, an “observer scale" can be generated by separating one of the D3-branes a distance r away from the rest of the stack. The mass of the string stretching this distance scales as r/ℓ_s^2 and the decoupling limit in the field theory side is $\ell_s \to 0$ but the mass of the stretched string is kept constant. So $r \ll \ell_s$, which for $Ng_s \gg 1$ implies $r \ll L$.

Combine the perspectives

We can now juxtapose the two perspectives:

- Open string perspective: $\mathcal{N} = 4$ SYM and type IIB SUGRA on $R^{9,1}$.

- Close string perspective: type IIB SUGRA on $AdS_5 \times S^5$ and type IIB SUGRA on $R^{9,1}$.

  And comparing this we arrive at:

- The conjecture that $\mathcal{N} = 4$ SYM is dual to type IIB SUGRA on $AdS_5 \times S^5$ (of course will all the extra identifications and limits stated at the beginning of this section).

In particular the AdS/CFT correspondence is a strong/weak coupling duality. It relates a difficult regime in the gauge theory (large 't Hooft coupling λ) to a simple regime in string theory (classical string theory). Thus one way of using the duality is to get predictions of strongly coupled gauge theories by doing computations in supergravity. One way to go beyond supergravity is to take into account stringy α' effects. They enable to compute subleading effects (for large λ) on the gauge theory side. The most difficult (and ultimately the most interesting) is to turn on interactions $1/g_s$ in string theory, which will translate to $1/N$ effects (i.e. non-planar) around the planar limit.

There have been thousands of papers written on aspects of AdS/CFT addressing the above issues, generalizing it to less supersymmetric cases, other dimensions, other gauge groups, field theories without Lorentz symmetries, higher spin theories, and not in the least, doing all kinds of non-trivial checks. (in cases where there are alternate routes to compute on the gauge theory side). There are also numerous interesting applications to particle physics, cosmology, condensed matter theories, hydrodynamics, black hole physics.

Check of the symmetries

When we have a duality, i.e. two distinct descriptions of the same physical system, the first check one should perform is to see if the symmetries implied by each of the two descriptions is the same.

Let us do this for the AdS/CFT correspondence: The (bosonic) global symmetries of $\mathcal{N} = 4$ SYM theory are $SO(2,4) \times SO(6)$. Here the first factor corresponds to the conformal group in four spacetime dimensions, which is present since $\mathcal{N} = 4$ SYM is a conformal field theory. The second factor is the R-symmetry group which is a consequence of the fact that it has extended supersymmetry. We have seen before how it acts on the fields. A necessary condition for the duality to hold is that we should be able to see the same symmetries on the string theory side. We have string theory on the background $\mathrm{AdS}_5 \times S^5$, so that the global symmetries will be isometries of this spacetime. Indeed this has exactly the same symmetry. There is the $SO(2,4)$ of AdS_5, which we have already seen before. In addition the S^5 factor has $SO(6)$ symmetry, which thus precisely coincides with the R-symmetry of the gauge theory. So far we have talked about the bosonic symmetries. If one includes also the fermionic symmetries, the full symmetry group is known as $PSU(2,2|4)$.

7.3 What next ?

Field-operator map and much more...

We have already seen in the part on pre-strings AdS/CFT the basic workings of the holographic dictionary between a bulk theory on anti-de Sitter space and a boundary conformal field theory. With the AdS/CFT correspondence obtained above we have a very concrete realization of this. In particular, one can now find a precise relation between operators in $\mathcal{N} = 4$ SYM and the spectrum of type IIB string theory on $\mathrm{AdS}_5 \times S^5$. An essential ingredient in this 1-1 map is the full symmetry group $PSU(2,2|4)$, which tells us on the one hand that the operators on the field theory should be in representations of this group, which should thus be mapped to string states in the same representation. This map is most easily established for the weak form of the conjecture, in which the latter are supergravity fields.

An important (simplest) subset is formed by the class of 1/2 BPS (or chiral) primary operators. These are gauge invariant composite operators formed by taking traces (over the $SU(N)$ indices) of the scalar fields Φ^I. They transfrom in representations of $SO(6)$. The corresponding fields on the supergravity side can be obtained by decomposing the type IIB supergravity fields into Kaluza-Klein towers on S^5, using appropriate spherical harmonics. By applying this technical machinery one can find the concrete dictionary for these 1/2 BPS states.

Next, one would like to compute correlations functions on both sides and compare them. For 1/2 BPS states it follows from supersymmetry that the three-point functions should coincide, due to the fact that they are protected operators, i.e. the weak and strong coupling results should be the same. A beautiful check of AdS/CFT is that this is indeed the case. The weak coupling computation uses (large N) methods to compute these three point functions using perturbative quantum field theory. At strong coupling, where we thus use AdS/CFT, we use the dual supergravity picture. This involves the calculation of so-called Witten diagrams. An important ingredient here is holographic renormalization, which is a procedure to get finite well-defined results on the gravity side.

This is just the tip of the iceberg and the start of a huge development in which one can explore numerous directions. The table of contents of the book essentially gives you already some idea, and beyond that there are still plenty more. The field is still a highly active research area, aiming to deepen, extend, generalize, apply etc. the correspondence.

References and further reading

Sections 5.1 and 5.2 of the AF book. Also section 5.3, in particular 5.3.4.

There is a substantial body of books, reviews and lecture notes on the gauge/gravity correspondence. You can check out the AdS/CFT Wiki

References

[1] M. Ammon and J. Erdmenger, *Gauge/gravity duality: Foundations and applications*. Cambridge University Press, Cambridge, 4, 2015.

[2] M. D. Schwartz, *Quantum Field Theory and the Standard Model*. Cambridge University Press, 3, 2014.

[3] P. Di Francesco, P. Mathieu and D. Sénéchal, *Conformal Field Theory*. Graduate Texts in Contemporary Physics. Springer-Verlag, New York, 1997, 10.1007/978-1-4612-2256-9.

[4] K. G. Wilson, *Confinement of Quarks*, *Phys. Rev. D* **10** (1974) 2445–2459.

[5] J. Ambjørn and J. L. Petersen, *Quantum Field Theory (lecture notes)*, 1998.

[6] G. 't Hooft, *A Planar Diagram Theory for Strong Interactions*, *Nucl. Phys. B* **72** (1974) 461.

[7] S. Coleman, *Aspects of Symmetry: Selected Erice Lectures*. Cambridge University Press, Cambridge, U.K., 1985, 10.1017/CBO9780511565045.

[8] D. Tong, *Gauge Theory (lecture notes)*, 2018.

[9] P. H. Ginsparg, *APPLIED CONFORMAL FIELD THEORY*, in *Les Houches Summer School in Theoretical Physics: Fields, Strings, Critical Phenomena*, 9, 1988, hep-th/9108028.

[10] R. Blumenhagen and E. Plauschinn, *Introduction to conformal field theory: with applications to String theory*, vol. 779. Springer, 2009, 10.1007/978-3-642-00450-6.

[11] J. D. Qualls, *Lectures on Conformal Field Theory*, 1511.04074.

[12] D. Simmons-Duffin, *The Conformal Bootstrap*, in *Theoretical Advanced Study Institute in Elementary Particle Physics: New Frontiers in Fields and Strings*, 2, 2016, 1602.07982, DOI.

[13] S. Rychkov, *EPFL Lectures on Conformal Field Theory in D>= 3 Dimensions*. SpringerBriefs in Physics. Springer, 1, 2016, 10.1007/978-3-319-43626-5.

[14] D. Tong, *General Relativity (lecture notes)*, 2019.

[15] A. Zee, *Einstein Gravity in a Nutshell*. Princeton University Press, New Jersey, 5, 2013.

[16] M. Blau, *Lecture Notes on General Relativity*, 2019.

[17] G. Compère and A. Fiorucci, *Advanced Lectures on General Relativity*, 1801.07064.

[18] A. Zaffaroni, *Introduction to the AdS/CFT correspondence*, 2009.

[19] K. Skenderis, *Lecture notes on holographic renormalization*, *Class. Quant. Grav.* **19** (2002) 5849–5876, [hep-th/0209067].

[20] M. Henningson and K. Skenderis, *The Holographic Weyl anomaly*, *JHEP* **07** (1998) 023, [hep-th/9806087].

[21] M. Bertolini, *Lectures on Supersymmetry*, 2021.

[22] A. Bilal, *Introduction to supersymmetry*, hep-th/0101055.

[23] M. F. Sohnius, *Introducing Supersymmetry*, *Phys. Rept.* **128** (1985) 39–204.

[24] P. C. Argyres, *An Introduction to Global Supersymmetry*, .

[25] S. J. Gates, M. T. Grisaru, M. Rocek and W. Siegel, *Superspace Or One Thousand and One Lessons in Supersymmetry*, vol. 58 of *Frontiers in Physics*. Addison-Wesley, 1983.

[26] M. T. Grisaru, W. Siegel and M. Rocek, *Improved Methods for Supergraphs*, *Nucl. Phys. B* **159** (1979) 429.

[27] N. Seiberg, *The Power of holomorphy: Exact results in 4-D SUSY field theories*, in *Particles, Strings, and Cosmology (PASCOS 94)*, 5, 1994, hep-th/9408013.

[28] G. T. Horowitz and A. Strominger, *Black strings and P-branes*, *Nucl. Phys. B* **360** (1991) 197–209.

[29] G. W. Gibbons and K.-i. Maeda, *Black Holes and Membranes in Higher Dimensional Theories with Dilaton Fields*, *Nucl. Phys. B* **298** (1988) 741–775.